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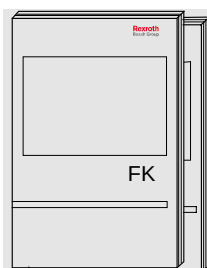
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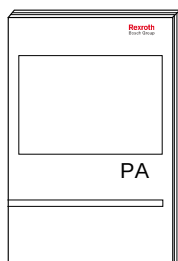
Summary of Documentation - Overview



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Help familiarize the user with SYNAX 200 and the functions of SYNAX 200

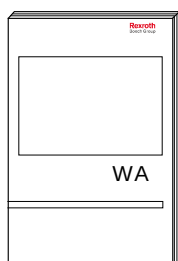
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Parameter Description:

Description of the SYNAX 200 system parameters

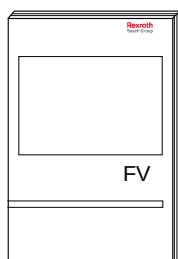
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Trouble Shooting Guide:

Explanation of the diagnostics states
How to proceed when eliminating faults

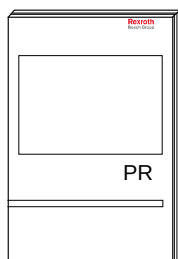
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Firmware Version Notes:

Description of the new and changed functions between SYNAX 200 version 12 and previous version 11

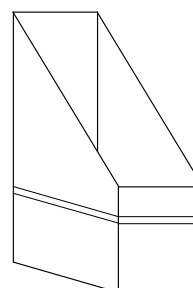
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Project Planning:

Selection of units and hardware components
Basic control in cabinet construction

Order designation:
DOK-SYNAX*-SY*-12VRS**-PR01-EN-P



Order designation:
DOK-SYNAX*-SY*-12VRS**-4001-EN-P

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1 General information about Fieldbus and HMI interfaces

1.1 Overview

SYNAX 200 offers interfaces for fieldbus communication and visualization purposes (HMI: Human Machine Interfaces). The following "open" communication interfaces are the base:

- Profibus master, Profibus slave
- DeviceNet master, DeviceNet slave
- Ethernet/IP (slave)
- Ethernet (slave)
- ARCNET (slave)
- 3964R interface with RK512 protocol
- RS232/422/485

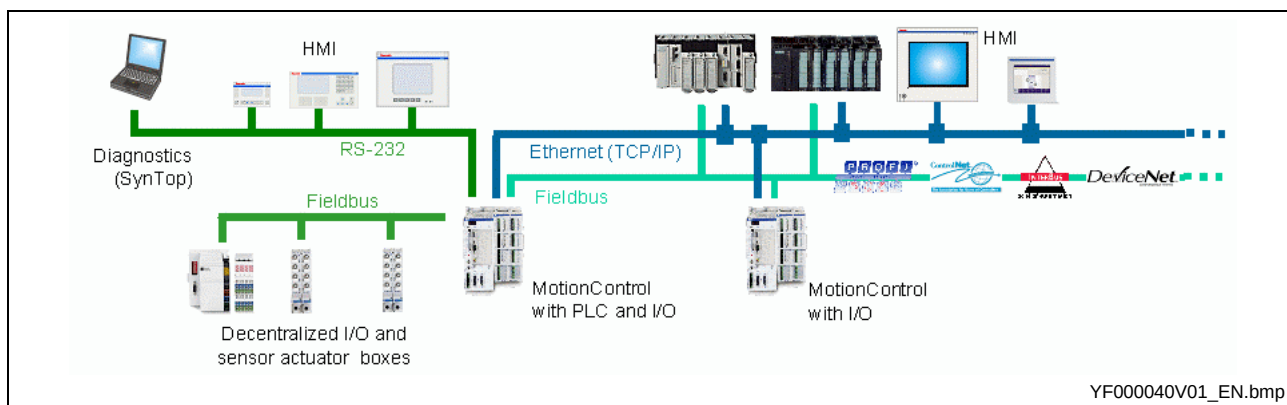


Fig. 1-1: Fieldbus/HMI coupling SYNAX 200

Fieldbus interfaces are used for communication of the motion control or automation system to superposed hierarchical levels (slave interfaces) and also to subordinate hierarchical levels (master interfaces). HMI interfaces are used for the communication of the motion control with process visualizations.

Often the data for process visualization is also transmitted via fieldbus.

In the following the fieldbus interfaces are shortly described. After that, the functions of the interfaces connected to the PPC are described (fieldbus slave interfaces and HMI interface Ethernet).

The interfaces connected to the integrated PLC (fieldbus master interfaces and HMI interfaces RS232/422/485) are described in detail in documentation "Rexroth SYNAX 200 - MotionLogic Version 12" (DOK-SYNAX*-IL*-12VRS**-AW01-EN-P).

Fieldbus master interfaces

Fieldbus master only with PLC

Fieldbus master interfaces are available in connection with the integrated PLC. Fieldbus master interfaces are option cards of the PLC.

The following fieldbus master interfaces are available:

- Profibus master (option card DPM01)
- DeviceNet master (option card DNM03)

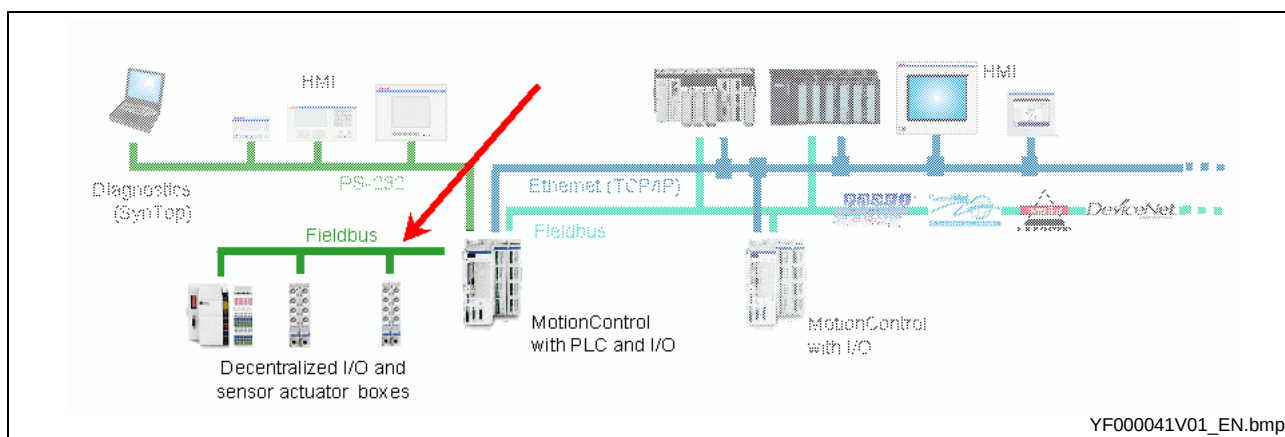


Fig. 1-2: Fieldbus master interfaces

Decentralized I/O or sensor actuator boxes are coupled to the PLC with the help of the fieldbus master interfaces (e.g., Rexroth Inline, Rexroth Fieldline). The information transmitted via the fieldbus is available in the PLC program.

Data client	Data server	transmitted data
PLC	decentralized I/O and sensor actuator boxes (Rexroth Inline, Rexroth Fieldline)	process signals (binary, analog)

Fig. 1-3: Communication connections fieldbus master interfaces

Detailed information can be found in documentation "Rexroth SYNAX 200 - MotionLogic Version 12" (DOK-SYNAX*-IL*-12VRS**-AW01-EN-P).

Fieldbus slave interfaces

Fieldbus slave is connected to the PPC

Fieldbus slave interfaces are available with and without integrated PLC. Fieldbus slave interfaces mostly are option cards of the MotionControl (PPC). At the PPC-R2x only an Ethernet interface exists on-board too.

The following fieldbus slave interfaces are available:

- Profibus slave (option card DPS01)
- Ethernet/IP slave (PPC-R0x and PPC-P11: option card ETH01, PPC-R2x: optional on-board)
- DeviceNet slave (option card DNS03)
- Ethernet (PPC-P11: option card ETH01, PPC-R2x: optional on-board)
- ARCNET with 7F protocol (option card DAQ03 or DAQ04)
- 3964R interface with RK512 protocol (via on-board serial interface of the PPC)

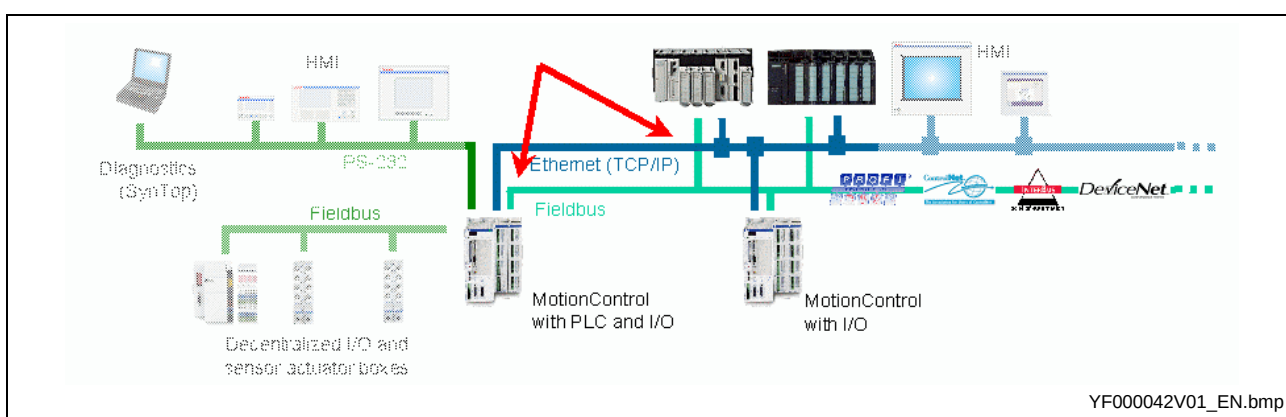


Fig. 1-4: Fieldbus slave interfaces

The SYNAX 200 system is coupled to superposed PLCs with the help of fieldbus slave interfaces. MotionControl parameters and PLC variables can be transmitted via fieldbus that means the superposed PLC can directly access to the MotionControl and the PLC.

Data client	Data server	transmitted data
superposed PLC	MotionControl (local I/O of RECO02, decentralized I/O of drive option cards)	MotionControl parameters, binary I/O data
superposed PLC	PLC (via MotionControl fieldbus channel)	PLC variables

Fig. 1-5: Communication connections fieldbus slave interfaces

HMI interfaces

HMI interfaces are available with or without integrated PLC. HMI interfaces are connected to the PPC.

The following HMI interfaces are available:

- Ethernet to PPC (PPC-P11: option card ETH01, PPC-R2x: optional on-board)
- RS232/422/485 (via existing serial interface)

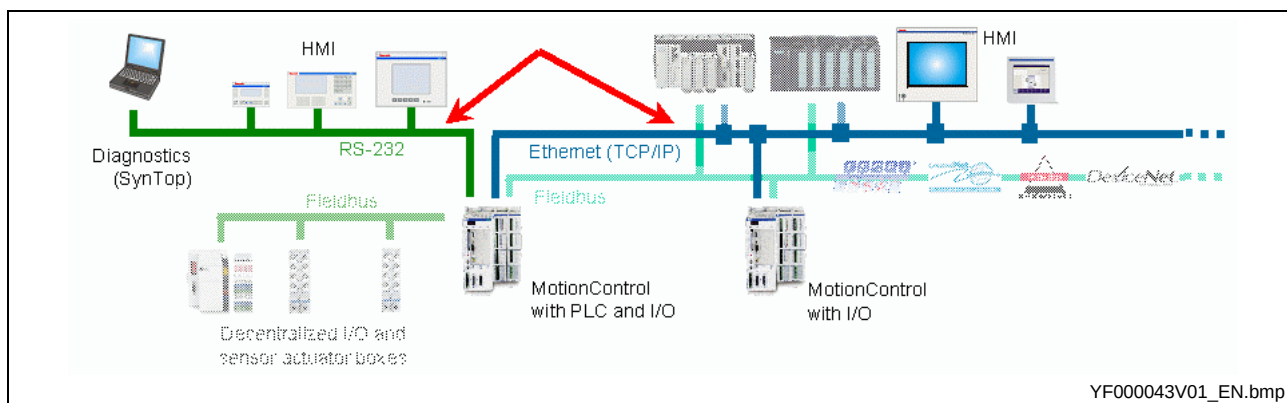


Fig. 1-6: HMI interfaces

The SYNAX 200 system is coupled to HMI systems with the help of the HMI interfaces.

Ethernet: access to PLC and MotionControl

MotionControl parameters and PLC variables can be transmitted via Ethernet interfaces, that means the superposed PLC can directly access to the MotionControl and the PLC.

PLC variables can be transmitted via RS232/422/485 interfaces. (Remark: With mapping the PLC variables to SYNAX parameters by means of the process data channel, it is possible to access indirect to SYNAX parameters.)

Data client	Data server	transmitted data
visualization (HMI)	MotionControl (local I/O of RECO02, decentralized I/O of drive option cards)	MotionControl parameters, binary I/O data
visualization (HMI)	PLC (via MotionControl)	PLC variables
visualization (miniature control panels)	PLC	PLC variables

Fig. 1-7: Communication connections HMI interfaces

1.2 Fieldbus transmission types: cyclic/non-cyclic transmission

The data transmission can be divided in two types:

- cyclic transmission,
- non-cyclic transmission.

The distinction refers to the efficiency and flexibility of the transmission. Depending on the fieldbus both transmissions are transmitted in the same frame or data telegram (e.g., parameter channel Profibus) or different telegrams for cyclic and non-cyclic transmission are used (e.g., explicit messaging at DeviceNet).

Cyclic transmission Cyclic transmission is more efficient, but less flexible.

The transmitted data are fixed and can be transmitted efficiently as a complete data block, an optional access to all data is not possible.

Non-cyclic transmission Non-cyclic transmission is more flexible, but less efficient.

The transmission is very flexible, because of the optional access mechanism to all data. The data are transmitted individually (protocol overhead).

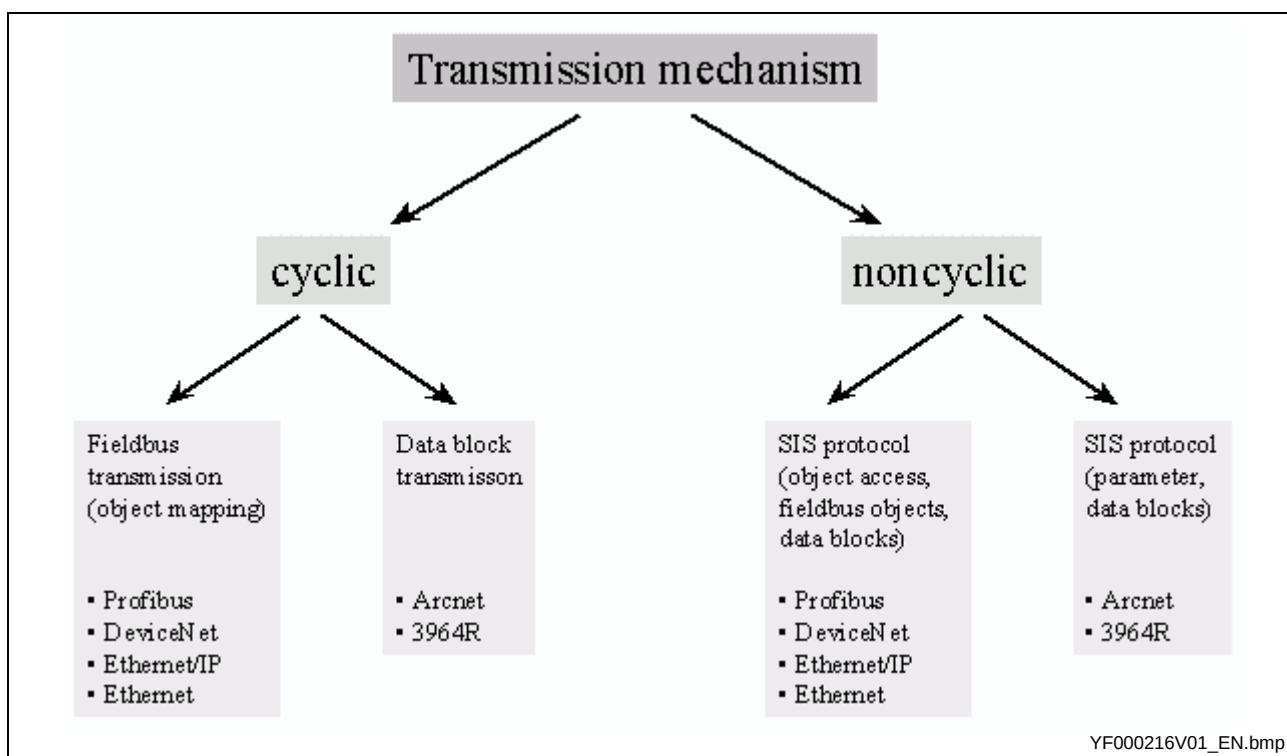


Fig. 1-8: Fieldbus transmission mechanism

1.3 Transmission mechanism of cyclic transmission

The transmitted data at cyclic transmission is fixed with SYNAX parameter (can not be changed in operating mode). Data is transmitted in groups, that means in blocks. The transmission in blocks is fieldbus-specific.

The following transmission mechanisms are available:

- Transmission in real time channel (Profibus, Ethernet/IP, DeviceNet, Ethernet)
- Transmission in data blocks (ARCNET, 3964R)

Up to 16 different data groups can be transmitted in blocks (in the so called multiplex operation or different data blocks).

Fieldbus	cyclic transmission	multiplex operation
Profibus	Fieldbus telegram (a maximum of 64 words)	up to 16 levels
Ethernet/IP	UDP/IP telegram (a maximum of 64 words)	up to 16 levels
DeviceNet	Polled I/O (a maximum of 64 words)	up to 16 levels
Ethernet	UDP/IP telegram (a maximum of 4 x 64 words)	up to 16 levels
ARCNET	Data blocks (DBs) in 7F protocol	up to 16 DBs
3964R	Data blocks (DBs) in RK512 protocol	up to 16 DBs

Fig. 1-9: Overview cyclic transmission

Transmission in the real time channel

Profibus, Ethernet/IP, DeviceNet, Ethernet

The real time channel is transmitted in every transmission cycle of the fieldbus. The transmitted data are defined in the parameterization mode and can not be changed in operating mode. Thus fixed data is transmitted in each transmission cycle.

Multiplex operation with 16 levels

A part of the real time channel can be used in the so called multiplex operation. This multiplex operation is optional and permits to allocate parts of the real time channel with up to 16 different data groups in both directions. The selection of the active multiplex levels (= active data group) is also transmitted in the real time channel.

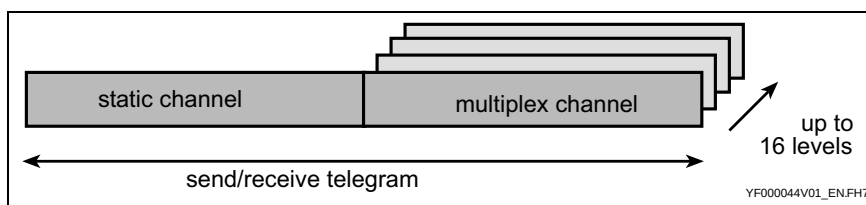


Fig. 1-10: Real time channel with static channel and multiplex channel

Transmission in data blocks

ARCNET, 3964R

The transmitted data block is selected via ARCNET or 3964R in every transmission cycle .

Data block transmission with 16 data blocks

For data transmission 16 different data blocks are available. The selection of the transmitted data group is made in the header of the data block.

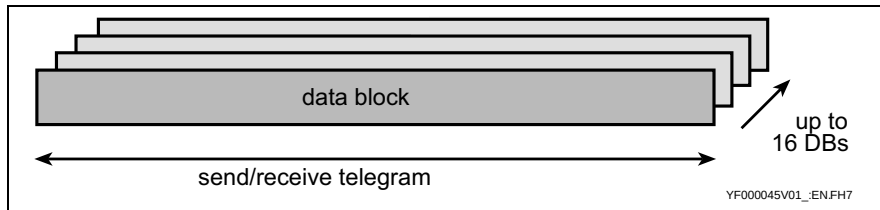


Fig. 1-11: Data block transmission

At fieldbusses the same transmission mechanism of the data blocks can be used acyclically.

Cyclic transmission in different operating modes

All parameters that are applied in the cyclic channel are transmitted in operating mode.

In all other modes (initialization mode, parameterization mode) the behavior can be configured with parameter "Fieldbus - control bits" (Y-0-0129):

- Cyclic channel is only processed in operating mode
- Cyclic channel is processed in all other modes in addition

Processing only in operating mode

If the cyclic channel is only processed in operating mode, the data that is transmitted in the cyclic channel

- is set to "0" in read direction (PPC -> fieldbus master) and
- is not updated in write direction (fieldbus master -> PPC) and is written not until operating mode is reached (cache function)

in all other modes (SERCOS phases smaller than 4).

Processing in all modes

If the cyclic channel is also activated in other modes (SERCOS phases smaller than 4),

- Y and C parameters are processed in write and read direction,
- A, S and P parameters are set to "0" in read direction,
- A, S and P parameters are not written in write direction until reaching the operating mode (cache function) and
- the multiplex channel function is active.

Memory mode of the transmitted data

It can be configured for the following parameters, if data is buffered at a switch off:

- "Fieldbus - External IO, PPC input word 1 – 32"
(Y-0-0200 - Y-0-0231)
This is the I/O data from a superposed PLC to the I/O logic or integrated IndraLogic.
- "Fieldbus - PLC input variable data container 1 (Y363 = 64)"
(Y-0-0300 - Y-0-0363)
This is the data that is routed from a superposed PLC directly to the integrated IndraLogic.

All other writable A/C/Y/S/P parameters are automatically buffered at a switch off.

It can be configured in parameter "PPC - configured options" (Y-0-0531), if the transmitted values

- are initialized with "0" at a switch on,
- are initialized with the last transmitted values before switching off at switch on (buffering).

1.4 Transmission of the non-cyclic transmission

The non-cyclic transmission permits the optional access to all SYNAX parameters. Depending on the transmission protocol or fieldbus, the non-cyclic transmission is transmitted within the cyclic transmitted telegrams, or some telegrams are used for non-cyclic transmission.

Different protocols are used for transmission:

- so called SIS protocol (Profibus, Ethernet/IP, DeviceNet, Ethernet)
- direct acyclic access via fieldbus objects (Profibus, Ethernet/IP, DeviceNet)
- so called protocol extension (ARCNET, 3964R)

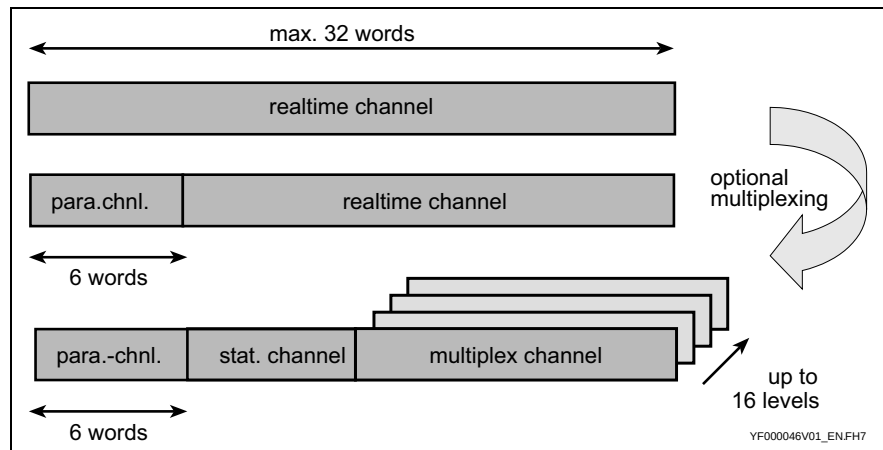
Fieldbus	Non-cyclic transmission
Profibus	transmission in the fieldbus telegram: - direct acyclic access via fieldbus objects - SIS protocol (single parameters and data blocks)
Ethernet/IP	transmission via explicit messaging: - direct acyclic access via class/instance/attribute - SIS protocol (single parameters and data blocks)
DeviceNet	transmission via explicit messaging: - direct acyclic access via class/instance/attribute - SIS protocol (single parameters and data blocks)
Ethernet	transmission in the TCP/IP telegram: - SIS protocol (single parameters and data blocks)
ARCNET	transmission in the ARCNET 7F protocol: - via protocol extension
3964R	transmission in the RK512 protocol: - via protocol extension

Fig. 1-12: Overview non-cyclic transmission

1.5 Mapping the transmission in the fieldbus slave or HMI interfaces

This section gives an overview about the mapping or integration of the data transmission to the fieldbus telegrams. Also see Fig. 1-9 and Fig. 1-12.

Profibus



para. chnl.: parameter channel

stat. channel: static channel of the real time channel

Fig. 1-13: Profibus telegram structure

The following telegram structures are possible with Profibus:

- only real time channel
- real time channel with parameter channel
- real time channel with multiplexing (not described in Fig. 1-13)
- real time channel with parameter channel and multiplexing

Ethernet/IP

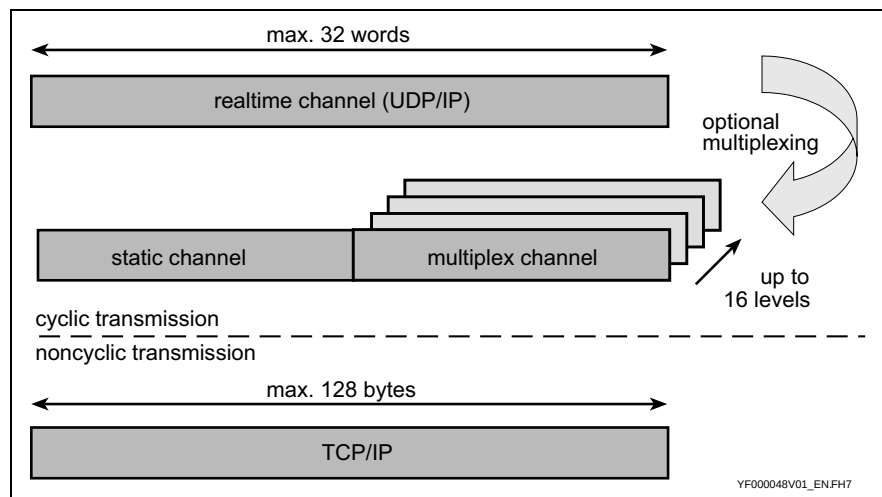


Fig. 1-14: Ethernet/IP telegram structure

The following telegram structures are possible with Ethernet/IP:

- only real time channel
- real time channel with multiplexing

Independently the non-cyclic transmission is handled via explicit messaging.

DeviceNet

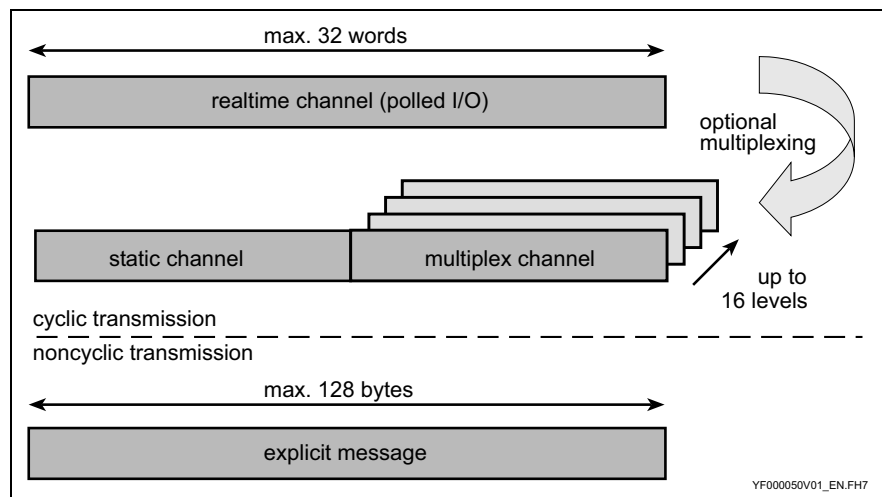


Fig. 1-15: DeviceNet telegram structure

The following telegram structures are possible with DeviceNet:

- only real time channel
- real time channel with multiplexing

Independently the non-cyclic transmission is handled via explicit messaging.

Ethernet

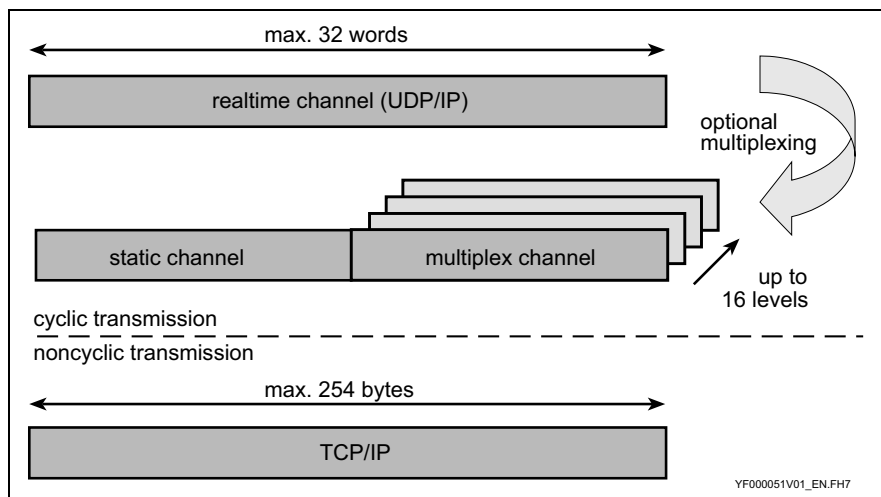


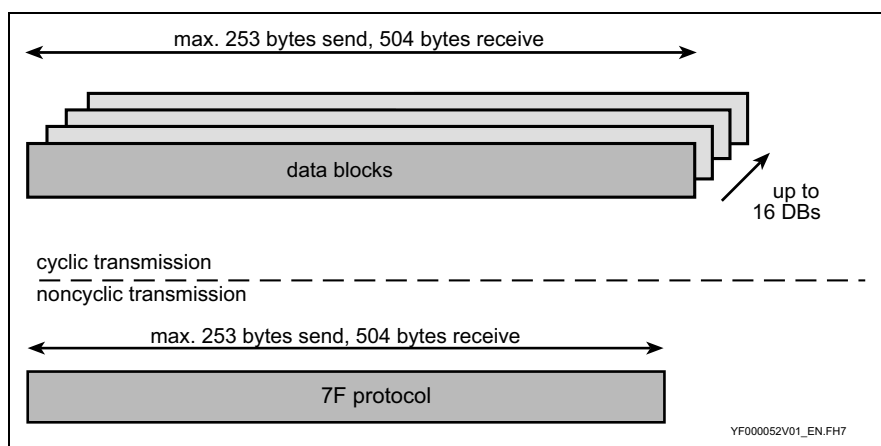
Fig. 1-16: Ethernet telegram structure

The following telegram structures are possible with Ethernet:

- only real time channel
- real time channel with multiplexing
- up to 4 fieldbus master with any own transmitted data

Independently the non-cyclic transmission is handled via TCP/IP (up to 4 active TCP/IP connections).

ARCNET



DBs: data blocks

Fig. 1-17: ARCNET telegram structure

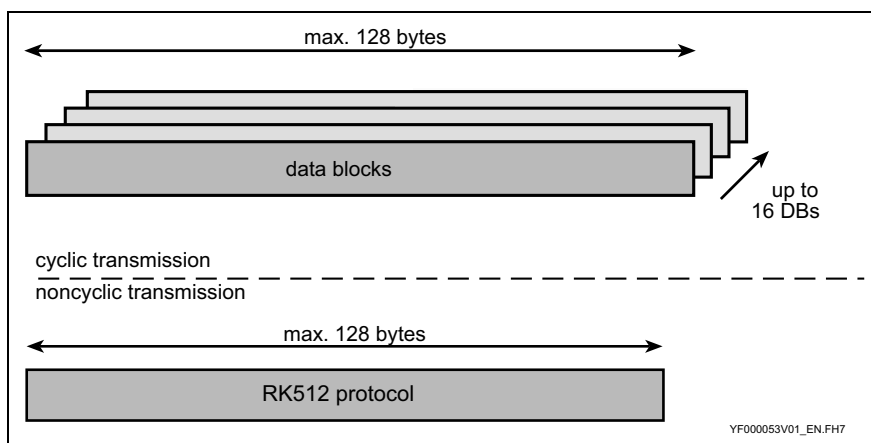
The following telegram structures are possible with ARCNET:

- up to 16 different data blocks

Any numbers of fieldbus master with up to 16 different data blocks are possible.

In addition the non-cyclic communication is handled via ARCNET 7F protocol (with any numbers of ARCNET master).

3964R



DBs: data blocks

Fig. 1-18: 3964R telegram structure

The following telegram structures are offered by the 3964R transmission:

- up to 16 different data blocks

In addition the non-cyclic transmission is handled via the RK512 protocol.

1.6 Parameter structure

SYNAX parameter structure

SYNAX parameters are always made up of 7 elements: the identity number, the name, the attribute and the operating data (parameter). Depending on the data type there can be a unit and a minimum and maximum input value.

SYNAX parameter structure:

Element 1	<Ident no.>	[see structure of identity number]
Element 2	<Name>	[name of the operating data, maximum 60 char.]
Element 3	<Attribute>	[see attribute structure]
Element 4	<Unit>	[operating data unit, maximum 12 char.]
Element 5	<Min. input value>	[depicted like operating data]
Element 6	<Max. input value>	[depicted like operating data]
Element 7	<operating data>	[see structure op. data]

Fig. 1-19: SYNAX parameter structure

Identity number structure

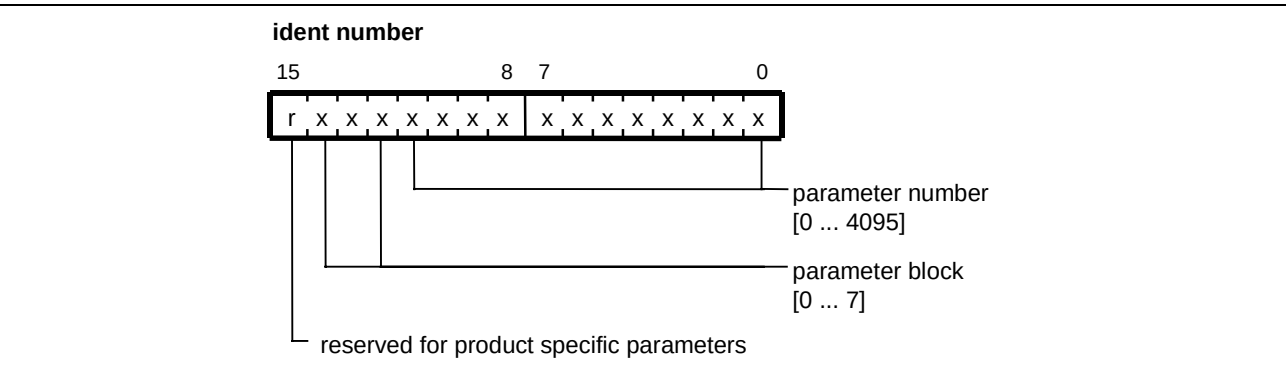


Fig. 1-20: Identity number structure

Attribute structure:

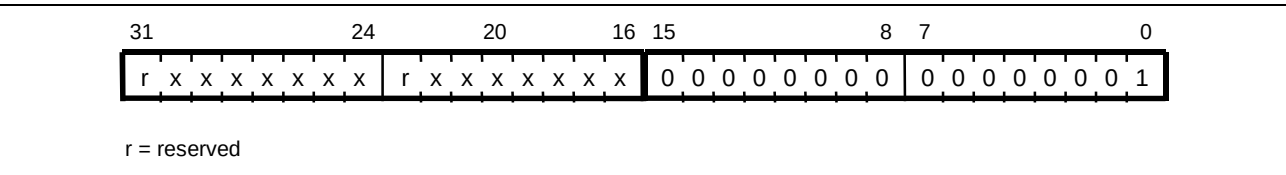


Fig. 1-21: Attribute Structure

Bit 15 - 0: Evaluation factor	The evaluation factor is a qualifying signal for integers and must be set at 1.	
Bit 18 - 16: Data length	0 0 0 - reserved 0 0 1 - Data is 2 bytes long 0 1 0 - Data is 4 bytes long 0 1 1 - reserved 1 0 0 - Data has a variable length of 1 bytes data 1 0 1 - Data has a variable length of 2 bytes data 1 1 0 - Data has a variable length of 4 bytes data 1 1 1 - reserved	
Bit 19: Function	0 - operating data or parameter 1 - command	
Bit 22 - 20: Data type and display format	Data type 0 0 0 - binary number 0 0 1 - unsigned, integer 0 1 0 - integer 0 1 1 - unsigned, integer 1 0 0 - ASCII character 1 0 1 - unsigned, integer 1 1 0 - float 1 1 1 - SYNAX identity number	display format binary decimal unsigned decimal signed hexadecimal text (MS-DOS) identity number 32 bit float with sign and exponent (single precision, ANSI/IEEE Std 754-1995) (see "Identification of SYNAX parameters, page 1-15)
Bit 27 - 24: Decimal places	0 0 0 0 - no decimal places ... 1 1 1 1 - 15 decimal places (maximum)	
Bit 28 - write protection phase 2	0 - no write protection in phase 2 (parameter mode) 1 - write protected in phase 2 (parameter mode)	
Bit 29 - write protection phase 3	0 - no write protection in phase 3 1 - write protection in phase 3	
Bit 30 - write protection phase 4	0 - no write protection in phase 4 (operating mode) 1 - write protection in phase 4 (operating mode)	

Operating data structure

The operating data has a data length of either

- 2 bytes, 4 bytes or
- a variable length of 0 to 65532 bytes.

The length of operating data with variable length is given by the first two bytes.

The third and fourth byte carry the maximum length of the operating data made available by the drive or PPC respectively. The data starts with the fifth byte.

Identification of SYNAX parameters

The identity number (element 1) of a parameter alone is not sufficient to clearly identify all SYNAX parameters. The information about data type, drive or master axis address are also necessary.

To configure parameters that should be transmitted in the cyclic channel of fieldbusses or via data blocks the identity number is expanded to a SYNAX identity number.

The SYNAX identity number is a 32 bit value to determine the unit address, the parameter type and the parameter number.

In the SYNAX identity number format the parameter number corresponds to the identity number of the parameter (element 1).

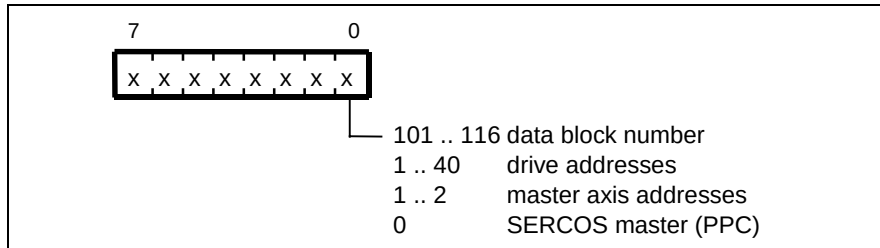


Fig. 1-22: Unit address

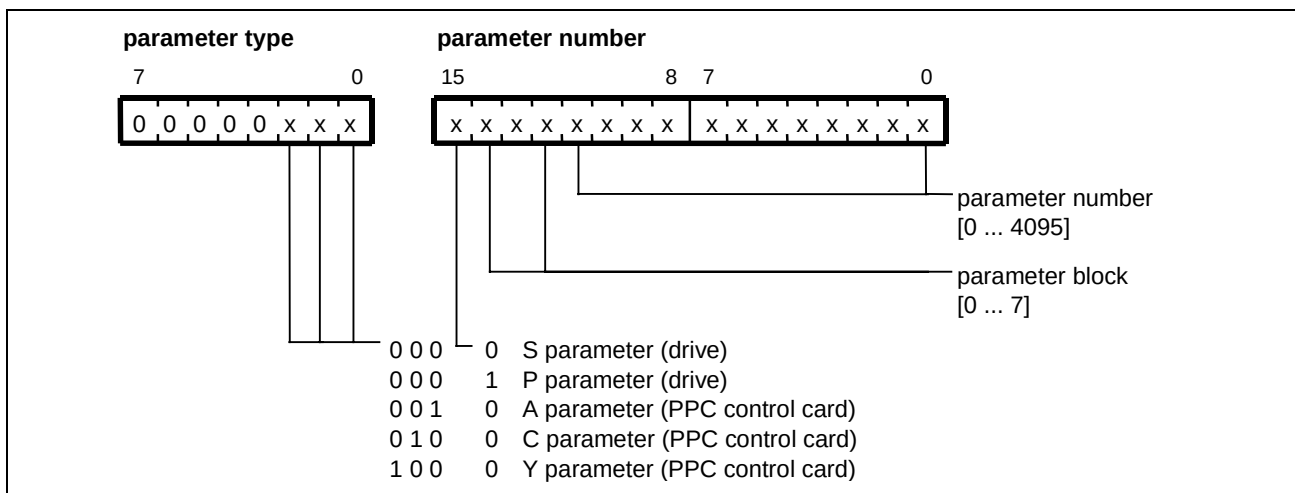


Fig. 1-23: Parameter identification (parameter transmission)

2 Fieldbus and Ethernet interfaces

2.1 Introduction

This section describes the general part of the fieldbus interfaces Profibus, Ethernet/IP and DeviceNet and the interface Ethernet-PLC and Ethernet-HMI of the PPC motion control.

Ethernet-PLC supports cyclic and acyclic data exchange as the fieldbusses and is handled in the following with the designation Ethernet just like a fieldbus.

Ethernet-HMI is an additional acyclic interface.

These interfaces can be used

- to transmit binary signals as a substitute for 24 volt I/Os,
- for the cyclical transmission (except Ethernet-HMI) of command and actual values while in operation, e.g., velocity command value / actual value of the virtual master axis,
- and non-cyclical transmission of parameters and phase progressions, e.g., a product-dependent reparameterization or read out of the diagnosis.

Note: Due to the internal running time of communication using a fieldbus a cyclical position control via the fieldbus is not possible.

The fieldbus mostly is connected via a plug-in card that is permanently linked to the PPC. The firmware of the plug-in card works like a fieldbus slave and internally exchanges data with the PPC via the Dual-Port-RAM. At the PPC-R2x only an Ethernet interface exists on-board too.

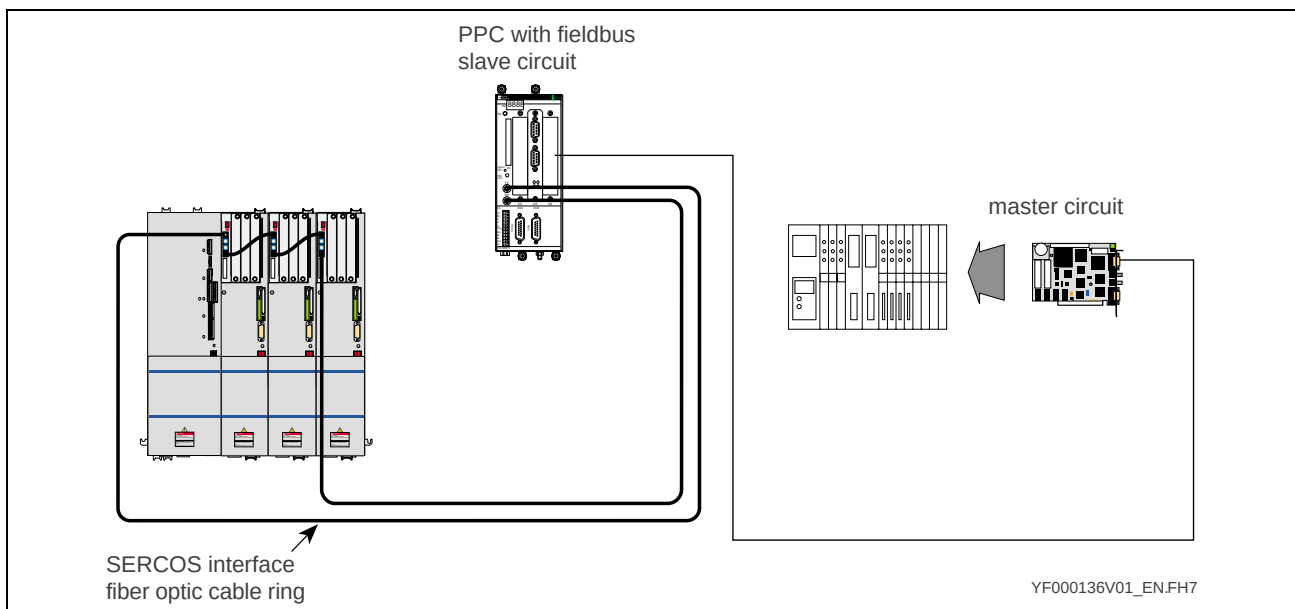


Fig. 2-1: Topological structure of master slave communication

Functional module To make it easier to connect SYNAX to an external PLC via a fieldbus Bosch Rexroth provides functional modules, e.g., for the Simatic-S7. Please contact the applications group whether there are examples for the PLC and the fieldbus which you use.

To connect SYNAX to a PLC, predefined function blocks are available for the acyclic communication channel.

2.2 Data objects

In non-cyclical transmission (communication channel) there are four data exchange objects. These can be used to R/W any parameter, (including list parameters and data blocks) and the PPC mode which is embedded in a special protocol. These four objects are byte arrays of various lengths (see "Fieldbus objects for data exchange", page 2-17).

The direct non-cyclic data exchange at fieldbusses (except Ethernet) takes place via parameter objects and only allows direct accesses to single parameters according their access rights (see "Fieldbus objects for data exchange", page 2-17).

Various transmission channels are available for transferring data depending on the fieldbus. These will be described in the following texts.

2.3 Transmission channels

Process data channel

The process data channel includes all words cyclically exchanged between the fieldbus master and the PPC and in real time. It is up to 64 words long with the Ethernet, Profibus, Ethernet/IP and the DeviceNet.

In the case of DeviceNet, Ethernet/IP and Ethernet the process data channel and the real time channel are identical. The process data channel of Profibus is made up of a real time channel and an optional parameter channel.

Real time channel

The real time channel is a part of the process data channel. In the real time channel the I/Os and the cyclical command and actual values are transmitted. The multiplex channel is optionally at the back. All data in the real time channel are processed by the PPC in operating mode only.

At start-up, this input and output data is parameterized via SynTop (Y-0-0127, Y-0-0128).

The static, non-multiplexed I/O data and parameters are entered in the first part of the configuration lists. The static real time channel is terminated with the entry Y00: Y-0-0197.

The further parts of the configuration lists can be used to configure the individual levels of the multiplex channel.

Multiplex channel

The multiplex channel starts after the static part and is configured with the requisite I/O data and parameters level by level, analogously to the static channel. Each layer is terminated with the entry Y00: Y-0-0198. The various levels can be different with regard to data structure and data length. The process data length automatically is the result of the longest multiplex level. The PPC automatically affixes the multiplex control word and the multiplex status word in the last word.

Depending on an index specified by a master, both the static data and the multiplex data are transmitted in either direction during operation, according to the multiplex level specified.

Communication channel

The communication channel is for non-cyclical transmission.

Direct data accessing to the contents of the container objects and the data exchange objects is not possible. Direct accessing to the other parameter objects is possible except of Ethernet.

Using the four data exchange objects it is possible to R/W indirectly

- any A, C, Y, S and P parameter (includes list parameters and data blocks) and
- of the PPC mode

embedded in a special protocol (Bosch Rexroth SIS protocol). (see "Fieldbus objects for data exchange", page 2-17).

The communication channel in Profibus to the Bosch Rexroth parameter channel, with DeviceNet and Ethernet/IP the Explicit Message and with Ethernet the acyclic TCP connection.

Parameter channel

The parameter channel is only available with Profibus-DP. The first six words in the process data channel are reserved for this purpose.

In order to transmit a single parameter embedded in SIS protocol in a PLC cycle, the abbreviated format 1 was additionally implemented (see "Short format 1 for parameters" in section 3-5).

Ethernet HMI channels

In parallel with the configured host communication interface (Y-0-0033), up to seven non-cyclic Ethernet HMI connections are provided automatically if an Ethernet board is detected during power-up of the PPC (optional board ETH01 or optional at the PPC-R2x on-board)..

2.4 Transmission time in the real time channel

The transmission of an I/O signal or a parameter in the real time channel, e.g., from PLC program to the point of effectiveness of the PPC, can be broken down into the following sections:

- The PLC puts the data in each PLC cycle into the dual port RAM between CPU and fieldbus master.
- The fieldbus master puts the data on the bus.
- The fieldbus interface card of SYNAX takes the data and puts it on the dual port RAM between interface card and PPC.
- The PPC copies the data out of the dual port RAM into a data buffer.
- I/O signals are processed in the I/O logic which is activated every 8 ms. The handler for the A, C, Y parameters writes all changed parameters in a 10ms cycle. The handler for the S, P parameters is activated every 200ms. It writes a changed drive parameter and reads one of the drive parameters that is not configured in the AT.

Note: The data handler needs ten SERCOS cycles for a write access to an S or P parameter (single parameters), and 10 SERCOS cycles for a read access to an S or P parameter if this parameter has not been configured in the drive telegram (S-0-0016). A pre-requisite is that the service channel of SERCOS is not occupied!

- The changed parameter value becomes effective once the value is read by the relevant PPC function the next time.

The transmission section between fieldbus circuit and PPC is displayed in Fig. 2-2: Temporal behavior when write accessing via real time channel.

This demonstrates that transmission time depends on

- the PLC cycle time,
- the fieldbus cycle time,
- the SERCOS cycle time,
- the number of the simultaneously changed parameters,
- the parameter type to be transmitted and
- the PPC load.

A general rule for a conservative projection is:

Time critical I/O signals with system reaction times < 100ms should not be transmitted via the fieldbus but instead, for example, via a RECO module.

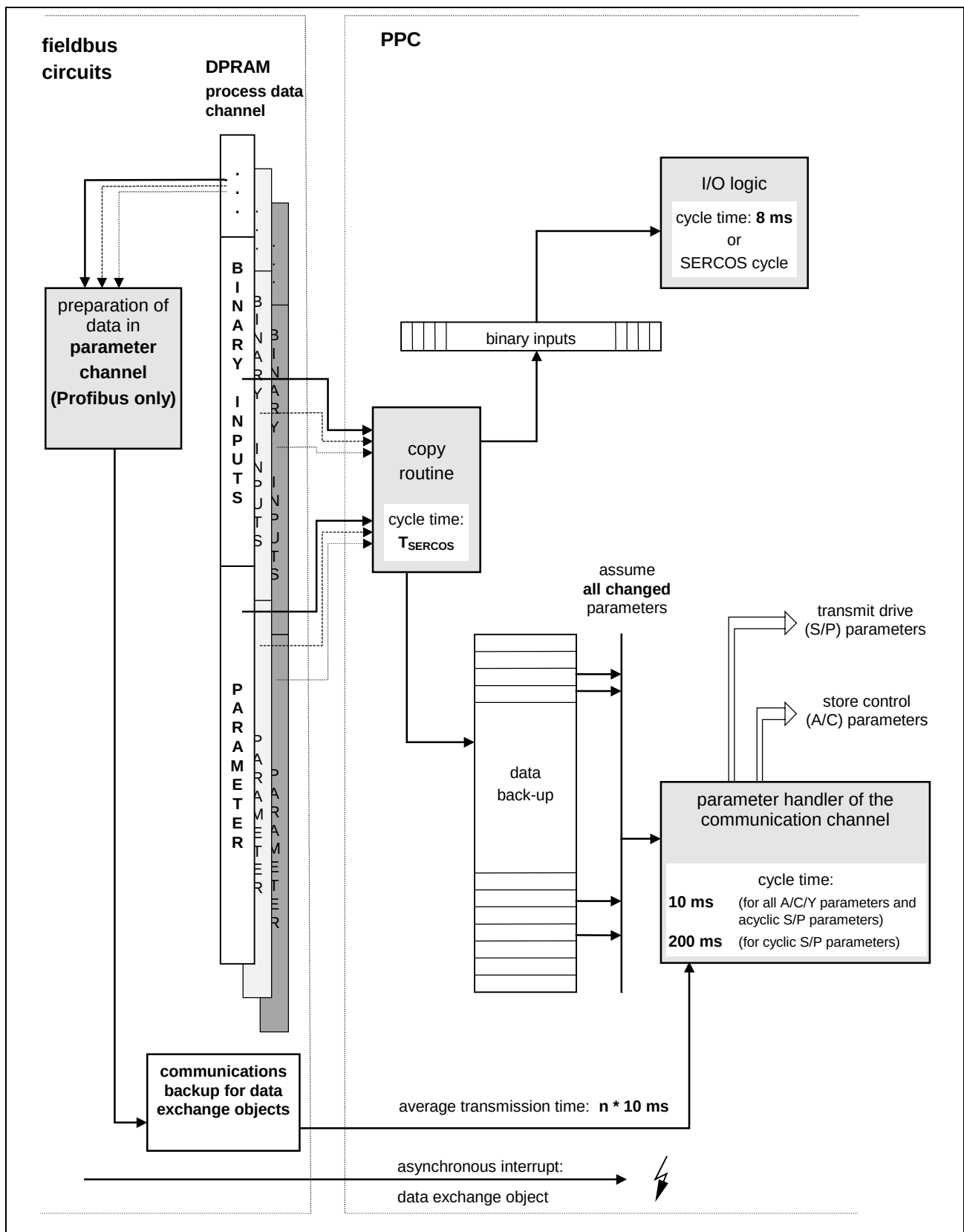


Fig. 2-2: Temporal behavior when write accessing via real time channel

2.5 Monitoring fieldbus transmission

Data safety

Transmitting data via the fieldbus is safe. If data are transmitted from master to the slaves or vice versa, then these are subjected to extensive checks. Data detected as faulty during transmission are discarded and thus ineffective. With Profibus the hamming distance = 4, in Ethernet/IP and DeviceNet = 6, i.e., in Profibus three bit errors are detected, with Ethernet/IP and DeviceNet 5 bit errors per message.

Behavior with bus failure

The fieldbus can fail. Failure can be caused by a cable break or slaves crashing. In this case, the once entered data is retained until the new data is received.

Watchdog function The fieldbus module has a bus monitor equipped with a watchdog timer and is thus able to immediately respond to any bus failure.

Error reaction The fieldbus module makes it possible to set different error reactions for the process data channel which can meet varying demands.

The behavior with bus failure is parameterized differently in each fieldbus.

2.6 Configuration of the real time channel

If a fieldbus interface has been parameterized in the "Host communication - control word" (Y-0-0033) then the fieldbus configuration always takes place with SynTop via the PPC.

Note: Bus configurations via the PPC mean that the bus master must not configure.

The number of parameters configured by SYNAX is much greater than the number of objects fixed for the fieldbus for transmission of parameters. To avoid limiting yourself with pre-selected parameters in the case of a SYNAX application with 40 drives, the fieldbus interface of the PPC can be freely configured.

Parameters for the fieldbus interface

The parameters shown in Fig. 2-3 are to be used for the fieldbus interface and for transmission in the real time channel. The parameters are all set via SynTop.

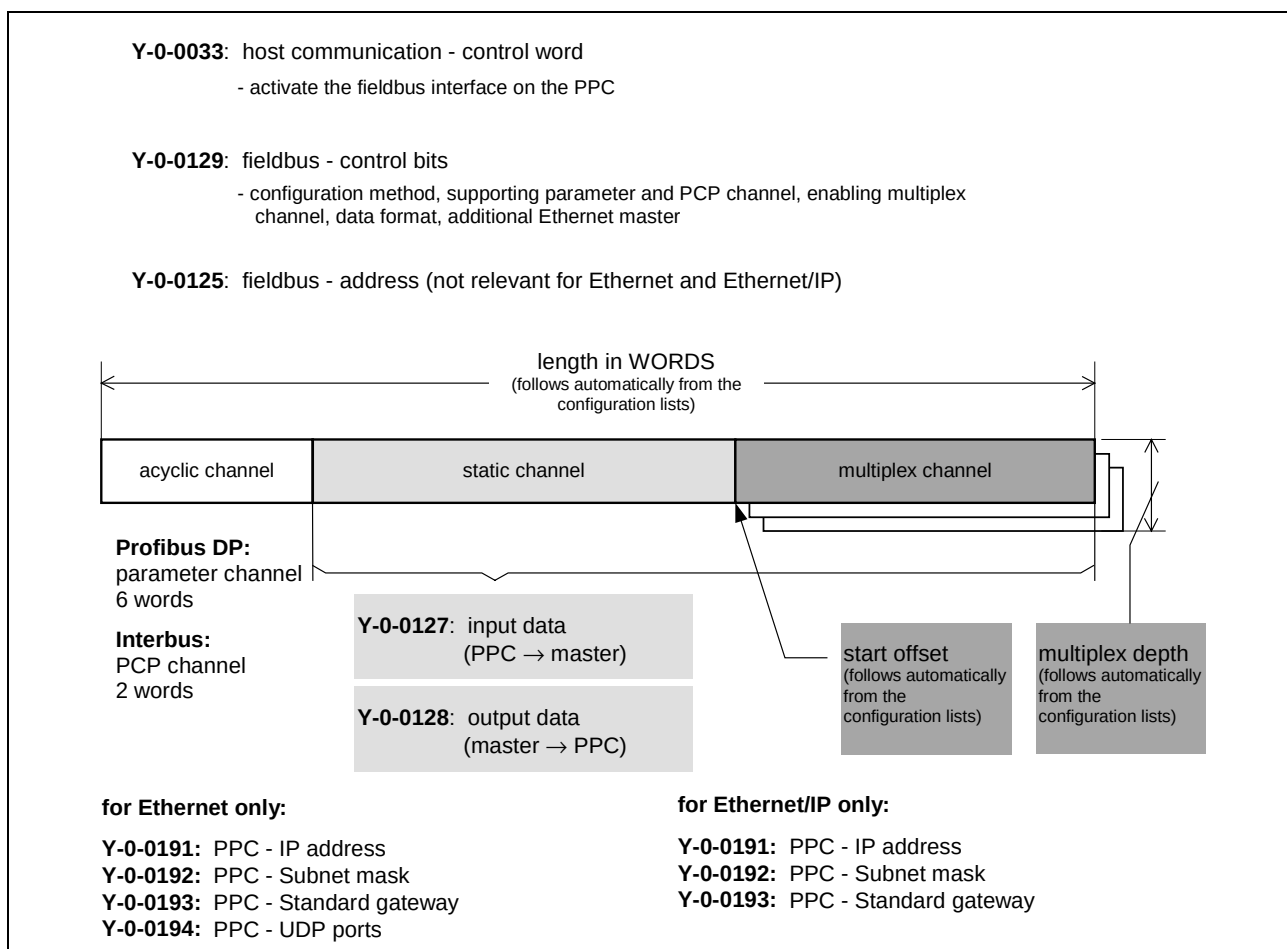


Fig. 2-3: Parameters for the fieldbus interface and the configuration of the real time channel

Note: The parameters are individually explained in the SYNAX 200 Parameter Description (DOK-SYNAX*-SY*-12VRS**-PA01-EN-P).

Bus configuration via the PPC

The slave is configured via SynTop fieldbus dialogs. The master does not have to undertake any other configurations.

The configuration can be broken down into five steps:

Step 1: Determining requirements

In the first step, determine the number of I/O signals and the parameters (command/actual values) to be cyclically transmitted when the machine is in operation. Using the attribute of the parameter determine whether the value has two or four bytes. This data is then later transmitted in the real time channel. List parameters cannot be transmitted in the real time channel.

Define all parameters via the fieldbus that might need to be R/W or with a product change. This data is transmitted - if possible - in the communication channel or parameter channel. Parameters write protected in operating mode can only be altered when in the communication or parameter channel.

	SPS → PPC	PPC → SPS
inputs/outputs $5 * 5 + 10 = 35$ inputs/outputs command value during production VM - speed command (C-0-0006) = 4 bytes setup position (A-0-0056) = 4 bytes adjustable parameter selection (A-0-0013) = 2 bytes monitoring and diagnosing VM - actual speed value (C-0-0067) = 4 bytes add. pos. comm. value (A-0-0004) = 4 bytes diagnostics text (Y-0-0047) = text string product specific data electronic gear (A-0-0170/A-0-0126) = parametr. mode! lead drive polarity (P-0-0108) = parametr. mode!	3 words for binary inputs 4 4byte objects 1 2byte object comm. channel/parameter channel	3 words for binary outputs 5 4 byte objects comm. channel/parameter channel

Fig. 2-4: Example of step 1: "determining requirements"

Step 2: Generating a concept

Put the parameters in the real time channel on the available data words. Start with the I/O signals.

If it is required to transmit more words than there is space in the real time channel, you have to decide which parameter you do not need to transmit each fieldbus cycle. These can be multiplexed in the back part (e.g., Fig. 2-5, word 12-15).

Reserve in this case an additional word for the multiplex control/status word.

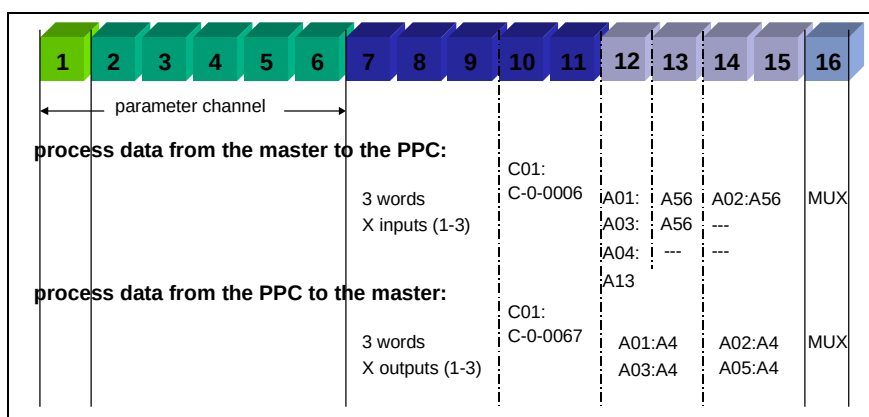


Fig. 2-5: Example (Profibus) of step 2 "generating a concept", the first six word parameter channel only with Profibus

Step 3: General settings

Activate the fieldbus via the SynTop dialog "PPC host communication basic settings" and set the parameters in the SynTop dialog "fieldbus settings".

Step 4: Configuring the assignment of the real time channel

On the basis of the concept of step 2 the real time channels is assigned with the designated parameters and the desired structure. The static part is terminated with Y00: Y-0-0197 and each multiplex level with Y00: Y-0-0198. The multiplex control/status word can not be found in the configuration list. The input lists in the SynTop dialog "Fieldbus process data channel" are stored on the PPC in parameters Y-0-0127, Y-0-0128.

Parameter	Name of parameter	Remarks
Y00: Y-0-0200	Fieldbus - External IO, PPC input word 1	static part: 1. word
Y00: Y-0-0201	Fieldbus - External IO, PPC input word 2	static part: 2. word
Y00: Y-0-0202	Fieldbus - External IO, PPC input word 3	static part: 3. word
C01: C-0-0006	VM - speed command 1	static part: 4./5. word
Y00: Y-0-0197	Configuration element process data channel - end of static	ID: end of static part
A01: A-0-0056	Stop position 0	Multiplex level 1: 1./2. word
A02: A-0-0056	Stop position 0	Multiplex level 1: 3./4. word
Y00: Y-0-0198	Configuration element process data channel - end of multiplex	ID: end of Mux level 1
A03: A-0-0056	Stop position 0	Multiplex level 2: 1./2. word
Y00: Y-0-0198	Configuration element process data channel - end of multiplex	ID: End of Mux level 2
A04: A-0-0013	Adjustable parameter selection	Multiplex level 3: 1. word
Y00: Y-0-0198	Configuration element process data channel - end of multiplex	ID: End of Mux level 3

Fig. 2-6: Example (Profibus) for step 4 "configuration of assignment of real time channel", data direction from master to PPC. (Six word parameter channel only with Profibus)

Parameter	Name of parameter	Remarks
Y00: Y-0-0250	Fieldbus -External IO, PPC output word 1	static part: 1. word
Y00: Y-0-0251	Fieldbus -External IO, PPC output word 2	static part: 2. word
Y00: Y-0-0252	Fieldbus -External IO, PPC output word 3	static part: 3. word
C01: C-0-0067	Output speed	static part: 4./5. word
Y00: Y-0-0197	Configuration element process data channel - end of static	ID: end of static part
A01: A-0-0004	Position command offset	Multiplex level 1: 1./2. word
A02: A-0-0004	Position command offset	Multiplex level 1: 3./4. word
Y00: Y-0-0198	Configuration element process data channel - end of multiplex	ID: end of Mux level 1
A03: A-0-0004	Position command offset	Multiplex level 2: 1./2. word
A04: A-0-0004	Position command offset	Multiplex level 2: 3./4. word
Y00: Y-0-0198	Configuration element process data channel - end of multiplex	ID: end of Mux level 2

Fig. 2-7: Example (Profibus) for step 4 "configuration of assignment of real time channel", data direction from PPC to master. (Six word parameter channel only with Profibus.)

All fieldbus objects for data exchange are listed in "Fieldbus objects for data exchange", page 2-17.

The configuration list has a variable length and a data width of 4 bytes. The parameter to be transported is entered according to the following syntax in each element of a list that is used (SYNTAX format):

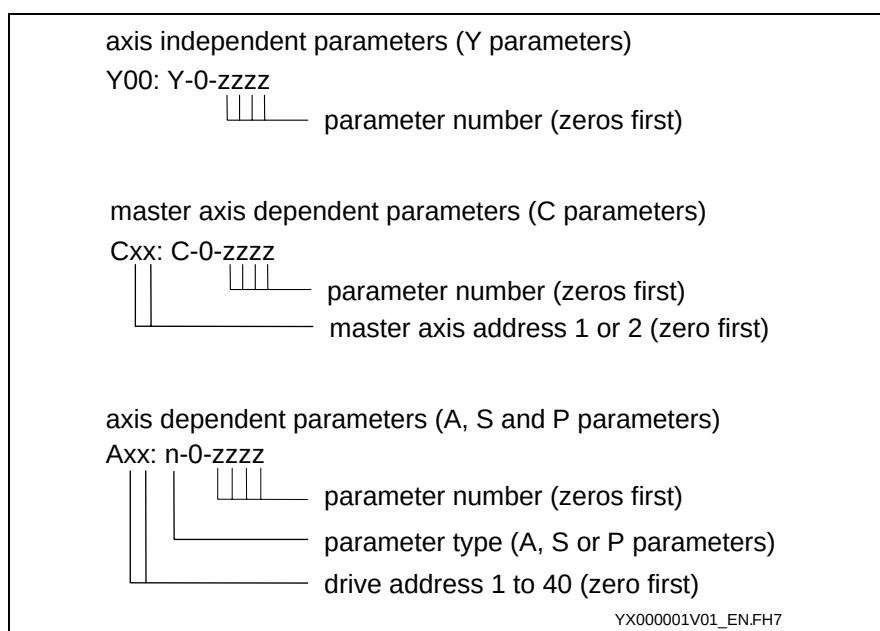


Fig. 2-8: Parameter structure in SYNAX identity number format

Control parameters The process variables are displayed in the A, C and Y parameters on the PPC. These are generally exchanged in real time between PLC and PPC. These variables can be configured as static parameters or multiplex parameters in the real time channel.

Drive parameters The S and P parameters are transmitted via the SERCOS fiber optic ring between the PPC control and the individual drives. Access times depend on whether a parameter is exchanged as "real time data" between PPC and drive or via a slower "service data channel". So as not to block the communication tasks on the PPC and the access via SynTop, the PLC should only read S and P parameters in the real time channel that have been cyclically configured via SERCOS (see S-0-0016). S and P parameters should generally not be written.

Note: When transmitting drive parameters, i.e., S and P parameters, do use the non-cyclical channel (communication / parameter channel).

Step 5: Activating the fieldbus

Switch into operating mode via SynTop to activate the fieldbus group.

2.7 Multiplex channel

If the required number of data words in the real time channel should exceed the available number, then multiplexing becomes necessary. In this case, the real time channel is made up of real time data and multiplex channel. The multiplex channel is always at the end of the real time channel.

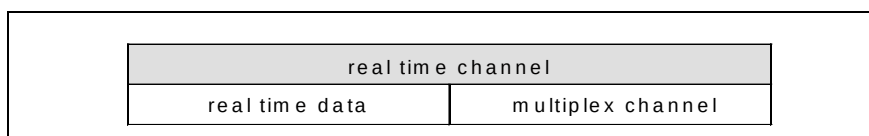


Fig. 2-9: Arrangement of the multiplex channel in the process data channel

The multiplex channel makes it possible for the user to configure a two-dimensional virtual matrix of I/O data and parameters in the process data channel. The fieldbus master determines the line (multiplex level) of the matrix which is located in the actual data telegram.

The multiplex data and the control or the status word form the multiplex channel. Up to 16 multiplex levels can be configured. If the process data channel has a length of e.g., 16 words, then 15 word fields (matrix columns) can be multiplexed. The data structure and the data length in the different multiplex levels can be different. The multiplex levels with less than 15 data words are automatically filled with dummy words. The 16th word of the process data channel is needed for the control or the status word..

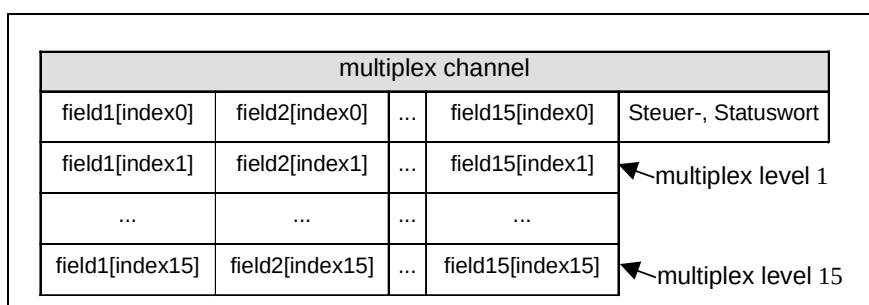


Fig. 2-10: Structure of the multiplex channel

The index in the control or status word determines which multiplex level is transmitted at present data telegram. Further bits in the control and status word control the handshake for multiplexing.

Multiplex control word / status word

The multiplex control word is transmitted from master to slave. The index_out_c displays which multiplex level is written from master to slave. Index_in_c displays which multiplex level is transmitted from slave to master.

Control word

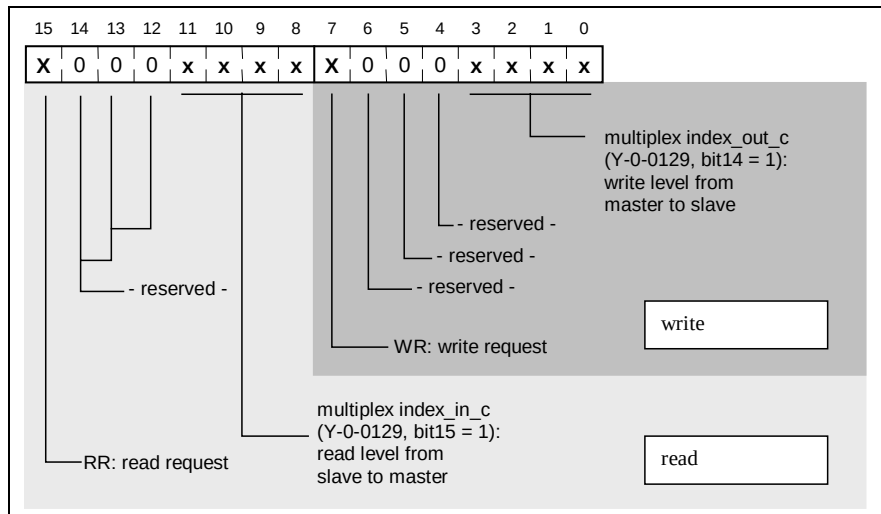


Fig. 2-11: Control word

The slave acknowledges data transmission and data request with the status word.

Status word

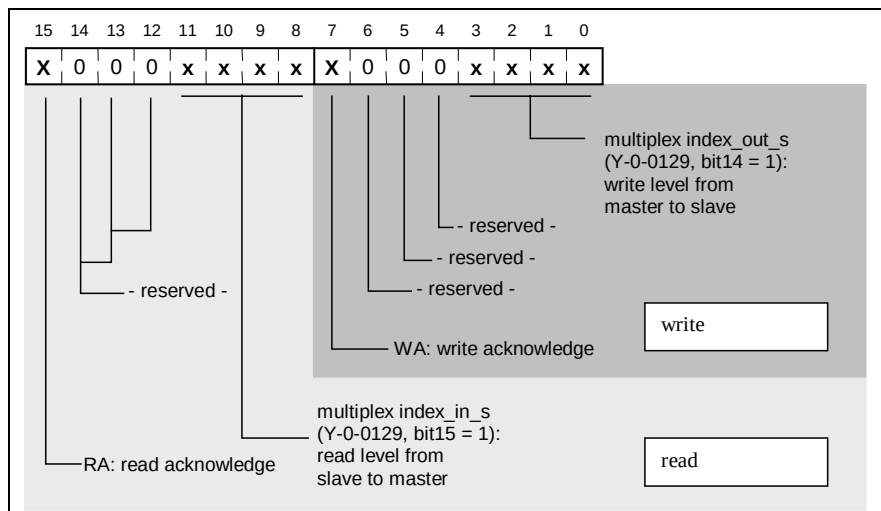


Fig. 2-12: Status word

The handshakes for reading and writing of data are completely independent of each other.

- Write data**
1. The master updates the multiplex data and the index_out_c in the control word.
 2. After that the master sets bit WR (write request) unequal WA (write acknowledge). Thus the multiplex data are declared for valid.
 3. Slave receives data.
 4. Bit WA sets bit WR equal by slave and acknowledges the take over of the data.
 5. Go to step 1.
- Read data**
1. The master sets index_in_c to multiplex level needed.
 2. After that the master sets bit RR (read request) unequal RA (read acknowledge).
 3. Slave puts data of the required level to the acknowledge telegram.
 4. Bit RA sets bit RR equal and signals, that the requested data are put in the multiplex channel.
 5. Master receives data.
 6. Go to step 1.

The following applies for reading and writing data: The master receives data from the fieldbus not before $RR = RA$. The slave receives data, if $WR = WA$.

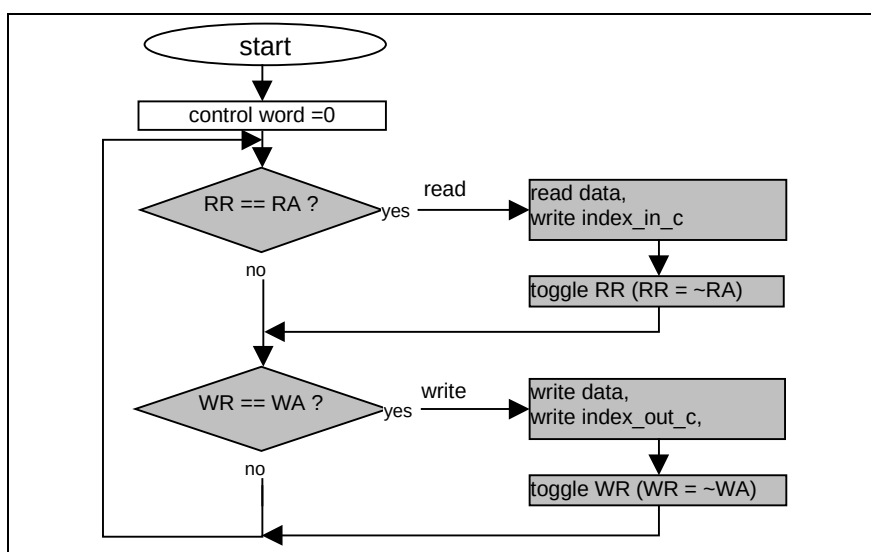


Fig. 2-13: Multiplex algorithm of the master

Multiplex depth

The depth of the multiplex channel follows automatically from the number of configuration elements Y00: Y-0-0198 in the corresponding configuration list.

The fieldbus slave checks the current setting via the master and changes the multiplex channel only if the index lies within the allowable value.

Start offset multiplex channel

The multiplexer of the fieldbus interface must be informed of the place in the process data channel where the multiplex channel starts. The end of the channel is always fixed by the control or status word.

The start offset of the multiplex channel follows automatically from the length of the static channel in the corresponding configuration list.

Enable of multiplex channel

Multiplexing is enabled for both data directions separately via two bits in the "Fieldbus - control bits" (Y-0-0129). If a number of actual values are multiplexed for visualization but only a few command values are transmitted by the master to the PLC, then it may not be necessary to multiplex the command values. Set the bit for the enable of the output data to zero at parameterization.

If a multiplex channel with several levels is configured, but not enabled in Y-0-0129, then only the first multiplex level is permissible. All other multiplex levels are ignored.

2.8 Notes on configuring the fieldbus interface

Notes on the fieldbus

- Process output data (Y-0-0128) and process input data (Y-0-0127) can have a different number of words (length of PD channel).
- Ensure that the fieldbus master supports the desired process data channel length.

Notes on SYNAX

- Each configuration list (Y-0-0127, Y-0-0128, Y-0-0185 - Y-0-0190) must be self-contained with configuration element Y-0-0197 or Y-0-0198.
- Parameters Y-0-0250 - Y-0-0281 (X outputs) are only allowed as process input data (Y-0-0127).
- Within the multiplex channel, SYNAX parameters may not be assigned to several process output data in Y-0-0128. Dummy parameter Y-0-0199 is the only exception.
- For computing time reasons, parameters in the process output data Y-0-0128 are only written if the operating data have changed from the operating data last transmitted at this point via the fieldbus. If a parameter is changed via the service interface (or the fieldbus parameter channel or communication channel) which is also cyclically overwritten via the fieldbus, then the cyclical change via the fieldbus could be lost.
(Ex.: in Y-0-0128: A01: A-0-0056; with A-0-0056= 30.0;
overwritten via SynTop A-0-0056 with 10.0;
10.0 remains in A-0-0056 until A-0-0056<>30.0
is transmitted via the fieldbus!)
- The real time data of the process data channel is only active in phase 4. This means that only parameters that can be changed in phase 4 can be changed in process output data in Y-0-0128. Against that the communication and the parameter channel are active in all phases.

2.9 Fieldbus objects for data exchange

The acyclic data exchange at fieldbusses (except: Ethernet) is based on objects which have been clearly identified by their object number and the subindex. Various object classes are available:

- objects for diagnosis
- objects for the control of the multiplex channel
- objects for data exchange with SIS protocol
- objects for the direct acyclic transmission of parameters

Note: The unit data of fieldbus Profibus is addressed via the object number (Index) and subindex. The data of a unit is illustrated in the specifications for Ethernet/IP and DeviceNet in terms of class, instance and attribute. The allocation of class, instance, attribute and SYNAX parameter is described in the specific section for DeviceNet interfaces (see section 4.4, "Object structure").

Object lists of the individual object classes

In the object lists of the individual object classes below, a few abbreviations are used. Thus

- R = read
- RW = read and write
- PD = process data channel

Objects for diagnoses

Designation:	Object no.:	Type	Access from master:	PD:
status fieldbus	5FF2	16 bit	R	No
diagnosis fieldbus	5FF0	16 bit	R	No
status PPC	5FF5	16 bit	R	No
fault code PPC	5FF6	16 bit	R	No

Fig. 2-14: Objects for diagnosis

Objects for the control of the multiplex channel

Designation:	Object no.:	Type	Access from master:	PD:
multiplex control word	5FFC	16 bit	R	No
multiplex status word	5FFD	16 bit	R	No
startoffset multiplex channel	5FFE	16 bit	R	No

Fig. 2-15: Objects for the control of the multiplex channel

Objects for the data exchange with SIS protocol

Designation: Data exchange objects	Object no.:	Type:	Access from master:	PD:
data exchange object 16 bytes	5E70	array	RW	No
data exchange object 32 bytes	5E71	array	RW	No
data exchange object 64 bytes	5E72	array	RW	No
data exchange object 128 bytes	5E73	array	RW	No

Fig. 2-16: Objects for the data exchange with SIS protocol

Note: Data are acyclically exchanged with these objects via the communication channel. There may not be direct access via object numbers, but rather indirectly via data requests by means of a telegram (see section "Transmission sequence via a data exchange object", page 2-38).

Objects for direct acyclic transmission of parameters

The object numbers (indices) 0x2000 - 0x5E6F and subindices 1 - 40 can be used to assign exactly one field bus object to each SYNAX parameter.

The object numbers are assigned to the various group Y/C/A/S/P parameters. The drive address is selected via the subindices.

This permits direct acyclic access to all single parameters of SYNAX via the communication channel of the field busses.

Note: Further on, the access to list parameters (e.g., cam, P-0-0072) is only possible via data exchange objects 0x5E70 - 0x5E73 with SIS telegram.

Allocation of fieldbus objects and SYNAX parameters for direct acyclic access

The following table contains the allocation of fieldbus objects and parameters.

(fieldbus object, subindex)	SYNAX identity number	Remark
(0x2000 + yyyy, xx)	Axx:S-0-yyyy	0 < yyyy < 4096, 0 < xx < 41
(0x3000 + yyyy, xx)	Axx:P-0-yyyy	0 < yyyy < 4096, 0 < xx < 41
(0x4000 + yyyy, xx)	Axx:A-0-yyyy	0 < yyyy < 2048, 0 < xx < 41
(0x4800 + yyyy, xx)	Cxx:C-0-yyyy	0 < yyyy < 4096, 0 < xx < 3
(0x5800 + yyyy, 1)	Y00:Y-0-yyyy	0 < yyyy < 1648

Fig. 2-17: Allocation of fieldbus objects and SYNAX parameters

2.10 Communication channel

The communication channel supports the exchange of data that do not have to meet real-time demands.

Examples are

- parameters newly set with product change,
- phase transitions (if a parameter is write protected in operating mode),
- read out of diagnoses,
- list parameter such as "Positioning speeds" (A-0-0099), "Cam shaft profile 1" (P-0-0072) can only be transmitted via the communication channel or in the case of Profibus the parameter channel,
- data blocks: transmission of several parameters with one data access (also see interfaces ARCNET and 3964R).

Accessing list parameters, data blocks or phase is embedded in specific protocols and transmitted via the data exchange objects 5E70 - 5E73 (see "Data storage protocol", page 2-19). Data exchange via these four objects is thus indirect.

All other data objects can be directly accessed via the communication channel (see "Direct access to data objects", page 2-40).

Fieldbus specific aspects of the communication channel

The specific communication channels are:

- Bosch Rexroth parameter channel at Profibus.
- Explicit Message at DeviceNet and Ethernet/IP,
- TCP port at Ethernet.

Data storage protocol (SIS protocol)

The data cannot be directly exchanged via data objects but indirectly via telegrams within data exchange objects 5E70 to 5E73. With these telegrams, the data demands are specified and the data transmitted. Securing the data transmission is the responsibility of the serial interface protocol (SIS protocol).

The following sections detail the telegram exchange process.

Telegram types

There are two types of telegrams:

Telegram type	Telegram sender
command telegram	master: of the "active" communication partner
reaction telegram	slave: of the "passive" communication partner

Fig. 2-18: Types of telegrams

There are also two types of master command telegrams:

Telegram type	Data direction
SEND telegram (write-request)	write access: data is sent to slave
FETCH telegram (read-request)	read access: data requested by slave

Fig. 2-19: Command telegram

A slave, correspondingly sends a reaction telegram (see "Protocol content (telegram content)", page 2-22).

Telegram content	Access type
transmission status	with error-free write access
transmission status and requested data	with error-free read access
transmission status and error code	with faulty read or write access

Fig. 2-20 Reaction telegram

Long data blocks in the individual types of telegrams must be broken down into several partial telegrams (see "Protocol content (telegram content)", page 2-22).

Note: Data blocks must be transmitted in one telegram, parts of telegrams are not possible.

Structure of the serial Bosch Rexroth protocol

The serial Bosch Rexroth protocol is a **binary protocol** with binary protocol contents.

The arrangement of the individual bytes of data from "Word" to "DWord" corresponds to the **Intel agreement**. The **IEEE** format applies to the floating point type.

The serial protocol is made up of three parts:

- protocol header (or telegram header)
- user data header
- user data

User data and user data header thereby create the protocol content (telegram content).

Command telegram structure

A **SEND telegram** (Write-Request) is made up of protocol header, user data header and user data.

A **FETCH telegram** (Read-request) is only made up of protocol and user data header (the user data header to specify read access can be eliminated if it is already specified by the service).

Reaction telegram structure

A **reaction telegram** by the recipient is made up of a protocol header as well as possibly a user data header and user data.

It has, with the exception of a few changes, see section "Protocol header (telegram header)", page 2-21, the same protocol header and may even contain sections of the user data header from the command telegram. The sender can thus clearly allocate a reaction telegram to his command telegram.

The **user data** vary depending on read or write access and a positive or negative result (see section "Protocol content (telegram content)", page 2-22).

Protocol header (telegram header)

The protocol header is made up of 8 bytes defined as follows:

Byte	Name	Definition of Individual Bytes																																						
1	StZ	Start symbol: STX (0x02)																																						
2	CS	This is the checksum byte. It is formed by adding all following telegram symbols as well as the start symbol StZ and then negated. This means the sum of all telegram symbols always equals "0" if the transmission was successful. The checksum check of an SIS telegram can be switched off with bit 2 in parameter "Fieldbus - control bits" (Y-0-0129). The transmission safety check is only assumed by the fieldbus itself in this case and not additionally by the SIS protocol.																																						
3	DatL	The length of the following user data is set here. DatL can be as big as the field length of the data exchange object used minus the 8 bytes for the protocol header. This equals the maximum values for DatL as follows: Object 5E70: 8 byte Object 5E71: 24 byte Object 5E72: 56 byte Object 5E73: 120 byte																																						
4	DatLW	This is the repeat of DatL. The telegram length equals DatLW and protocol header, i.e.: Telegram length = DatLW + 8.																																						
5	Cntrl	Bit 0 - 3: reserved, Bit 4: 0 => command telegram 1 => reaction telegram, Bit 5 - 7: Status of reaction telegram: <table><tr><td>000</td><td>no error, request could not be processed</td></tr><tr><td>001</td><td>presently reserved</td></tr><tr><td>010</td><td>presently reserved</td></tr><tr><td>011</td><td>presently reserved</td></tr><tr><td>100</td><td>presently reserved</td></tr><tr><td>101</td><td>presently reserved</td></tr><tr><td>110</td><td>global error (see Y-0-0046, Y-0-0047 and Y-0-0048)</td></tr><tr><td>111</td><td>presently reserved</td></tr></table> Note: for a command telegram it always applies that Cntrl = 0	000	no error, request could not be processed	001	presently reserved	010	presently reserved	011	presently reserved	100	presently reserved	101	presently reserved	110	global error (see Y-0-0046, Y-0-0047 and Y-0-0048)	111	presently reserved																						
000	no error, request could not be processed																																							
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011	presently reserved																																							
100	presently reserved																																							
101	presently reserved																																							
110	global error (see Y-0-0046, Y-0-0047 and Y-0-0048)																																							
111	presently reserved																																							
6	Service	Specifies service that the sender is requesting from receiver or which is executed. Global services: <table><tr><td>0x00</td><td>reserve</td></tr><tr><td>0x01</td><td>data transmission terminated</td></tr><tr><td>0x02 ... 0x0F</td><td>reserved</td></tr><tr><td>0x10</td><td>read parameter</td></tr><tr><td>0x11</td><td>read a list element</td></tr><tr><td>0x12</td><td>read current SERCOS phase</td></tr><tr><td>0x13 ... 0x1C</td><td>still available</td></tr><tr><td>0x1D</td><td>activate phase switching with SERCOS target phase</td></tr><tr><td>0x1E</td><td>writing a list element</td></tr><tr><td>0x1F</td><td>writing a parameter</td></tr><tr><td>0x20 ... 0x8F</td><td>reserved</td></tr></table> Group specific services (SYNTAX): <table><tr><td>0x90</td><td>read parameter</td></tr><tr><td>0x91</td><td>read a list element</td></tr><tr><td>0x92</td><td>read current SERCOS phase</td></tr><tr><td>0x93 ... 0x9C</td><td>still available</td></tr><tr><td>0x9D</td><td>activate phase switching with SERCOS target phase</td></tr><tr><td>0x9E</td><td>writing a list element</td></tr><tr><td>0x9F</td><td>writing a parameter</td></tr><tr><td>0xA0 ... 0xFF</td><td>reserved</td></tr></table>	0x00	reserve	0x01	data transmission terminated	0x02 ... 0x0F	reserved	0x10	read parameter	0x11	read a list element	0x12	read current SERCOS phase	0x13 ... 0x1C	still available	0x1D	activate phase switching with SERCOS target phase	0x1E	writing a list element	0x1F	writing a parameter	0x20 ... 0x8F	reserved	0x90	read parameter	0x91	read a list element	0x92	read current SERCOS phase	0x93 ... 0x9C	still available	0x9D	activate phase switching with SERCOS target phase	0x9E	writing a list element	0x9F	writing a parameter	0xA0 ... 0xFF	reserved
0x00	reserve																																							
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0x02 ... 0x0F	reserved																																							
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0x11	read a list element																																							
0x12	read current SERCOS phase																																							
0x13 ... 0x1C	still available																																							
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0x91	read a list element																																							
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0x9E	writing a list element																																							
0x9F	writing a parameter																																							
0xA0 ... 0xFF	reserved																																							
7	AdrS	Address of the transmitter: station number (0 - 127)																																						
8	AdrE	Address of the receiver: station number (0 - 126, 128)																																						

Fig. 2-21: Protocol header structure

Protocol content (telegram content)

Data format

User data are transmitted in Intel format (see section "Data storage protocol (SiS protocol)", page 2-19).

User data header

The user data header describes the type of request. The elements of the user data header are

- control byte
- unit address
- parameter number and type (command telegrams only)
- list offset (command telegrams only)
- status byte (reaction telegrams only)

Control byte The control byte specifies how a data block element of a parameter is accessed. The transmission of a following telegram is controlled with bit 2 (lists are written in several steps).

The control byte is read out of the command telegram and the reaction telegram copied. The transmission of a following telegram is controlled with bit 2 (lists read in several steps).

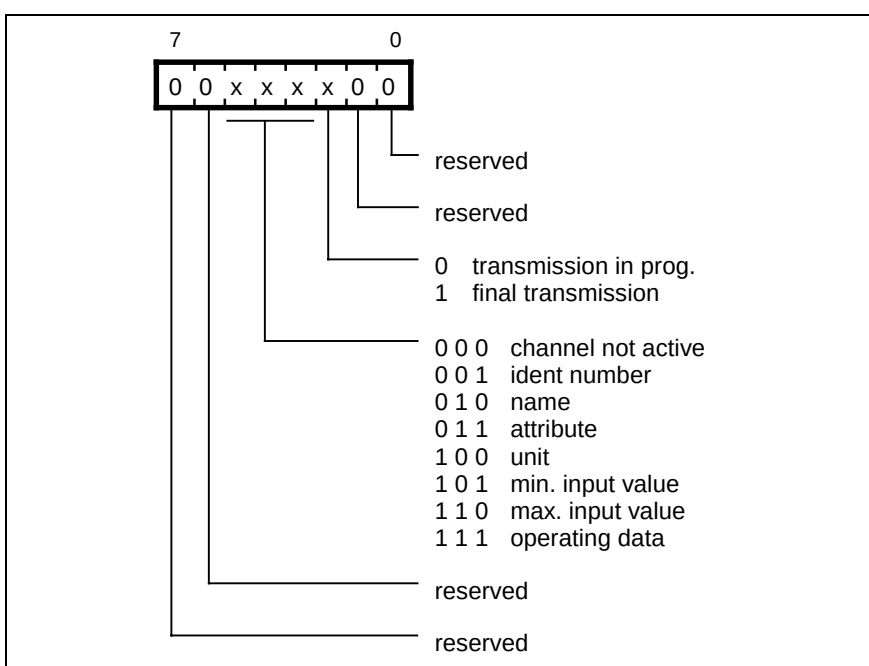


Fig. 2-22: Control data structure

- Write access**
- of the operating data
 - and the identity number (data status read, compare SERCOS interface specification 5.1.3.8) is possible.

Read access of all elements that describe a parameter is possible. These elements are:

- the attributes,
- minimum and maximum input values,
- the name,
- the unit and
- the operating data.

Unit address The unit address is read in the command telegram and copied into the reaction telegram.

The serial Bosch Rexroth interface controls with the device address accesses via the PPC on

- drive parameters,
- system parameters,
- master axis parameters,
- data blocks.

Given communication with a SERCOS master (e.g., PPC) implementing a fieldbus interface, then the master must be informed as to which unit the request relates to. This unit can be the SERCOS master itself, a master axis organized by the SERCOS master or any of the drives it controls.

The address set at the drive controller, a master axis address or "0" are transmitted.

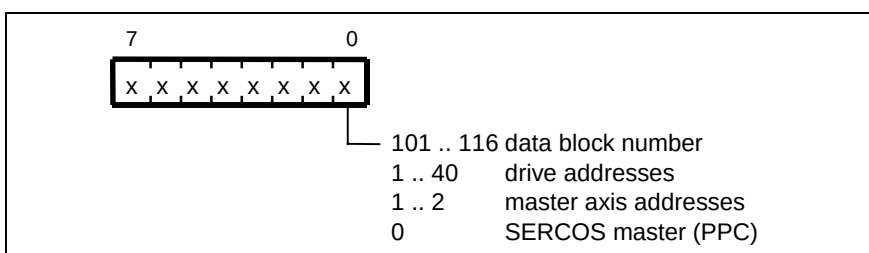


Fig. 2-23: Unit address

Parameter number and type The parameter number has the form determined in the SERCOS interface specification. To be able to also address control parameters, one byte is set ahead of the address to identify the parameter type.

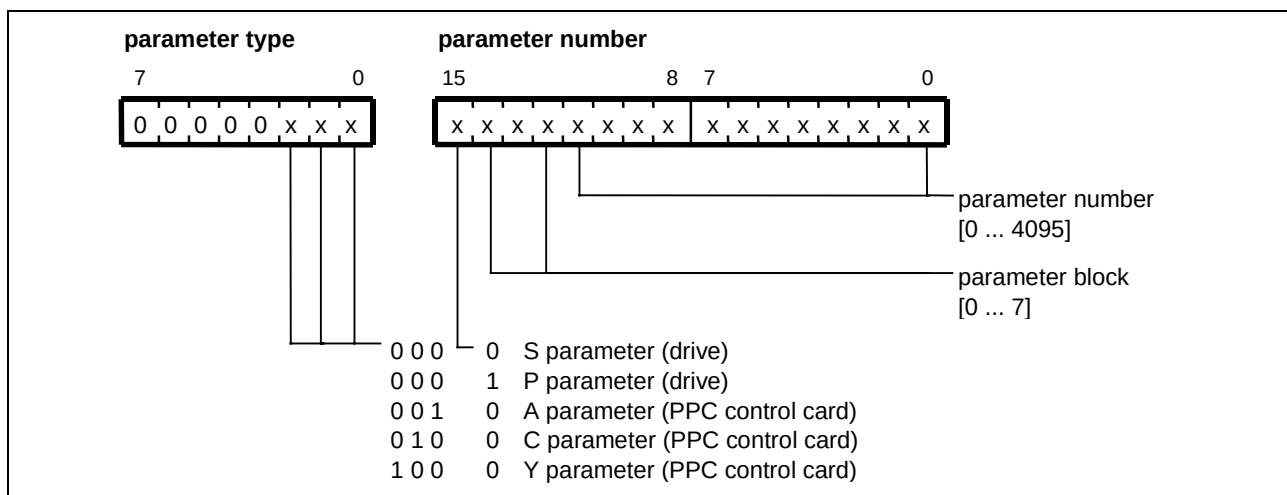


Fig. 2-24: Parameter identification (parameter transmission)

Status byte The status byte supplies the results of a transmission in the form of a code number.

It generally applies:

Error type	Error code
transmission error-free	0x00
protocol error	0xF0 ... 0xFF
execution error	0x01 ... 0xEF

Fig. 2-25: Error types

whereby

Protocol error	Code number	Error description
"Invalid service"	0xF0	The service requested is not specified or is not supported by the participant addressed.
"Invalid telegram"	0xF1	The telegram cannot be evaluated. Either an incorrect start character or a reaction telegram has arrived.
"Incorrect telegram length"	0xF2	The lengths in the telegram header do not agree.
"Incorrect checksum"	0xF4	The checksum received does not agree with internal calculations.
"Incorrect following telegrams"	0xF8	The address of the master, the control byte, the axis address of the parameter has changed in the following telegram.

Fig. 2-26: Protocol error

Operating error	Code number	Description of error
"Error during parameter transmission"	0x01	An error occurred while reading or writing a parameter (see below "execution error with parameter transmission")
"Error during phase switching"	0x02	The specified target phase was not achieved (see below "Operating error when accessing SERCOS phase")

Fig. 2-27: Operating error

List offset The list offset is only specified with the transmission of a single element of a parameter list. It sets the number of bytes the desired element is to be shifted in contrast to the first element within the list.

User data

User data are data to be transmitted. User data elements are:

- PPC mode (target mode of current mode)
- parameter value
- error word with operation error (with reaction telegram only)

Operating error during parameter transmission

Errors occurring during the transmission of a parameter generate an error word in the reaction telegram with a specified error code. In case of a read access this error code will then be issued in lieu of the requested data in the user data field.

Error code	Error message in serial protocol
0x0000	No error
0x0001	Service channel not open
0x0009	Incorrect access to element 0
0x0080	"Master axis address not permitted" The master axis address is larger than permitted or is 0
0x00A0	"Non-permissible request" e.g., an access to S/P parameters in initialization mode
0x00B0	"Non-permissible element" e.g., write access only with element operating data
0x00C0	"Drive address not permitted" The drive address is larger than permitted or the drive in the SERCOS ring is not active (deactivated or not there)
0x00F0	"Fatal software error" A PPC internal error occurred during parameter transmission (see Y-0-0041). It has affected data exchange.
0x1001	No IDN available
0x1009	Incorrect access to element 1
0x2001	No name available
0x2002	Name transmission too short
0x2003	Name transmission too long
0x2004	Name cannot be changed
0x2005	Name presently write-protected
0x3002	Attribute transmission too short
0x3003	Attribute transmission too long
0x3004	Attribute cannot be changed
0x3005	Attribute presently write-protected
0x4001	No unit available
0x4002	Unit transmission too short
0x4003	Unit transmission too long
0x4004	Unit cannot be changed
0x4005	Unit presently write-protected
0x5001	No minimum input value available
0x5002	Minimum input value transmission too short
0x5003	Minimum input value transmission too long
0x5004	Minimum input value cannot be changed
0x5005	Minimum input value presently write-protected
0x6001	No maximum input value available
0x6002	Maximum input value transmission too short
0x6003	Maximum input value transmission too long
0x6004	Maximum input value cannot be changed
0x6005	Maximum input value presently write-protected

0x7002	Data transmission too short
0x7003	Data transmission too long
0x7004	Data cannot be changed
0x7005	Data presently write-protected
0x7006	Data smaller than minimum input value
0x7007	Data greater than maximum input value
0x7008	Data not correct
0x7009	Data write protected with password
0x700A	Data cyclic configured
0x700B	"Invalid list element" The list offset set in SIS services 0x91 or 0x9E exceeds range of list or does not show the start address of a list element.
0x700C	Data presently write protected because of other settings (e.g., parameter, operating mode, drive enable)
0x710B	Data changeable by motion control parameter only
0x710C	"Data exceeds numeric range" The transmitted value is smaller than zero or greater than the "Modulo value" (S-0-0103) in the case of a modulo axis.
0x710D	"Data length cannot presently be changed" The data length in current mode cannot be changed.
0x710E	"Data length cannot be changed" The length of the data is permanently write protected.
0x8001	"Service channel presently busy (BUSY)" The desired access was not concluded within a timeout (programmed via Y-0-0124) because the service channel, for example, was (still) busy. Data transmission is not conducted.

Fig. 2-28: Operational errors during parameter transmission

Operational errors when accessing PPC mode

Errors occurring during phase switching generate an error word with a specific error code in the reaction telegram. It follows the current phase in the user data field.

Error code	Error message in serial protocol
0x8004	Incorrect phase specified via serial protocol
0xD005	"Phase switching still active" A phase switching presently not possible as one is still active
0xD006	"Phase switching with drive enable not possible" Set for at least one drive - "AF"
0xD007	"Phase switching with rotating master axis not permitted"
0xD008	"Cannot switch phase" Master axis functionality enabled

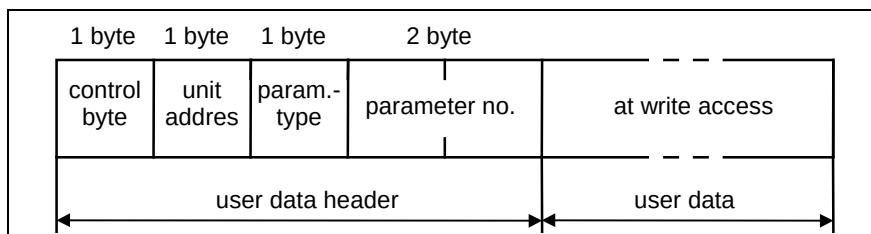
Fig. 2-29: Operational error during phase switching

Transmitting parameters

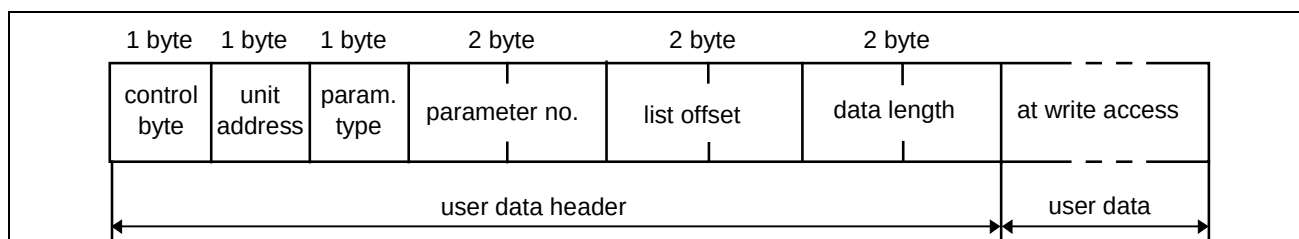
Protocol content when transmitting parameters

The following protocol contents is fixed for parameter transmissions:

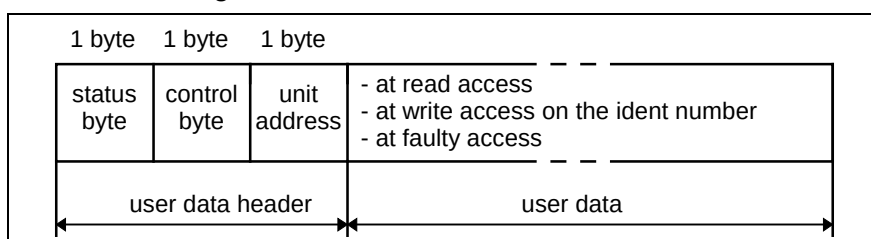
- **Command telegram (standard telegram for parameter transmission):**



- **Expanded command telegram (to transmit individual list elements):**



- **Reaction telegram:**



The data to be transmitted are entered in the user data field. Maximum user data length depends on telegram type:

- **Command telegram:**
Length of data exchange object - 8{protocol header} - 5{user data header} = max. user data length
- **Reaction telegram:**
Length of data exchange object - 8{protocol header} - 3{user data header} = max. user data length

Example of reading a parameter (service 0x90)

Parameters or elements with a length exceeding maximum user data length of the data exchange object are read in steps. Bit 2 in the control byte identifies the current transmission step as a **running** or **final** transmission. A list of four byte data is read. The final data is 0x05F5E100. Data exchange object 5E73 is used.

The control word for a transmission is several steps is listed below. The final data of the list is explicitly depicted to illustrate the Intel format of four byte data.

1. Step:

Write-Request of the master with parameter request.

	3C	..	..	..	..
protocol header	control byte	unit address	param.-Type	parameter -Nr. (LSB)	(MSB)

After the first Read-Request of the master, the PPC sends a reaction telegram:

	..	38	..	..	..	..	..	..	..
protocol header	status byte	control byte	unit address	116 data bytes					

2nd step:

Next Write-Request of the master with the parameter request.

	3C	..	..	..	..
protocol header	control byte	unit address	param.-Type	parameter -Nr. (LSB)	(MSB)

After the first Read-Request of the master, the PPC sends the next reaction telegram (1st following telegram):

	..	38	..	..	..	..	..	..	..
protocol header	status byte	control byte	unit address	116 data bytes					

...

Final step

Final Write-Request of the master with the parameter request.

	3C	..	..	..	..
protocol header	control byte	unit address	param.-Type	parameter -Nr. (LSB)	(MSB)

After the final Read-Request of the master, the PPC sends the final reaction telegram (final following telegram):

	..	3C	..	..	..	00	E1	F5	05
protocol header	status byte	control byte	unit address	1 ... 112 data bytes		(LSB)	4-byte data		(MSB)
							...	...	

Example for writing into a parameter (Service 0x9F)

Parameters or elements with a length exceeding maximum user data length of data exchange objects are transmitted in several steps. Bit 2 in the control byte identifies the current transmission steps as either **in progress** or the **final transmission**.

The "Cam shaft profile 1" (P-0-0072) of drive 1, a list of 4 byte data with 1024 table points is to be written. The final data is 0x000186A0. Data exchange object 5E73 is used.

The control word for a transmission is several steps is listed below. The final data of the list is explicitly depicted to illustrate the Intel format of four byte data.

At list parameters two data words are prefixed to the list elements that specifies the actual list length (actual length) and the maximum possible list length (max. length). The lengths are specified as number of data bytes.

The following applies in our example:

$$\text{actual length} = \text{max. length} = 1024 \times 4 = 4096$$

1st step:

Write-Request of the master with the first data block

	38	01	00	48	80	00	10	00	10	
protocol header	control byte	unit address	param. type	parameter no. (LSB)	parameter no. (MSB)	actual length (LSB)	actual length (MSB)	max. length (LSB)	max. length (MSB)	108 data bytes

The transmission of data can optionally be checked. The master sends a Read-Request to this end. The PPC responds with a reaction telegram.

	00	3C	01
protocol header	status byte	control byte	unit address

2nd step:

Write-Request of the master with additional data

	38	01	00	48	80	
protocol header	control byte	unit address	param. Type	parameter no. (LSB)	parameter no. (MSB)	112 data bytes

The transmission of data can optionally be checked. The master sends a Read-Request for this purpose. The PPC responds with a reaction telegram.

	00	3C	01
protocol header	status byte	control byte	unit address

Final step:

Write-Request of the master with the final data block (final following telegram):

	3C	01	00	48	80		A0	86	01	00
protocol header	control byte	unit address	param.-Type	parameter no. (LSB)	parameter no. (MSB)	1 ... 64 data bytes	(LSB)	4-byte data ...	...	(MSB)

After the concluding Read-Request, the master sends the reaction telegram to the PPC:

	00	3C	01
protocol header	status byte	control byte	unit address

Note: As long as the SYNAX internal data transmission has not been activated, the read request of a master will be negatively acknowledged as the operational results of the write access has not yet been generated.

Example for reading the data status of a parameter

To check a procedure command execution, it is necessary to read the data status. This is supplied in an SIS reaction telegram (as per SERCOS interface) when write requesting the ID number of a parameter in the form of user data (service 0x9F).

The write request is performed with the identity number (2-byte data) of the procedure command.

The following is an example of a check of the data status of the procedure command "C300 command set absolute measurement" (P-0-0012, drive 1).

Command telegram:

	0C	01	00	0C	80	0C	80
Protocol header	control byte	unit address	param. type	parameter no. (LSB)	parameter no. (MSB)	user data	
user data header							

After the master's read request the PPC sends the reaction telegram:

	00	0C	01	03	00
protocol header	status byte	control byte	unit address	data status	
user data header					

Data status 0x0003 shows that the procedure command is set, enabled and executed successfully.

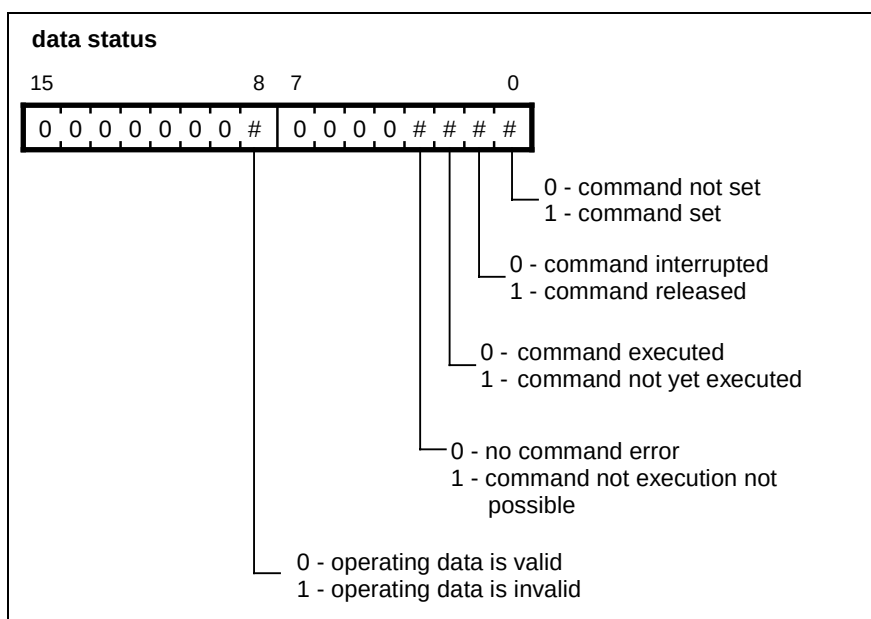


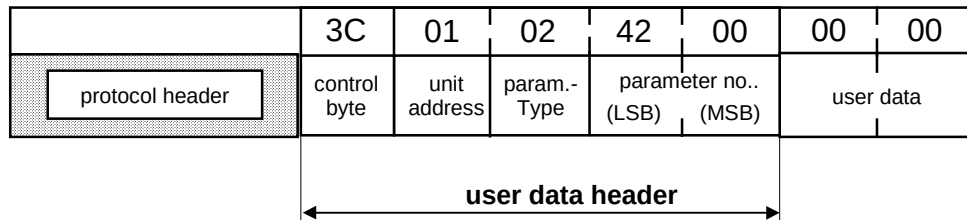
Fig. 2-30: Data status

Example of faulty parameter accessing

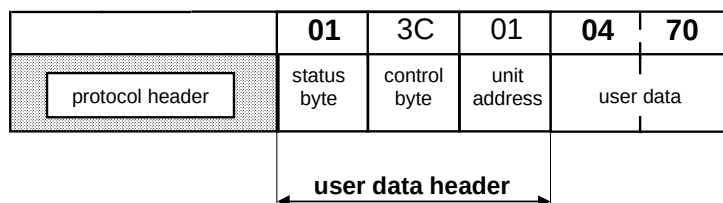
Write access to write-protected PPC parameter "Output position" (C-0-0066) of the master axis 1:

The master is trying to write a value of 0 into the parameter. SYNAX acknowledges with error message 0x7004 ("data cannot be changed").

Command telegram:



Reaction telegram:



Example of an abort of a write access with following telegrams (service 0x01)

Command telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	Adr.E
02	F9	01	01	00	01	00	02	00

Fig. 2-31: Command telegram "abort", 8 byte protocol header, 1 byte user data, CS, AdrS and AdrE are evaluated on the PPC+

This command terminates a write access not yet fully completed. The incomplete data block is discarded. The PPC expects no following telegrams and makes itself ready for new data transfer.

Reaction telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	Adr.E
02	E9	01	01	10	01	02	00	00

Fig. 2-32: Reaction telegram "abort", 8 byte protocol header, 1 byte user data

Note: A read access to list parameters is aborted with the same telegram.

Example of reading a list element (Service 0x91)

The eleventh element of a list of 4 byte data is to be read. The data is 0x05F5E100. Data exchange object 5E71 must be used.

The following is a control word for a transmission in one step. The list offset equals 40 bytes (= 10 elements). The data of the list is explicitly illustrated to clarify the Intel format of four-byte data.

Write-request of the master with parameter request:

	3C	..	..	..	..	0x28	0x00	..	..
protocol header	control byte	unit address	para.-type	parameter - (LSB)	parameter - (MSB)	list offset (LSB)	list offset (MSB)	any	

After the read request of the master, the PPC sends the reaction telegram:

	..	3C	..	00	E1	F5	05
protocol header	status byte	control byte	unit address	4 byte data (LSB) ... (MSB)			

Example of writing a list element (service 0x9E)

The second element of a line of 2-byte data is to be written. The data is 0x86A0. The list offset equals 2 bytes (= 1 element). Data exchange object 5E71 is used.

The following is the control word for a transmission in one step. The final data of the list is explicitly illustrated to clarify the Intel format of the two byte data.

Write-request of master with the value for the list element:

	3C	..	..	..	..	02	00	..	..	A0	86
protocol header	control byte	unit address	para.-type	parameter - (LSB)	parameter - (MSB)	list offset (LSB)	list offset (MSB)	any		2 byte data (LSB)	2 byte data (MSB)

Optionally, the transmission of the data is checked. The master sends a read request to the master for this. The PPC responds with a reaction telegram:

	..	3C	..
protocol header	status byte	control byte	unit address

Configurable data blocks

On the PPC 16 data blocks can be freely configured for a fieldbus interface.

The configuration is effected via the 8 parameters "Configuration list data block 101" (Y-0-0058) to "Configuration list data block 108" (Y-0-0065). This corresponds to data block addresses 101 to 108. The 8 parameters "Configuration list data block 109" (Y-0-0078) to "Configuration list data block 116" (Y-0-0085) configure the data blocks with the addresses 109 to 116.

Y-0-0058 "Configuration list data block 101"	————>	data block no. 101
		...
Y-0-0065 "Configuration list data block 108"	————>	data block no. 108
Y-0-0078 "Configuration list data block 109"	————>	data block no. 109
		...
Y-0-0085 "Configuration list data block 116"	————>	data block no. 116

These parameters are lists of variable length. Data width equals four bytes. Every element in the list designates a parameter in the associated data block and has the following structure (SYNTAX format).

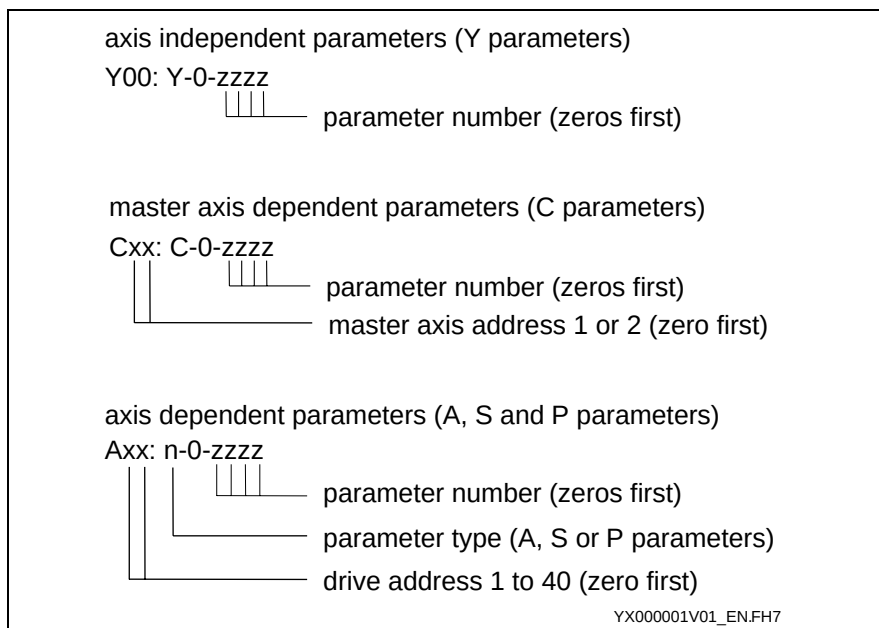


Fig. 2-33: Parameter structure in SYNAX identity number format

Note: Data blocks must be transmitted in one SIS telegram. Data exchange object 5E73 is recommended for transmission. The maximum user data length of all parameters is 116 bytes when reading or 114 bytes when writing.

Example of reading a data block (service 0x90)

Data block 101 is to be read. It consists of 29 parameters with 4 bytes each. The final data 0x05F5E100 is explicitly illustrated to clarify the Intel format of a four byte data.

Write-Request of the master with data block 101 (0x65) with object 5E73:

	..	65	..	..	..
protocol header	control byte	unit address	param.-Type	parameter - (LSB)	parameter - (MSB)

After the Read-Request of the master, the PPC sends the reaction telegram:

	..	..	65		00	E1	F5	05
protocol header	status byte	control byte	unit address	1 ... 112 data bytes	(LSB)	4-byte data (MSB)		

Example for writing into a data block (Service 0x9F)

Data block 116 is to be written. It consists of 28 parameters with 4 bytes each. The final data 0x000186A0 is explicitly illustrated to clarify the Intel format of a four byte data.

Write-Request of the master with the data block to write for data block 116 (0x74) with object 5E73:

	..	74	..	..	..		A0	86	01	00
protocol header	control byte	unit address	param.-Type	parameter no. (LSB)	parameter no. (MSB)	1 ... 108 data bytes	(LSB)	4-byte data (MSB)		

After the Read-Request, the master sends the reaction telegram to the PPC:

	..	..	74
protocol header	status byte	control byte	unit address

Note: As long as the SYNAX internal data transmission has not been activated, the read request of a master will be negatively acknowledged as the operational results of the write access has not yet been generated.

Transmission of PPC mode

Protocol contents during phase switching

The serial protocol can be used to switch between parameterization and operating mode. This progression is triggered by setting a communication phase. It applies:

communication phase 2= parameter mode
communication phase 4= operating mode

The following protocol contents has been set for phase switching:

- **Command telegram:**
The command telegram has no user data header. One byte is transmitted in the user data. This byte contains the desired phase.
- **Reaction telegram:**
The reaction telegram contains no user data. Only the status byte is transmitted.

Note: A successfully started phase transition cannot be aborted with a general service termination (0x01).

Example: Switching into operating mode (Service 0x9D)

Progression from parameter into operating mode is done by setting communication phase 4.

Command telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	User data
02	5B	01	01	00	9D	00	00	04

Fig. 2-34: Command telegram "phase 4 transition"

The command is acknowledged in the first reaction telegram, not the execution of the progression.

Reaction telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	Status byte
02	4F	01	01	10	9D	00	00	00

Fig. 2-35: Reaction telegram "Phase 4 transition"

Right after the first reaction telegram, the master detects with "polling", whether and when the progression is concluded. The current communication phase is repeatedly read (see example for reading a current phase: Service 0x04), until the operating mode or the error is signaled during phase switching.

Example: Switching into parameter mode (service 0x9D)

Switching from operating into parameter mode is done by setting communication phase 2.

Command telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	User data
02	5D	01	01	00	9D	00	00	02

Fig. 2-36: Command telegram "Phase 2 transition"

The command is acknowledged in the first reaction telegram, not however the execution of the progression.

Reaction telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	Status byte
02	4F	01	01	10	9D	00	00	00

Fig. 2-37: Reaction telegram "Phase 2 transition"

Right after the first reaction telegram, the master detects whether and when the progression is concluded by means of "polling". The current communication phase is repeatedly read (see example for reading a current phase: service 0x04), until the operating mode or the error is signaled during phase switching.

Protocol contents when reading the current phase

The following applies to reading the current PPC mode:

- **Command telegram:**
The command telegram does not contain user data.
- **Reaction telegram:**
The reaction telegram contains a status byte and no more than three user data bytes. Protocol or operational acknowledgement are transmitted in a status byte (compare section "Siemens S5 coupling 3964R", section 8.3). The current communication phase and possibly an error code are in the user data.

Example when reading the current phase (Service 0x92)

Command telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E
02	6C	00	00	00	92	00	00

Fig. 2-38: Command telegram "Read phase"

Reaction telegram:

StZ	CS	DatL	DatLW	Cntrl	Service	Adr.S	Adr.E	Status byte	User data
02	56	02	02	10	92	00	00	00	02

Fig. 2-39: Reaction telegram "Read phase"

The current communication phase is transmitted in the user data (example: Phase "2").

After phase switching set (see example of phase switching: service 0xF9) the status word displays the status of the progression. The master must repeat read access to the communication phase until the set phase or an error is signaled in the status word.

If the set phase switching cannot be executed, then "error with phase switching" is signaled in the status word of the reaction telegram. The current communication phase and error code are transmitted in the user data. With a non-permitted phase (phase > 4), error code 0x8004 is set. The reaction telegram then looks like this:

	02	04	04	80
protocol header	status byte	current phase	error code (LSB) (MSB)	

The error code in the reaction telegram illustrates the Intel format of 2-byte data.

Transmission sequence via a data exchange object

Selecting the data exchange object

The four data exchange objects 5E70 to 5E73 are available to the master for indirect data exchange via telegrams. These objects each represent byte arrays of varying lengths.

Data exchange object	Data length (in bytes)
5E70	16
5E71	32
5E72	64
5E73	128

Fig. 2-40: Length of the data exchange objects

Depending on the telegram length, any suitable data exchange object can be selected.

Note: The entire data length of the data exchange object must always be transmitted even if the telegram is shorter.

Data request specification with a write-request from the master

As no parameter objects are configured for the transmission of lists and no data object for the PPC mode has been determined, data accessing this types must first be specified. This takes place in a command telegram.

In this command telegram are for a parameter transmission apart from the service also parameter number, type, etc. defined. With write access, the (first) user data are transmitted in the command telegram (see "Protocol header (telegram header)", page 2-21 and "Protocol content (telegram content)", page 2-22).

The master writes this command telegram using a **write-request** into a data exchange object of suitable length. This makes the data request available to the PPC and it can then be processed. Upon processing, the PPC places a reaction telegram into this data exchange object.

Note: Read access is standardly requested with data exchange object 5E70. Write access requests depend on user data length. With a cam (P-0-0072) this is typically data exchange object 5E73.

Reading the reaction telegram with a read-request

With a **read-request** to a data exchange object the master can read the reaction telegram. The data exchange object for the individual Read-Requests can be of varying lengths.

The maximum length of a reaction telegram corresponds to the length of the data exchange object previously used by the master. This means that there are critical cases when read accessing must be noted.

Read-Request after read request

After a read request, the requested data must be picked up. Reaction telegram, in this case, is read by the PPC via a Read-Request.

Read requests with follow-up telegrams necessitate a renewed write request (identical command telegram) with each further reaction telegram which must then be fetched by a read request.

In the **control byte** of the reaction telegram (see section "Protocol header (telegram header)", page 2-21) it is identified as to whether it is an **in-progress** or **final** reaction telegram

In the **status byte** of the reaction telegram (see "Protocol header (telegram header)", page 2-21) both **protocol** and **operational errors** can be marked. If an operational error has occurred, then user data can no longer be picked up.

Note: If the read requests implement different data exchange objects, then the new data exchange object may never be smaller than the previous one. Otherwise, data would be lost with a read request with shorter field lengths as a master picks up fewer user data than the PPC has available to it. Especially important is that the data exchange object for the first read request when reading long lists is not smaller than that one used in the write request!

Read-Request after a write request

After a write request it is necessary to check whether the request was successful. The reaction telegram prepared by the PPC supplies this information.

In the case of write access with following telegrams, a read request for the running command telegrams is optional. The reaction telegram only informs about the error at protocol level. The execution in the PPC simply takes the form of the collection of user data.

After the final telegram, operational errors in the reaction telegrams are also considered as the internal data transmission is executed. This means that the master, after the final telegram, should run a read request in every case with write access.

Note: The data exchange object 5E70 always suffices for a read request in the case of write access.

Examples

Examples for read and write access to parameters and the PPC mode are detailed in section "Protocol content (telegram content)", page 2-22

Note: As long as no reaction telegram of the PPC is available, the PPC denies read requests from the master. The internal data transmission with a cam (P-0-0072) takes at least 4.2 seconds at 2ms SERCOS cycle time.

If the PPC denies a read request, then in object 5FF6 ("Fault code PPC") bit 11 (0x0800) is set, (see "Diagnosis on the fieldbus interface", page 2-41). If this object is configured in the process data channel, then the master can recognize the status of the communication channel in real-time.

Direct access to data objects

There is direct access to all objects defined on the fieldbus except for data exchange objects via the communication channel. It must thus be noted that:

- it is not possible to write access data objects already written into via the process data channel
- write access to PPC inputs, not used in the I/O logic (see [section "Control signals \(binary inputs/outputs\)"](#), SYNAX 200 - Functional description, DOK-SYNAX*-SY*-12V*1/2-FK01-EN-P), may be permitted but have no effect
- read access to PPC outputs, not used in the I/O logic (see [section "Control signals \(binary inputs/outputs\)"](#), SYNAX 200 - Functional description, DOK-SYNAX*-SY*-12V*1/2-FK01-EN-P) are permitted but only supply a base load value

2.11 Diagnosis on the fieldbus interface

There are a total of four 16 bit objects available for diagnostic handling on the fieldbus interface. The PPC (objects 5FF5 and 5FF6) and the fieldbus card (objects 5FF0 and 5FF2) each update two of these 16 bit fields.

Using these objects, the fieldbus master can detect the status of the fieldbus interface of the PPC and the PPC can generate its diagnosis.

Note: Objects 5FF0, 5FF2, 5FF5 and 5FF6 can only be accessed via communication channel.

Bit assignment of diagnostic objects 5FF5 and 5FF6

Bit field "Status PPC" (object 5FF5)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	x3	---	x1	x0

x0 Status bit for the **Process data channel (PD channel):**

= 0 --> the process data channel is OK, the data are valid,

= 1 --> there is an error pending for the process data channel, the data are invalid. A precise diagnosis is possible with the help of the 8 lower bits of the bit field "fault code PPC" (object 5FF6).
No interference bit for the PD channel is set here. The data are thus not valid because the slave is (still) not ready.

x1 Status bit for the **Communication channel:**

= 0 --> the communication channel is OK, the previous data access was successful.

= 1 --> No error has occurred in the communication channel (data transmission error).
A more precise diagnosis can be obtained with the help of the 8 higher bits of bit field "fault code PPC" (object 5FF6).

x3 Status bit for a **global SYNAX error:**

= 0 --> The PPC shows no global SYNAX error.

= 1 --> There is a global SYNAX error. Specific information is listed in the diagnostic parameters Y-0-0046, Y-0-0047 and Y-0-0048.

--- Free status bit.

Bit field "Fault code PPC" (object 5FF6)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
---	---	---	y4	y3	y2	y1	y0	---	---	---	x4	x3	x2	x1	x0

The interference bits in object 5FF6 are broken down in 8 bits each for the process data and the communication channel. The lower 8 bits are for the process data channel, the higher for the communication channel.

The interference bits for the process data channel are written first:

The errors in the process data channel are bit coded and show a faulty configuration of the process data or multiplex channels (MUX-). The slave recognizes them by means of the initialization or after a reconfiguration and keeps them pending until a new correct configuration as no sensible cyclical data exchange is possible.

- x0** Interference bit for **support of process data**:
 - = 0 --> All process data is supported in the process data channel.
 - = 1 --> At least one process data is not available via the process data channel.

- x1** Interference bit for **process data length**:
 - = 0 --> Access to the process data is correct.
 - = 1 --> At least one process data has been incorrectly accessed (read or write access not permitted).

- x2** Interference bit for **object length of process data**:
 - = 0 --> The configured process data lengths for the process data channel are permissible.
 - = 1 --> The length of a minimum of one process data channel exceeds the permissible process data length (64 words).

- x3** Interference bit for **configuration elements**:
 - = 0 --> The allocation of the configuration elements in all configuration lists is permissible.
 - = 1 --> In a minimum of one configuration list a configuration element is missing or too many multiplex levels are configured.

- x4** Interference bit for **process output data**:
 - = 0 --> Each parameter of the process output data is only configured once.
 - = 1 --> In a minimum of one configuration list, at least one parameter is multiplexed in the process output data.

The interference bits for the communication channel are described below. They remain pending until the next correct access to a data object.

If the diagnostic object 5FF6 ("fault code PPC") is not configured in the process data channel, then note the following when diagnosing problems:

Note: After an error in communication channel, immediately read diagnostic object 5FF6 ("fault code PPC") before further diagnoses to avoid clearing interference bits with next valid access.

- y0** Interference bit for **support of parameters**:
 - = 0 --> The parameter is supported in the communication channel.
 - = 1 --> The parameter is not available via the communication channel.

y1 Interference bit for **access to parameters**:

= 0 --> The access to parameters was correct.

= 1 --> Incorrect access to parameters (write request cannot be read).

y2 Interference bit for **write-request with data exchange objects**:

= 0 --> Access to data exchange object is correct.

= 1 --> A write request is still active via a data exchange object meaning that a new access is not permitted. This conflict can only occur in a multi-master system.

y3 Interference bit for **read-request with data exchange objects**:

= 0 --> Access to data exchange object is correct.

= 1 --> The data for the read request via data exchange object have not yet arrived

Possible causes are:

- interference in slave system
- faulty receiver address in data telegram (see point y4),
- the read request was too early (e.g., with P-0-0072).

In the last case, the read request must be repeated.

y4 Interference bit for **data content in data exchange object**:

= 0 --> The data telegram has been sent to the correct recipient.

= 1 --> The receiving address in the data telegram protocol does not agree with the physical bus address. The telegram is discarded. No reaction telegram can be read (see point y3).

--- Free interference bit.

Bit assignment of diagnostic objects 5FF0 and 5FF2**Bit field "status fieldbus" (Object 5FF2)**

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x15	x14	---	x12	---	---	---	---	---	---	x5	x4	x3	---	x1	x0

x0, x1 Status bit for **internal (DPR) communication** of the fieldbus slave:

Bit 1	Bit 0	
0	0	a reset has been conducted on the DPR
0	1	the DPR has only been initialized by the fieldbus card
1	0	the DPR is complete, i.e., the PPC has also initialized it
1	1	the internal communication of the fieldbus slave, especially the watchdog function, is working correctly

Fig. 2-41: Status bit for the internal (DPR) communication of the fieldbus slave

- x3** Status bit for the **active bus capabilities** of the fieldbus slaves:
 = 0 --> The fieldbus slave is not (yet) ready for data exchange.
 = 1 --> The fieldbus slave can actively participate on the bus.
- x4** Status bit for **communication channel**:
 = 0 --> The communication channel can not (yet) be used.
 = 1 --> The communication channel is available.
- x5** Status bit for **reconfiguring the process data channel**:
 = 0 --> The process data channel on the fieldbus card is configured.
 = 1 --> The process data channel is being reconfigured on the fieldbus card at this moment.
- x12** Status bit for **non-supported SYNAX firmware**:
 = 0 --> SYNAX firmware is supported.
 = 1 --> SYNAX firmware is not supported. The slave modules support the fieldbus but not the synchronous mechanisms with the PPC.
- x14** Status bit for the **process output data**:
 = 0 --> Process output data are invalid.
 = 1 --> process output data are valid.
- x15** Status bit for the **multiplex channel**
 = 0 --> The multiplex channel is not (yet) active.
 = 1 --> The multiplex channel is active
 --- Free status bit.

Bit field "diagnosis fieldbus" (Object 5FF0)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x15	x14	x13	x12	---	---	---	---	x7	x6	x5	x4	x3	x2	x1	x0

x0 - x7 Interface specific **fault code**:

x12 - x15 Specification of the **fieldbus interface**:

Bit 15	Bit 14	Bit 13	Bit 12	
0	1	0	0	Profibus
0	0	1	1	DeviceNet
0	1	1	1	Ethernet and Ethernet/IP with ETH01 at PPC-R0x
1	0	0	1	Ethernet and Ethernet/IP on-board at PPC-R2x

Fig. 2-42: Fieldbus interface specification

All other bit combinations x12 to x15 are still free.

--- Free interference bit.

PPC diagnoses

The PPC diagnoses the following failures and errors, which may occur in connection with the field bus interface, thus permanently preventing the bus communication:

PPC output "Field bus - real-time channel active"

If the PPC is in the operating mode and data is not received via the field bus within the field bus timeout, the PPC clears the output "Fieldbus - real-time channel active" (_A:C01.08).

This output can be used to selectively switch off the drive enable signal of individual axes in case of a bus failure in the combinational logic system.

Initialization error

The PPC monitors the initialization of the field bus communication task and the device driver, in order to detect a fatal failure of the field bus interface.

If problems arise during the initialization of the field bus communication task, which prevent the field bus slave connection from functioning properly, the PPC displays the following diagnosis after a waiting time of no more than 12 seconds:

F33 (150) "Communication via the field bus is impossible"

Thereafter, the run-up will be stopped; the PPC remains in the parameterization mode.

After the field bus communication task has been started successfully, the PPC continues to run up until it enters the operating mode. Here, it is checked whether the device driver of the slave connection is able to correctly access the hardware. If the address or interrupt setting on the field bus board is not correct, the PPC displays the following diagnosis:

F33 (151) "Fieldbus: incorrect jumper setting on field bus board"

In either case, it is not possible to operate the SYNAX application via the field bus interface. It is not possible to clear the diagnostic messages. Please contact the Bosch Rexroth Service to eliminate the failure.

Errors in the configuration of the process data

If the configuration of the process data is impermissible, the PPC displays one of the following diagnoses:

F15 (148) "List of UDP ports in Y-0-0194 is too short (see Y-0-0129)"

F15 (149) "Y-0-0194: UDP port number exists twice"

F15 (153) "Parameter channel: Is supported by Profibus only (Y-33/129)"

F15 (280) "Parameter Y-0-0127 not correct. Line nbr.: xxx"

F15 (281) "Parameter Y-0-0188 not correct. Line nbr.: xxx"

F15 (282) "Parameter Y-0-0189 not correct. Line nbr.: xxx"

F15 (283) "Parameter Y-0-0190 not correct. Line nbr.: xxx"

F15 (284) "Parameter Y-0-0128 not correct. Line nbr.: xxx"

F15 (285) "Parameter Y-0-0185 not correct. Line nbr.: xxx"

F15 (286) "Parameter Y-0-0186 not correct. Line nbr.: xxx"

F15 (287) "Parameter Y-0-0187 not correct. Line nbr.: xxx"

The PPC stops running up and remains in the parameterization mode until the configuration is a permissible one.

Note:	This error state can also be detected by the field bus master by evaluating the diagnostic objects 5FF5 and 5FF6.
--------------	---

3 The fieldbus interface Profibus

3.1 Introduction

Section 2 describes the basics that apply to all fieldbusses within SYNAX. This sections outlines the bus-specific features and functions of the Profibus.

The connection of SYNAX to Profibus uses PPC plug-on assembly BGR RECO-CONTROL P2 LK KA-PB01. In the following the shortening DPS01 (Profibus **D**ecentralized **P**eriphery **S**lave) is used for this assembly.

3.2 Functional features and GSD file

GSD file The features of the Profibus slave are filed in a specific file called "Geräte-Stamm-Daten-Datei" Indm04B0.gsd. It can be objected from Bosch Rexroth for DPS01, installed in the projecting tool and automatically checked during project planning.

Functional features Module DPS01 is outfitted with the following features:

- It is possible to integrate into the Profibus networks the categories Profibus-DP as per DIN 19245-3.
- All data rates are supported as per DIN 19245-3 to 12 Mbit.
- Freely configurable process data channel up to 64 words in both directions (configurable via the PPC). Data consistency over the entire length of the PD channel.
- Monitoring of the process data channel (watchdog function).
- LED diagnostic field in front panel of DPS01 for easy diagnosis of the bus functions and important communication links between DPS01 and PPC.
- Implementation of an object structure for simple accessing of variables and parameters of control module PPC and drive.
- Upload / download functions via four arrays of 16 to 128 byte data lengths.

3.3 Monitoring the fieldbus transmission-behavior with bus failure

As described in section 2.5 the DPS01 monitors the bus behavior with a watchdog timer so that it can specifically response to a bus failure.

Watchdog function

The PLC determines at run up with service "parameter data" whether work is with or without monitoring on the DPS01. The PD monitoring time is, depending on the master used either calculated by the master automatically or set by user and is also set at runup phase.

Note: When setting the PD monitoring time note that fieldbus cycles could fail under normal operating conditions.

Error reactions with bus failures

The DPS01 freezes the most recently received data in the event of a bus failure and deletes bit 3 in object 'Status fieldbus' (5FF2). This bit is displayed in operating mode on the PPC at output "Fieldbus - real time channel active" (`_A:C01.08`). The user can use this bit in the I/O logic (VKL) for a specific error reaction.

With bus return, output "Fieldbus - real time channel active" (`_A:C01.08`) is set.

3.4 Configuration of the process data channel

In section 2.6 the configuration of the real time channel was generally discussed. If the parameter channel in Profibus is activated, then the first six words in the process data channel are reserved for this.

Fig. 3-1 shows typical bus assignments for SYNAX applications.

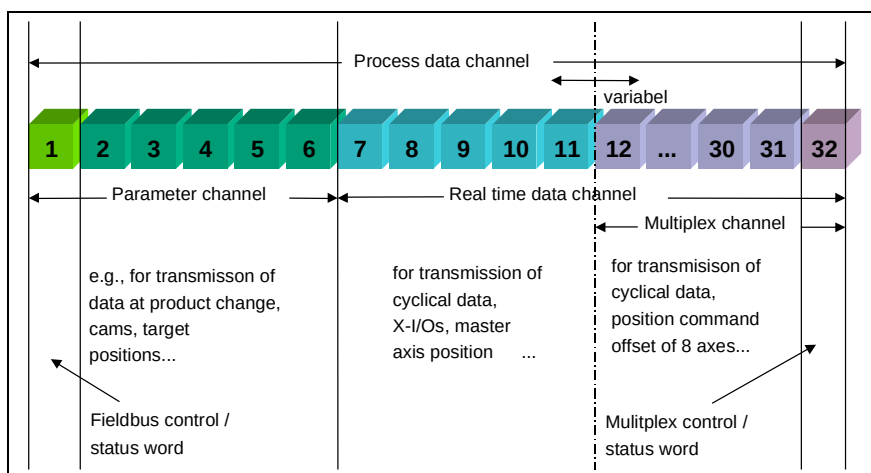


Fig. 3-1: Typical assignment of the 32 words in the process data channel

3.5 Parameter channel on the process data channel

The Rexroth parameter channel replaces communication channel (FMS). Six words in the process data channel are reserved for this purpose.

The concept in the PPC for data administration is based on data exchange objects 5E73. In the parameter channel the telegrams (=contents of 5E73) are broken down into blocks of five words, whereby the first word in the parameter channel (fieldbus control word/status word) coordinates transmission.

For the transmission of a single parameter within a PLC cycle, the SIS protocol was shortened in the SIS protocol to six words, see "Data storage protocol (SIS protocol)", page 2-19. A telegram was also created for accessing fieldbus objects directly.

Therefore three transmission formats are used:

- "SIS short format 1" for A/C and S/P parameters (theoretically is a transmission in SIS possible).
- "SIS Format" for phase transitions and service "abort".
- "Short format 3" for direct access to objects (parameters) of the motion control.

Note: The Motorola format must be used for the protocol shell in the parameter channel, Intel for the Bosch Rexroth SIS protocol. Words and double words were transmitted either swapped or not swapped depending on the protocol they originate from.

If data in SIS format or short format 1 are transmitted, then the rules of indirect access via data exchange object of the communication channel apply. This means that the response of the slave to a write request (fieldbus status word bit 12=0) for "reading" or "writing" only acknowledges the correct receipt of a telegram.

If a read request follows this (fieldbus status word bit 12=1), then the master must pick up the data for "reading" or check for correct execution when "writing".

Note: The response of the PPC to write parameter or phase transition does not confirm correct execution. It is thus highly recommended to follow up the transaction directly with a read request as to whether the new value has been correctly accepted by SYNAX after the internal check (see example Fig. 3-8 and Fig. 3-9).

Fieldbus control word and status word

The master uses the fieldbus control word to control the entire transmission procedure via the parameter channel. The slave shows the current status of the transmission in the fieldbus status word.

Idle state The slave must be brought into its idle state prior to each transmission in the parameter channel. Regardless of the definition of the individual bits, the master specifies the values for the individual bits, i.e., 0x000F. The slave acknowledges the idle state or sets the error bits (see below) as needed.

	FBControl / FBStatus	
M	00 0F	idle state
S	00 0F	acknowledge idle state

Fig. 3-2: Setting basic state; S = slave (PPC), M = master (PLC)

The structure of the FB control word and FB status word is as follows:

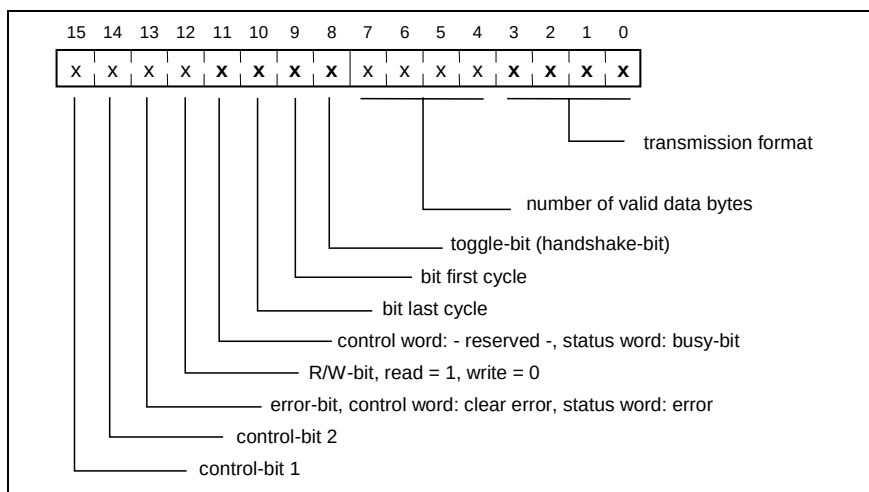


Fig. 3-3: Assigning fieldbus control and status word

The bits have the following meaning:

Control-Bit 1 / Control-Bit 2: These bits are reserved for expansions of the multiplex channel and must be zero.

Error-Bit Via the error bit in the status word the PPC signals an error when transmission format 1 is applied. Given a transmission error in the standard SIS protocol, the error bit is only set if the error has occurred during the processing sequence within the parameter channel, i.e., in the protocol shell of the parameter channel. (See also "Error codes in the parameter channel", page 3-15)

The master sets the error bit to clear an error in the slave.

	FBControl / FBStatus	
S	2x xx	slave has set error bit
M	27 00	clear error
S	01 00	error cleared
M	00 0F	idle state
S	00 0F	acknowledge idle state

Fig. 3-4: Example of clearing an error with set toggle bit; S = slave (PPC), M = master (PLC)

Busy-Bit If the master requests data from the slave, then these are not immediately available on the PPC with S and P parameters. If the master wants to fetch the data while the PPC is collecting data via the SERCOS ring, then SYNAX acknowledges the query with a telegram in error format and sets the busy bit. After the idle state is set, the master tries to again fetch the data. This is repeated until the busy bit is not set when fetching data.

Note: General rule of thumb for 4 byte values: The busy bit is only set if PLC cycle time is less than 20 SERCOS cycles.

First cycle, last cycle To clearly determine the entire length of a telegram and detect the start and end of a complete transmission, the first and last cycle are displayed via one bit each. This clearly defines data transmissions that extend over several cycles.

Transmission type	L Bit	F Bit
first transmission	0	1
last transmission	1	0
intermediate step	0	0
first and final transmission; one step	1	1

Toggle bit (handshake bit) To identify a new transmission step and confirm the receipt of data the control and status words include by a toggle bit.

The master toggles the toggle bit if new data are available for the slave during a WRITE access or if the slave requests new data during a READ access.

The slave acknowledges the toggle bit when the slave has fetched the data during a WRITE access or if the data are ready for the master during a READ access.

Regardless of how the toggle bit was previously set, it is cleared by the master once it specifies the idle state.

Number of valid data bytes The number of valid data bytes are all bytes after the control or status words during a data cycle in the parameter channel.

Note: In contrast to this, byte 3,4 in the SIS protocol header references user data within the entire telegram (compare with section "Protocol header (telegram header)").

Transmission format By specifying the transmission format it is determined in which telegram mode (or telegram type) the parameters are sent via the bus or are received.

These formats are coded as follows:

Format	Coding
S-/P parameters in short format 1	0000
A parameters in short format 1	0001
C parameters in short format 1	0010
Y parameters in short format 1	0100
fieldbus objects in short format 3	1100
transmission in SIS protocol	1011

Short format 1 for parameters

Drive and control parameters are generally accessed via "short format 1". In the first data block, words 2, 3 and 4 are transmitted in the telegram and user data header and the user data in words 5 and 6. This sends a 2 or 4 byte data in one transmission.

In the case of lists, user data is available in the following blocks starting with word 2.

Data format: MOTOROLA						Format: INTEL	
FBControl / FBStatus	Cntrl	service	control byte	unit address	parameter number	user data	user data

Fig. 3-5: Serial protocol "short format 1"

In comparison to the standard SIS protocol irrelevant information such as start symbol and sender address is dropped, a part of the information has been built into the fieldbus control and status words.

If list data are transmitted that are greater than the data exchange object 5E73, then the data is broken down into steps. Into one step, it is possible to write up to 114 or read up to 116 data bytes. (This equals 128 bytes minus telegram and user data header of the standard SIS protocol). These bytes are transmitted in single blocks whereby in the first block of a step only telegram and user data headers in short format 1 may be sent. This means that in the first block of a step, only 4 data bytes may be sent, in each additional block ten data bytes. The sender of the data displays in bit "Last_cycle" of a block that a step has been completely toggled. In the first block of the last step the final transmission bit is set in the control byte. With bit "Last_cycle" in the last block of the last step, the receiver recognizes that all data has been received. The idle state is always set between the individual transmission steps.

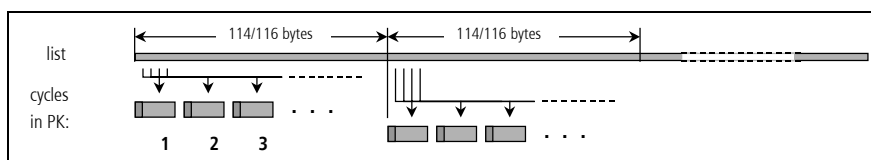


Fig. 3-6: Transmitting a long list

The answer of the slave in short format 1

Two cases must be differentiated for the answer of the slave:

- error-free transmission and
- faulty transmission.

Error-free transmission

Given an error-free transmission, the error bit in the FB status word is cleared (E=0) and possible requested data are following at word offset 2.

Faulty transmission

Given a faulty transmission, the PPC sets the error bit or the busy bit in the FB status word and sends a telegram in the following format:

Word 1	Word 2	Word 3	Word 4	Word 5	
status word	parameter no.	error class 0x06	0x00	additional error code	Ctrl
fieldbus control	error message				

The transmission faults in the parameter channel are discussed separately in section "Error codes in the parameter channel" on page 3-15.

Example 1: Read single data

PPC parameter, data length 4 bytes

Parameter: Y-0-0046 "SYNAX - error source"**Unit address:** 0**Results:** Value of parameter: 0x00000001 = address 1

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit addr.	Parameter number	User data	User data	
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	07 64	00	90	3C	00	00 2E	xx xx	xx xx	start "read parameter" service
S	01 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request
S	17 A4	10	90	3C	00	00 2E	01 00	00 00	transmission

Fig. 3-7: Example: read parameter Y-0-0046

Example 2: Write single data

Master axis parameter, data length 4 bytes

Parameter: C-0-0006 "VM - speed command 1"**Data:** 0x000186A0 = 10.0000 Rpm**Unit address:** Master axis address; in the example: address 2**Results:** Error message "data larger than maximum input value"

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit addr.	Parameter number	User data	User data	
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	07 A2	00	9F	3C	02	00 06	A0 86	01 00	write parameter
S	01 02	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 02	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request

Slave sends execution acknowledge with error message:

	FBStatus	Parameter number	Error class	00	Additional-Error-Code	Ctrl	Status	
S	37 81	00 06	06	00	70 07	D0	01	xx xx

Fig. 3-8: Example: Write parameter C-0-0006 with error message "data greater than maximum value"

Example 3: Write list data

Drive parameter, data length 4x4 bytes (short list)

Parameter: A-0-0099 "Positioning speed"**Unit address:** Drive address; in example: address 3**Data:** 0x0010 (actual length of list: 16 bytes), 0x0010 (max. length: 16 bytes)

0x000186A0 = 10.0000 Rpm 0x000186A0 = 10.0000 Rpm

0x000186A0 = 10.0000 Rpm 0x000AAE60 = 70.0000 Rpm

Results: Data successfully written

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
M	03 A1	00	9F	3C	03	00 63	10 00	10 00	start: "write parameter" service
S	01 01	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge

	FBControl / FBStatus	User data	User data	User data	User data	User data	
M	00 A1	A0 86	01 00	A0 86	01 00	A0 86	transmission element 1-3
S	00 01	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	05 61	01 00	60 AE	0A 00	xx xx	xx xx	transmission element 3-4
S	01 01	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx xx	xx xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge idle state

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
M	17 01	xx	xx	xx	xx	xx xx	xx xx	xx xx	start: "read parameter" service
S	17 61	10	9F	3C	03	00 63	xx xx	xx xx	acknowledge, write successful

Fig. 3-9: Example: Write parameter A-0-0099

Example 4: Read list data

Drive parameter, data length 1024x4 byte (long list)

Parameter: P-0-0092 "Cam shaft profile 2"**Unit address:** Drive address; in the example: address 5**Results:** 0x1000 (actual length: 4096 bytes), 0x1000 (max. length: 4096 bytes)

List values: 0, 1,3

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	07 60	00	90	3C	05	80 5C	xx xx	xx xx	start: "read parameter" service
S	01 00	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 00	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request

Busy signal until list collected on PPC:

	FBControl / FBStatus	Parameter number	Error class	00	Additional error code	Ctrl	Status		
S	1F 80	80 5C	06	00	00 08	00	00	xx xx	busy signal
M	00 0F	xx xx	xx	xx	xx xx	xx	xx	xx xx	establish idle state
S	00 0F	xx xx	xx	xx	xx xx	xx	xx	xx xx	acknowledge idle state
M	17 00	xx xx	xx	xx	xx xx	xx	xx	xx xx	read request

... etc....

Slave sends min./max. values and first data

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
S	13 A0	10	90	38	05	80 5C	00 10	00 10	transmission of the list length, 1 st block, running transmission
M	16 00								next read request

	FBControl / FBStatus	User data	User data	User data	User data	User data	
S	10 A0	00 00	00 00	01 00	00 00	04 00	element 1,2 and half of element 3
M	17 00	xx xx	xx xx	xx xx	xx xx	xx xx	read request

... etc....

Slave sends final data in penultimate step and first data in the last step.

	FBControl / FBStatus	User data	User data	User data	User data	User data	
M	17 00	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	15 20	02 00	xx xx	xx xx	xx xx	xx xx	bytes 115 and 116 per step
M	00 0F	xx xx	xx xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge idle state

M	07 60	00	90	3C	05	80 5C	xx xx	xx xx	start: "read parameter" service
S	01 00	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 00	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
S	13 A0	10	90	3C	05	80 5C	00 00	37 00	next data, 1st block, last transmission
M	16 00	xx	xx	xx	xx	xx xx	xx xx	xx xx	next read request

... etc....

Slave sends last data in the last step (end)

	FBControl / FBStatus	User data	User data	User data	User data	User data	
M	17 00	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	15 60	00 00	03 00	00 00	xx xx	xx xx	last 6 bytes of data

Fig. 3-10: Example: read parameter P-0-0092

Example 5: Read data block 101

master axis parameter, data length 12 x 4 bytes

Parameter: Y-0-0001 (permanent for all data blocks)

Data block (unit address): 101 (configured with Y-0-0058)

Results: The parameters have the following values

0x000186A0 = 10.0000 degree	0x00030D40 = 20.0000 degree
0x000493E0 = 30.0000 degree	0x00061A80 = 40.0000 degree
0x0007A120 = 50.0000 degree	0x000927C0 = 60.0000 degree
0x000AAE60 = 70.0000 degree	0x000C3500 = 80.0000 degree
0x000DBBA0 = 90.0000 degree	0x000F4240 = 100.0000 degree
0x0010C8E0 = 110.0000 degree	0x00124F80 = 120.0000 degree

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	07 64	00	90	3C	65	00 01	xx xx	xx xx	start: "read parameter" service
S	01 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request
S	13 A4	10	90	3C	65	00 01	A0 86	01 00	first SF1 partial telegram, transmission of element 1

	FBControl / FBStatus	User data	User data	User data	User data	User data	
M	16 04	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	10 A4	40 0D	03 00	E0 93	04 00	80 1A	running SF1 partial telegram, transmission of element 2, 3 and half element 4
M	17 04	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	11 A4	06 00	20 A1	07 00	C0 27	09 00	running SF1partial telegram, transmission of half element 4, element 5 and 6
M	16 04	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	10 A4	60 AE	0A 00	00 35	0C 00	A0 BB	running SF1 partial telegram, transmission of element 7, 8 and half element 9
M	17 04	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	11 A4	0D 00	40 42	0F 00	E0 C8	10 00	running SF1 partial telegram, transmission of half element 9, element 10 and 11
M	14 04	xx xx	xx xx	xx xx	xx xx	xx xx	read request
S	14 44	80 4F	12 00	xx xx	xx xx	xx xx	final SF1partial telegram, transmission of element 12

Fig. 3-11: Example: read data block 101

Example 6: Write data block 116

drive parameter, data length 12 x 4 Byte

Parameter: Y-0-0001 (permanent for all data blocks)**Data block (unit address):** 116 (configured with Y-0-0085)**Results:** The parameter was written successfully.

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	03 A4	00	9F	3C	74	00 01	A0 86	01 00	start: "write parameter" service, first SF1 partial telegram, transmission of element 1
S	01 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge

	FBControl / FBStatus	User data	User data	User data	User data	User data	
M	00 A4	40 0D	03 00	E0 93	04 00	80 1A	running SF1 partial telegram, transmission of element 2, 3 and half element 4
S	00 04	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	01 A4	06 00	20 A1	07 00	C0 27	09 00	running SF1 partial telegram, transmission of half element 4, element 5 and 6
S	01 04	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	00 A4	60 AE	0A 00	00 35	0C 00	A0 BB	running SF1 partial telegram, transmission of element 7, 8 and half element 9
S	00 04	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	01 A4	0D 00	40 42	0F 00	E0 C8	10 00	running SF1 partial telegram, transmission of half element 9, element 10 and 11
S	01 04	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge
M	05 44	80 4F	12 00	xx xx	xx xx	xx xx	final SF1 partial telegram, transmission of element 12
S	01 04	xx xx	xx xx	xx xx	xx xx	xx xx	acknowledge

	FBControl / FBStatus	Cntrl	Service	Control byte	Unit add.	Parameter number	User data	User data	
M	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx xx	xx xx	xx xx	acknowledge idle state
M	17 04	xx	xx	xx	xx	xx xx	xx xx	xx xx	read request
S	17 64	10	9F	3C	74	00 01	xx xx	xx xx	write access successful

Fig. 3-12: Example: write data block 116

Bosch Rexroth SIS format (phase switching)

The "Bosch Rexroth SIS format" transmission format is needed for phase progressions and the "abort" service. (Theoretically, a transmission of A, C and Y and S and P parameters in SIS format is also possible, but the short format 1 has been designed for this.)

The SIS telegram is broken down into blocks of up to five words which are transmitted with the help of the fieldbus control and status words.

The general structure of the SIS format is described in section "Data storage protocol (SIS protocol)", page 2-19. This is why this section restricts itself to telegrams for phase transitions and terminate services.

Note: The PPC only acknowledges the receipt of phase progression requests. To detect whether the transition has been properly executed or not, the current mode must be polled until the target phase is reached, an error is displayed in the status byte or a timeout has run out.

The reaction telegram

It is necessary to differentiate between three cases for the PPC reaction telegram, namely,

- transmission without fault,
- faulty transmissions with error in the PK protocol shell and
- faulty transmission with error message in the SIS telegram.

Transmission without fault

Given an error-free transmission, the error bit in the FB status word is cleared (E=0) and possible requested data are following at word offset 2

Faulty transmission with error in the PK protocol shell

Given a faulty transmission with an error in the PK protocol shell, the PPC sets the error bit in the FB status word and sends a telegram in the following format:

Word 1	Word 2	Word 3		Word 4	Word 5	
Status word	Parameter no.	Error class 0x06	0x00	Additional error code	Ctrl	Status
Fieldbus control	Error message					

Faulty transmission with error message in the SIS telegram

Given a faulty transmission with error message in the SIS telegram and the error bit has not been set in the FB status word, then the error is displayed in the status byte.

The errors during a transmission in the parameter channel are discussed separately in section "Data storage protocol (SIS protocol)", page 2-19.

Example 1: Switching into parameter mode

	FBControl / FBStatus	StZ	CS	DatL	DatLW	Cntrl	Service	Adr	Adr	User data		
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	07 9B	02	5D	01	01	00	9D	00	00	02	xx	transition request
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge

Fig. 3-13: Example: Switching into parameter mode

Example 2: Switching into operating mode

	FBControl / FBStatus	StZ	CS	DatL	DatLW	Cntrl	Service	Adr	Adr	User data		
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	07 9B	02	5B	01	01	00	9D	00	00	04	xx	transition request
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge

Fig. 3-14: Switching into operating mode

Example 3: Reading current phase (phase switching into operating mode active)

	FBControl / FBStatus	StZ	CS	DatL	DatLW	Cntrl	Service	Adr	Adr	Status byte	User data	
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	07 8B	02	6C	00	00	00	92	00	00	xx	xx	start: "read mode" service
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	17 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	read request
S	17 AB	02	54	02	02	10	92	00	00	00	03	actual mode: e.g. "3"
M	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
												
												
M	07 8B	02	6C	00	00	00	92	00	00	xx	xx	start: "read mode" service
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	17 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	read request
S	17 AB	02	54	02	02	10	92	00	00	00	04	actual mode: "4"

Fig. 3-15: Switching into operating mode: read current mode

Example 4: Reading current phase (phase transition not possible)

In example: Error "D007", phase progression with turning master axis not permitted.

	FBControl / FBStatus	StZ	CS	DatL	DatLW	Cntrl	Service	Adr	Adr	Status byte	User data	
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	07 8B	02	6C	00	00	00	92	00	00	xx	xx	start: "read mode" service
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	establish idle state
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	17 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	read request
S	13 AB	02	77	04	04	10	92	00	00	02	04	status byte <> 0 (=2), phase 4
M	01 60	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	fetch 2 nd part

	FBStatus	Additional error code										
S	14 2B	07 D0	xx	xx	xx xx	xx	xx	xx xx				error "D007"

Fig. 3-16: Switching into parameterization mode: read the current mode with error message

Example 5: Abort

The abort service terminates all running services. It is needed, e.g., if the transmission of a cam profile was prematurely stopped and a new query has to be started.

	FBControl / FBStatus	StZ	CS	DatL	DatLW	Cntrl	Service	Adr	Adr	User data		
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	07 9B	02	F9	01	01	00	01	02	00	00	xx	terminate service
S	01 0B	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge
M	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	idle state
S	00 0F	xx	xx	xx	xx	xx	xx	xx	xx	xx	xx	acknowledge idle state
M	17 0B	02	5B	01	01	00	9D	00	00	04	xx	reading the execution acknowledgement
S	17 9B	02	E9	01	01	10	01	02	00	00	xx	service successfully completed

Fig. 3-17: "Abort" service

Error codes in the parameter channel

Overview

Types of errors

There are three different types of errors:

- Error in the sequence of the parameter channel (e.g., "Idle state was not established").
- Error in the SIS protocol sequence (e.g., "Invalid service"). The error is displayed in the status byte, value 0 is in the error code.
- Execution error during phase or parameter transmission (e.g., "Error with parameter transmission") The error is displayed in the status byte, further information is in the error code.

Fig. 3-18 (Table for determining error codes) lists how to determine the error out of the status byte, the error flag in the field bus control word and the error code of the telegram. The slave answers with each error either in the same format as the request, or if the error flag is set, with a telegram in error format.

Status byte	Errorflag	Error code	Evaluated error code
= 0	0	don't care	no error
don't care	1	<> 0	error code (error in parameter channel or execution error)
0x1 or 0x2	0	<> 0	error code (execution error)
<> 0	1	= 0	status byte (SIS protocol error)
<> 0	0	= 0	status byte (SIS protocol error)

Fig. 3-18: Table for determining error codes

For individual transmissions this means:

Drive and control parameters

If an error occurs during a parameter transmission in "**short format 1**", then an **error bit is set** and the error code is in word 4 in **Motorola format**.

See Fig. 3-19: Error codes in Motorola format with an error in the sequence of the parameter channel and

Fig. 3-20: Error codes in Motorola format with an execution error during parameter transmission.

Fig. 3-22: Value in the status byte with an SIS protocol error

Phase transition

If an error occurs during phase transition in "**Standard SIS format**"

- then an error telegram is generated if it is an error in the parameter channel sequence (e.g., idle state not established). The **error bit is set** and the error code is in word 4 in **Motorola format**

See Fig. 3-19: Error codes in Motorola format with an error in the sequence of the parameter channel.

- a standard SIS data telegram is generated if it is an error during phase transition (e.g., phase transition with turning master axis not allowed) or an error in the SIS protocol. The **error bit is not set**, the status byte not equal to 0 and in word 2 of the slave telegram, there is an additional error code in **Intel format**.

See Fig. 3-21: Execution error in Motorola format with phase transition.

See Fig. 3-22: Value in the status byte with an SIS protocol error.

Error codes of the parameter channel

Error code	Error description
0x0004	Write access was attempted while the PPC was waiting during an old write access for further data (slave telegrams).
0x0008	Busy signal: data requested not yet on the PPC. Busy bit is set. Errorflag not set.
0x000C	Combination of error code 0x0004 and 0x0008. Busy bit and Errorflag set.
0x0080	No error.
0x0081	New data being sent, F-bit not yet set.
0x0083	More data expected, but new data sent (F-bit set during transmission of several data blocks).
0x0085	Data length too long (here >128 bytes).
0x0086	Idle state not established.
0x0088	An error occurred during the data transmission of the data from/to the PPC.
0x0089	Incorrect checksum detected upon receipt of an SIS telegram from the master.
0x008A	A relayed object number of a field bus object is not defined (valid range: 0x5e74 to 0x5fff).
0x008B	Format specified incorrect or changed between writing/reading of an SIS telegram.
0x008C	The length set in the control byte greater than parameter channel.
0x0090	The format and/or bits 11 to 15 of the control word were changed while transmitting several data blocks.
0x0095	No data present with a write command.
0x0098	Error during collection of data blocks. Length 0 was set.

Fig. 3-19: Error codes in Motorola format with an error in the sequence of the parameter channel

Error codes with an execution error of a parameter transmission

Error code	Error messages in the serial protocol
0x0000	No error
0x0001	Service channel not open
0x0009	Incorrect access to element 0
0x0080	"Master axis address not permitted" The master axis address is larger than permitted or is 0
0x00A0	"Non-permissible request" e.g., an access to S/P parameter initialization mode
0x00B0	"Non-permissible element" e.g., write access only with element operating data
0x00C0	"Drive address not permitted" The drive address is greater than permitted or the drive in SERCOS ring not active (deactivated or not available)
0x00F0	"Fatal software error" During a parameter transmission a PPC internal error occurred (see Y-0-0041) and it effected data exchange
0x1001	No IDN available
0x1009	Incorrect access to element 1

0x2001	No name available
0x2002	Name transmission too short
0x2003	Name transmission too long
0x2004	Name cannot be changed
0x2005	Name presently write-protected
0x3002	Attribute transmission too short
0x3003	Attribute transmission too long
0x3004	Attribute cannot be changed
0x3005	Attribute presently write protected
0x4001	No unit available
0x4002	Unit transmission too short
0x4003	Unit transmission too long
0x4004	Unit cannot be changed
0x4005	Unit presently write protected
0x5001	No minimum input value available
0x5002	Minimum input value transmission too short
0x5003	Minimum input value transmission too long
0x5004	Minimum input value cannot be changed
0x5005	Minimum input value presently write protected
0x6001	No maximum input value available
0x6002	Maximum input value transmission too short
0x6003	Maximum input value transmission too long
0x6004	Maximum input value cannot be changed
0x6005	Maximum input value presently write protected
0x7002	Data transmission too short
0x7003	Data transmission too long
0x7004	Data cannot be changed
0x7005	Data presently write protected
0x7006	Data smaller than minimum input value
0x7007	Data greater than maximum input value
0x7008	Data not correct
0x7009	Data write protected with password
0x700A	Data cyclic configured
0x700B	"Invalid list element" The list offset set in SIS services 0x91 or 0x9E exceeds range of list or does not show the start address of a list element.
0x700C	Data presently write protected because of other settings (e.g., parameter, operating mode, drive enable)
0x710B	Data changeable by motion control parameter only
0x710C	"Data exceeds numeric range" The transmitted value is less than zero or greater than the "modulo value" (S-0-0103), given a modulo axis
0x710D	"data length cannot be changed presently" The length of the data cannot be changed in the current mode

0x710E	"Data length cannot be changed" The length of data permanently write-protected.
0x8001	"Service channel presently busy (BUSY)" The desired access had not been completed during a timeout period (programmable via Y-0-0124). The service channel was, e.g., (still) busy. Data transmission cannot be executed.

Fig. 3-20: Error codes in Motorola format with an execution error during parameter transmission

Error codes with an execution error of a phase transition

This error code follows the current phase.

Error code	Error messages in the serial protocol
0x8004	Incorrect phase specified via serial protocol
0xD005	"Phase transition still active" phase transition presently not possible as another is active
0xD006	"Phase transition with drive enable not allowed" "AF" set for at least one drive
0xD007	"Phase transition with turning master axis not permitted"
0xD008	"Cannot switch phase" Master axis functionality enabled

Fig. 3-21: Execution error in Motorola format with phase transition

Values in the Status Byte

With a standard SIS telegram and a telegram in error format, the status byte signals whether there is an error or not. The detailed error codes are signaled separately.

Protocol error	Value in the status byte	Error description
"invalid service"	0xF0	The service requested is not specified or is not supported by the participant addressed.
"invalid telegram"	0xF1	The telegram cannot be evaluated. Either an incorrect start character or a reaction telegram has arrived.
"incorrect telegram length"	0xF2	Both lengths specified in the telegram header do not agree.
"incorrect checksum"	0xF4	The checksum received does not agree with internal calculations.
"incorrect following telegrams"	0xF8	The address of the master, the control byte, the axis address or the parameter number has changed in the following telegram.

Fig. 3-22: Value in the status byte with an SIS protocol error

Execution error	Value in the status byte	Error description
"error with parameter transmission"	0x01	During read or write of a parameter, an error occurred (see below "Execution error with parameter transmission").
"error with phase transition"	0x02	The specified target phase was not reached (see below "Execution error when accessing SERCOS phase").

Fig. 3-23: Value in the status byte with an SIS execution error

3.6 Short format 3 (direct parameter access)

The parameter channel formats described above encapsulate the SIS protocol for the transmission of lists or parameters.

The short format 3 refrains from encapsulating the SIS protocol and optimizes the handshake of the control and status word.

This permits the user to directly access single parameters (words, doublewords) of the controller or drives in an easier and quicker way.

Object parameter mapping

In case of the format 3, the SERCOS and control parameters are addressed via the index and subindices. The object parameter mapping can only be used to address the operating datum (SERCOS element 7).

To map the parameters to index and subindex, the following formulas must be applied:

Index = SercosIDNumber + Z (Fig. 3-24, column 1)

Subindex = DriveNumber (Fig. 3-24, column 2)

SercosIDNumber = Index - Z (Fig. 3-24, column 3)

DriveNumber = Subindex (Fig. 3-24, column 4)

For Y parameters, the value of the subindex is always 1.

Z is dependent on the parameter type and can be read from the table below.

Index hex	Subindex	Sercos IDNumber	Drive Number	
0x5E6F	1	1647	-	Y parameter Z=0x5800
...	1		-	
0x5802	1	2	-	
0x5801	1	1	-	
0x57FF	Master axis-No	4095	1 to 2	C parameter Z=0x4800
...	Master axis-No		1 to 2	
0x4802	Master axis-No	2	1 to 2	
0x4801	Master axis-No	1	1 to 2	
0x47FF	Drive-No	2047	1 to 40	A parameter Z=0x4000
...	Drive-No	...	1 to 40	
0x4002	Drive-No	2	1 to 40	
0x4001	Drive-No	1	1 to 40	
0x3FFF	Drive-No	4095	1 to 40	P parameter Z=0x3000
...	Drive-No	...	1 to 40	
0x3002	Drive-No	2	1 to 40	
0x3001	Drive-No	1	1 to 40	
0x2FFF	Drive-No	4095	1 to 40	S parameter Z=0x2000
...	Drive-No	...	1 to 40	
0x2002	Drive-No	2	1 to 40	
0x2001	Drive-No	1	1 to 40	

Fig. 3-24: Mapping of SERCOS parameters to index and subindex.

User interface

The format 3 user interface is defined according to the client-server model and supports the following functions:

- ReadRequest(index, subindex)
- ReadConfirmationPos(*data length, *data)
- ReadConfirmationNeg(*error)
- WriteRequest(index, subindex, data length, *data)
- WriteConfirmationPos()
- WriteConfirmationNeg(*error)
- ResetRequest()
- ResetConfirmationPos()

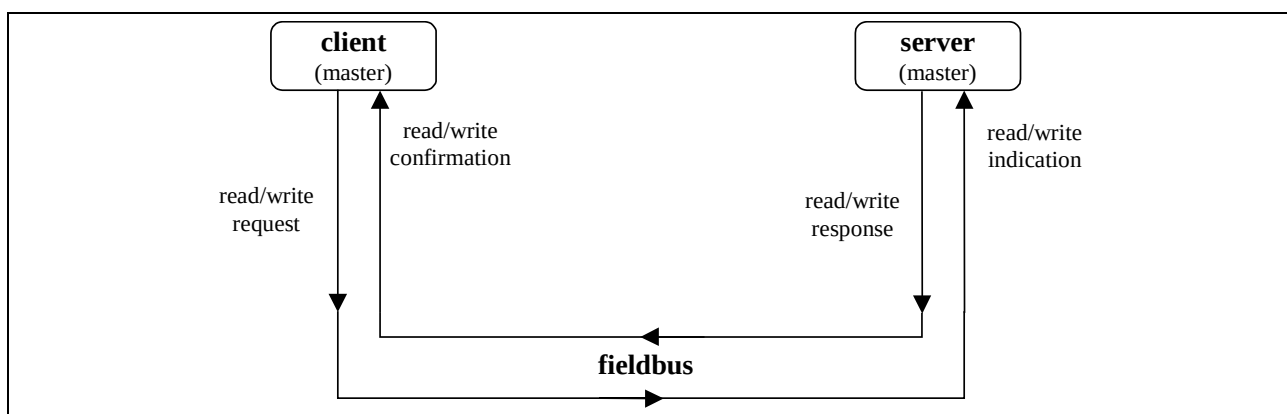


Fig. 3-25: Client server model

The client transmits a **ReadRequest** to the server and, using the index and subindex, informs the server of the object and/or parameter it intends to read. The server generates the value of the parameter and transmits the response to the client. If the reading process is completed successfully, the master receives a positive **ReadConfirmation**.

In order to write a parameter, the client generates a **WriteRequest**. Using this request, the client informs the server of which object it intends to write with which value. After the write request has been processed successfully in the server, the client receives a positive **WriteConfirmation**.

The client has the possibility to execute a **ResetRequest** of the parameter channel. As a result, the server resets its internal state machine and is ready for new requests. The slave acknowledges the reset by a **ResetConfirmationPos()**.

The **index** and **subindex** are used to address the SERCOS parameters (see "Object parameter mapping"). The index is a 16-bit value and the subindex an 8-bit value.

The **data length** specifies how many bytes of data (useful data) should be transmitted or received. The maximum data length is 128 bytes.

Telegram structure

The data of the header and the data field is transmitted in the Big Endian (Motorola) format. The maximum telegram length is 12 bytes. The first two bytes of a request telegram (telegram from client to server) represent the control word. The first two bytes of the confirmation telegram are the status word.

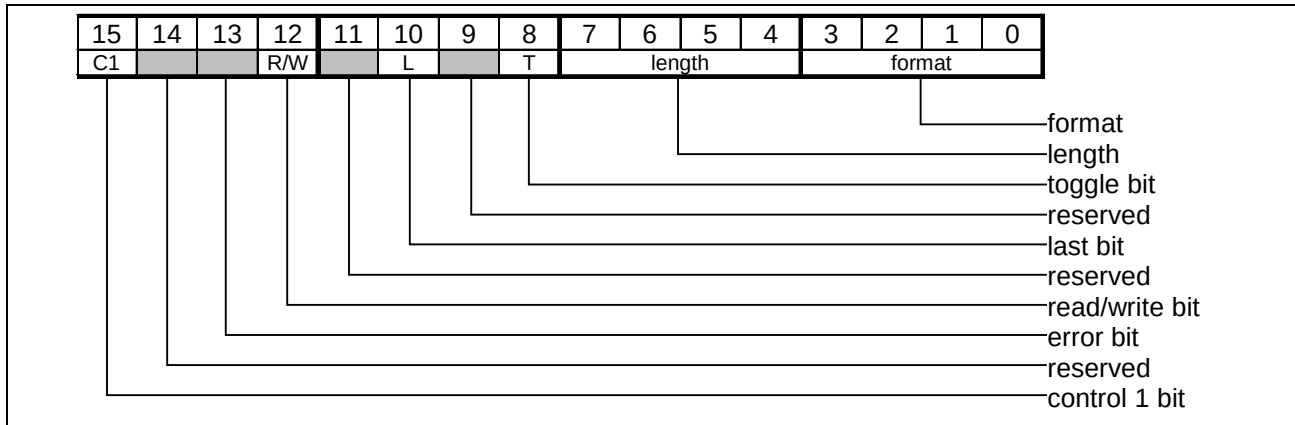


Fig. 3-26: Control word

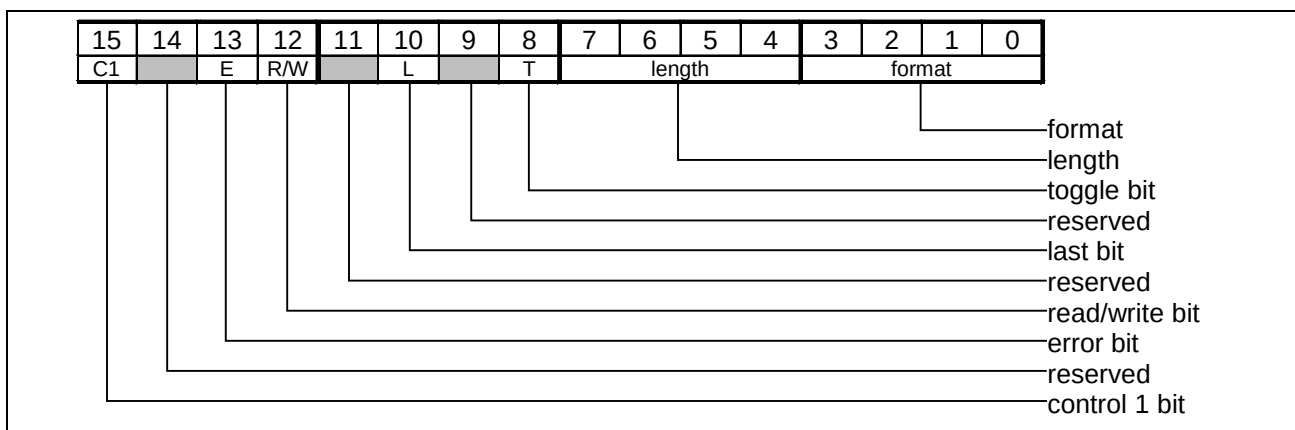


Fig. 3-27: Status word

The various bits have the following meaning:

- Format:** These bits describe the meaning of the following data of the parameter channel. The value in the format 3 is 0xC.
- Length:** These bits describe the number of valid bytes of the parameter channel without the two bytes of the control and status word. The values of the residual bytes of the parameter channel are not defined.
- Toggle bit:** This bit is toggled upon each new telegram. The client is allowed to toggle this bit only if the toggle bit of the control word is equal to the toggle bit of the status word. The server is allowed to toggle this bit if it is not equal to the toggle bit of the control word.
- Last bit:** This bit is set when the last telegram of a fragmentation arrives.

Read / Write bit: R/W=1: indicates that the client transmits a ReadRequest.

R/W=0: the client transmits a WriteRequest. The value of this bits is reflected in the status word.

Error bit: The server sets the error bit in the status word if an error has occurred. Thus, the server's telegram is a negative confirmation telegram.

Control 1 bit: This bit is always set to 1 in the short format 3.

H byte ¹	L byte	H byte	L byte	H byte	L byte	-	-	-	-	-	-
control word		index		-	sub-index	-					

Fig. 3-28: Read request

H byte	L byte	-	-	-	-	-	-	-	-	-	-
status word		user data									

Fig. 3-29: Read confirmation positive

H byte	L byte	H byte	L byte	-	-	-	-	-	-	-	-
status word		error code		-							

Fig. 3-30: Read confirmation negative

H byte	L byte	H byte	L byte	H byte	L byte	-	-	-	-	-	-
control word		index			sub-index	user data					

Fig. 3-31: Write request

H byte	L byte	-	-	-	-	-	-	-	-	-	-
control word		-									

Fig. 3-32: Write confirmation positive

H byte	L byte	H byte	L byte	-	-	-	-	-	-	-	-
status word		error code		-							

Fig. 3-33: Write confirmation negative

H byte	L byte					-	-	-	-	-	-
control word											

Fig. 3-34: Reset request

H byte	L byte					-	-	-	-	-	-
status word											

Fig. 3-35: Reset confirmation positive

¹ H byte = high byte; L byte = low byte

Error Codes

If it detects a defective message, the slave generates a negative confirmation. This negative confirmation contains a 16-bit error code. If the high byte = 0x1F, the low byte contains an error code of the SYNAX application. All other error codes represent errors in the handling of the parameter channel or object access.

Error code	Description
0x0085	The total of data of all fragments collected is too long.
0x008A	The index transmitted by the master does not exist.
0x008B	The format is unknown.
0x008C	The length of valid data specified in the status word exceeds that of the parameter channel itself.
0x0090	The format has changed during collection of the various fragments.
0x0095	The length in case of a ReadRequest is incorrect (not equal to 4 bytes).
0x009C	The subindex specified does not exist.
0x009D	The parameter is write-protected.
0x009E	The parameter is read-protected.
0x00A1	The data length of the parameter datum has been exceeded during a write access.
0x00A2	The data length of the parameter data has been fallen below during a write access.
0x00A3	The maximum value of the parameter data has been exceeded during a write access.
0x00A4	The minimum value of the parameter data has been fallen below during a write access.
0x00A5	A non-valid datum has been written during a write access.
0x1FXX	Error of the SYNAX application. An application error XX is not yet known.

Fig. 3-36: Error codes of the parameter channel

Examples

Example 1: Read access to parameter Y00:Y-0-0001

In this example, the switch-on delay time is read. The slave responds with 2 bytes of data 0x0FA0 (= 4000 ms).

First of all, the parameter must be converted into index and subindex.

The following applies according to Fig. 3-24:

Index = SercosIDNumber + Z

with SercosIDNumber = 1; Z = 0x5800, because of Y parameter (see Fig. 3-37)

This results in

Index = 0x0001 + 0x5800 = 0x5801

Subindex = 1, because of Y parameter

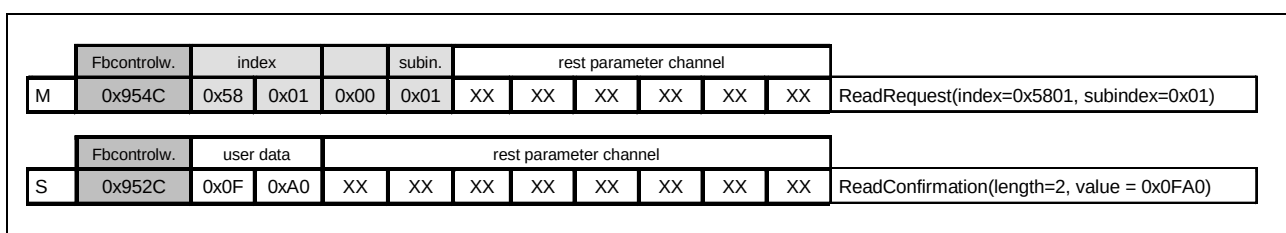


Fig. 3-37: Read parameter Y00:Y-0-0001

Example 2: Read access to parameter A02:S-0-0057

In this example, the positioning window of drive 2 is read. The slave responds with 4 bytes of data 0x000003E8 (= 1000*10⁻³ degrees).

First of all, the parameter must be converted into index and subindex.

The following applies according to Fig. 3-24:

Index = SercosIDNumber + Z

with SercosIDNumber = 57(0x39); Z = 0x2000, because of S parameter (see Fig. 3-38)

This results in

Index = 0x0039 + 0x2000 = 0x2039

Subindex = 2, because of drive 2

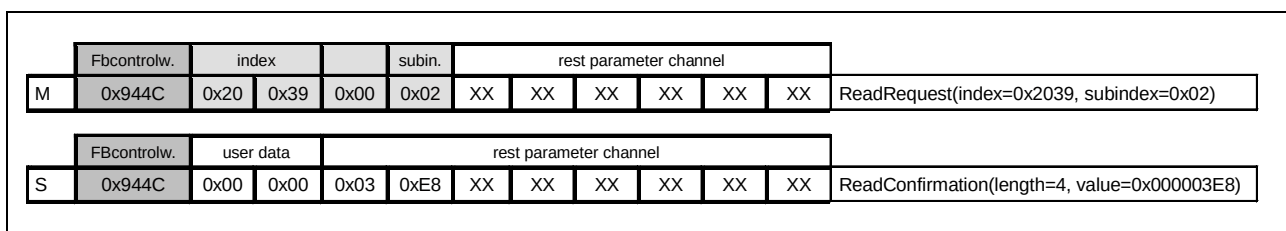


Fig. 3-38: Read parameter A02:S-0-0057

Example 3: Write access to parameter A03:P-0-0040

In this example, the weighting of the speeds of the analog output channel 1 of drive 3 is written with the value 0x0001 (= 1 rpm/10 V).

First of all, the parameter must be converted into index and subindex:

The following applies according to Fig. 3-24:

Index = SercosIDNumber + Z

with SercosIDNumber = 40(0x28); Z = 0x3000, because of P parameter (see Fig. 3-39)

This results in

Index = 0x0028 + 0x3000 = 0x3028

Subindex = 3, because of drive 3

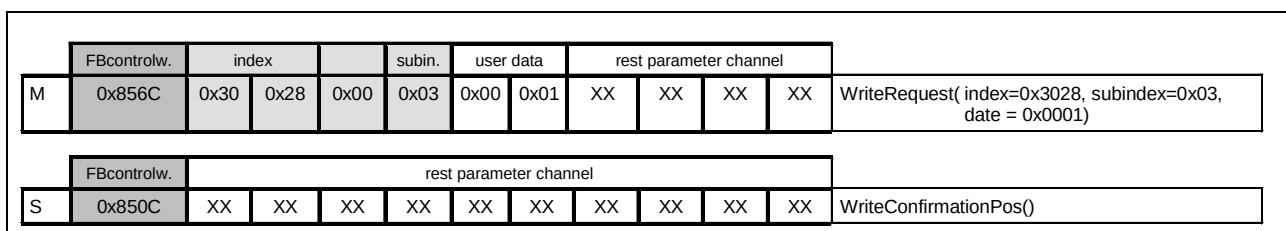


Fig. 3-39: Read parameter A03:P-0-0040

Example 4: Write access to parameter A04:A-0-0030

In this example, the process controller proportional gain 1 of drive 4 is written with the value 0x12345678.

First of all, the parameter must be converted into index and subindex.

The following applies according to Fig. 3-24:

Index = SercosIDNumber + Z

with SercosIDNumber = 30(0x1E); Z = 0x4000, because of A parameter (see Fig. 3-40)

This results in

Index = 0x001E + 0x4000 = 0x401E

Subindex = 4, because of drive 4

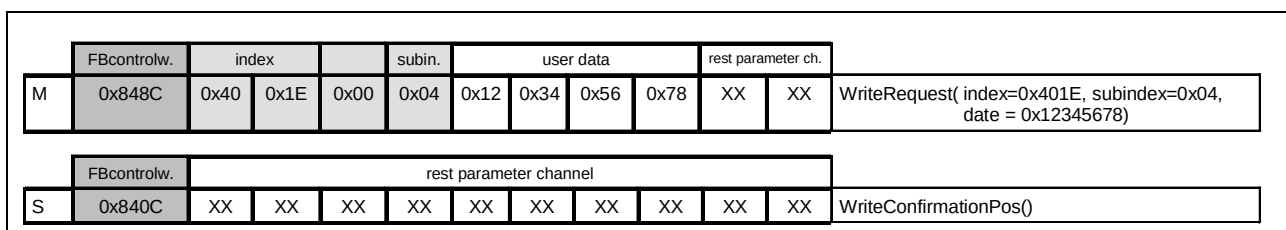


Fig. 3-40: Write parameter A04:A-0-0030

Example 5: Error after write access

In this example, the output position of master axis 2 (C02:C-0-0066) is written with the value 0x12345678. The slave acknowledges the WriteRequest with a negative WriteConfirmation. The error code 0x009D indicates that the parameter is write-protected.

First of all, the parameter must be converted into index and subindex:

The following applies according to Fig. 3-24:

Index = SercosIDNumber + Z

with SercosIDNumber = 66(0x42); Z = 0x4800, because of C parameter (see Fig. 3-41)

This results in

Index = 0x0042 + 0x4800 = 0x4842

Subindex = 2, because of master axis address 2

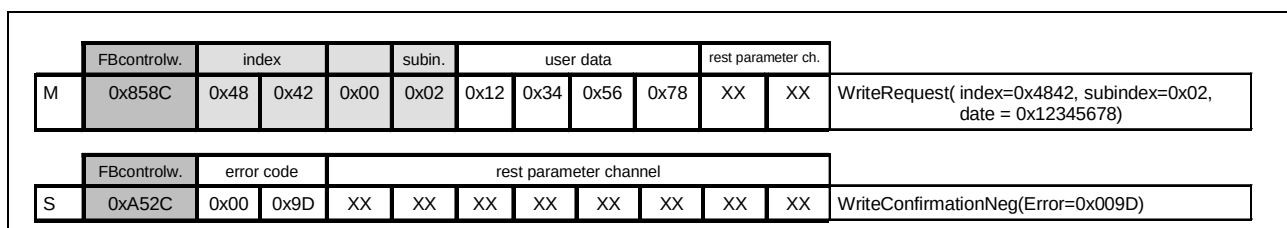


Fig. 3-41: Write parameter C02:C-0-0066. The slave responds with error code 0x009D parameter write protected.

Example 6: Reset of the parameter channel

In this example, the parameter channel is reset. The slave resets its internal state machine to the basic state and is ready for new requests. This may become necessary if, for instance, it is not possible to complete a fragmented WriteRequest. In this case, the state machine of the slave must be reset. To execute a reset, the master must set the control word = 0x000F.

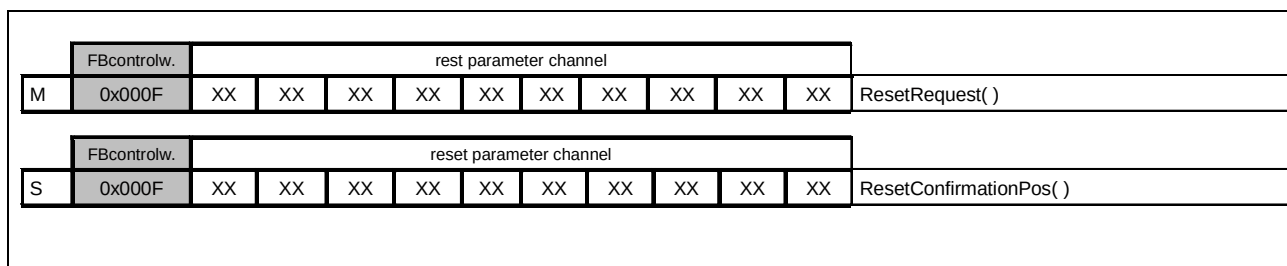


Fig. 3-42: Reset of the parameter channel

3.7 DPS01 board hardware

Front view of the DPS01

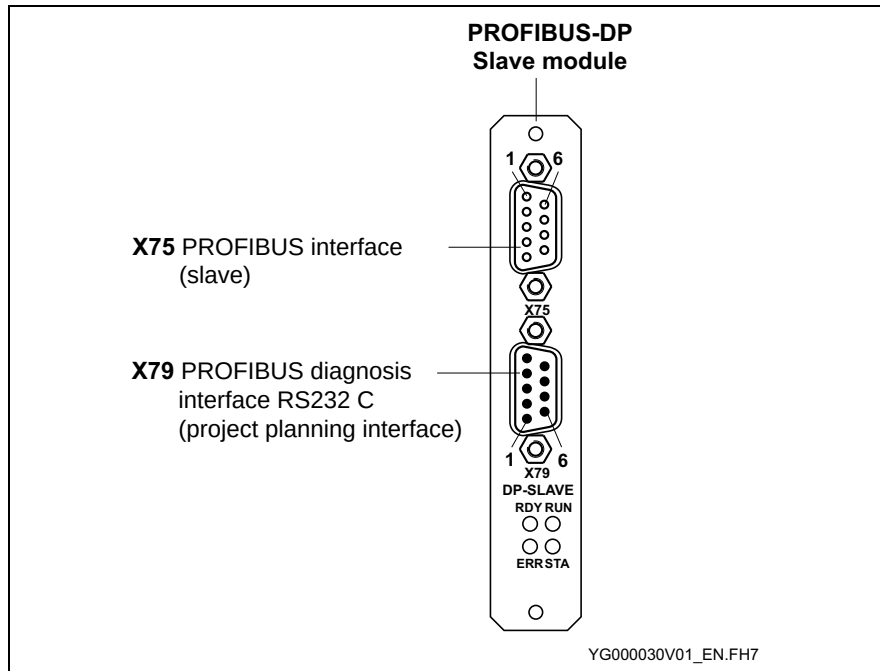


Fig. 3-43: Front view of the DPS01

DPS01 structure

The DPS01 board is conceived as a PC104 bus board so that it can be directly mounted to the PPC control board.

The PC-104 bus is the interface to the PPC and also the power supply for the DPS01 card.

The DPS01 has the following interfaces:

Profibus interface (X75)

The Profibus interface meets DIN 19245, sec. 1 requirements for power connections with category A or B lines as per DIN 19245, sec. 3. This supports both type A and B as per DIN 19245, sec. 3.

To ensure EN standards for EMC safety the Profibus interface has been completely galvanically isolated.

As per DIN 19245, sec. 1, module DPS01 is equipped with a 9-pin D-subminiature plug-in connector to accommodate the Profibus.

Signal configuration X75, Profibus connections

X 75	RS 485 reference	Signal according to DIN 19245 sec. 3	Definition
1			
2			
3	B / B'	RxD / TxD-P	receive/send data - P
4			
5	C / C'	DGND	data reference potential
6		VP	supply voltage plus (P5V)
7			
8	A / A'	RxD / TxD-N	receive / send data - N
9			

Fig. 3-44: Signal configuration X75, Profibus connection

Profibus diagnostic interface X79

The Profibus diagnostic interface X79 is not available for the user.

DPS01 diagnoses

LEDs on the front panel

The DPS01 has four diagnostic LEDs on the front. They enable the diagnosing of current states of the Profibus and the communication between DPS01 and PPC.

Arrangement of diagnostic LEDs

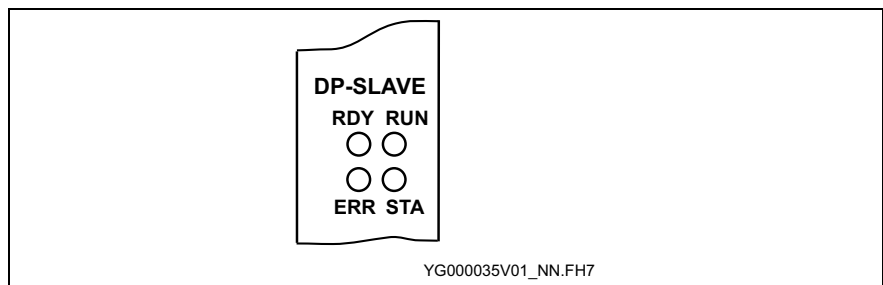


Fig. 3-45: Arrangement of LED diagnoses

LED diagnostic definitions

LED		Status	Definition
RDY		on	DPS01 ready, hardware test and software test completed successfully
Ready		flashing cyclic	bootstraploader active, there is no firmware on the DPS01
Yellow		flashing acyclic	hardware or system error
		off	hardware defective
RUN		on	bus communication is active
Run		flashing cyclic	no bus communication
Green		flashing acyclic	missing bus configuration
		off	no communication
ERR		on	Profibus error
Error		flashing cyclic	---
Red		flashing acyclic	---
		off	no error
STA		on	data exchange with master
Status		flashing cyclic	---
Yellow		flashing acyclic	---
		off	no data exchange with master

Examples:

RDY			RUN	hardware error
ERR			STA	
RDY			RUN	bootstraploader active, there is no firmware on the DPS01
ERR			STA	
RDY			RUN	no configuration, the DPS01 was not yet configured via PPC
ERR			STA	
RDY			RUN	short circuit on Profibus
ERR			STA	
RDY			RUN	watchdog between DPS01 and PPC has operated
ERR			STA	
RDY			RUN	communication runs without an error
ERR			STA	

4 The fieldbus interface DeviceNet

4.1 Introduction

Section 2 describes the basics that apply to all fieldbusses within SYNAX. This sections outlines the bus-specific features and functions of the DeviceNet.

The connection of SYNAX to DeviceNet uses BGR RECO-CONTROL V2 LK DNS03. In the following shortening DNS03 is used for the DeviceNet assembly.

4.2 Functional features

Module DNS03 has the following features:

- DeviceNet Slave per ODVA specification 2.0
- DeviceNet – circuit with completely galvanically isolated interface. Uses a Phoenix connector.
- LED diagnostic field that corresponds to ODVA guidelines on the front panel.
- Freely configurable process data channel with 1 to 64 words data width at the BUS.
- Monitoring of both process data channel and Explicit Messages.
- Data exchange to control module PPC via the dual port memory.
- Hardware and software synchronization with control module PPC.
- Implementation of an object structure for simple accessing of variables and parameters of the control module and drive.
- Upload / download function via four array of 16 to 128 bytes implemented (Explicit Message).
- The module uses the predefined master/slave connection set and works as group only two server.

4.3 Setting baudrates, MAC-IDs and data formats

The DeviceNet baudrate is set via a SynTop dialog "PPC (host) communications basic settings".

The DeviceNet interface of the SynTop dialog "Fieldbus - settings" should be used to specify MAC-IDs and data formats.

Baudrate DeviceNet permits baudrates of 125, 250 and 500 kbps. The rate is written via SynTop in parameter "DeviceNet - Baudrate" (Y-0-0151). The default value is 125kbps (Y-0-0151 = 125000).

MAC-ID The MAC-ID contains the participant address of the slave in DeviceNet. It is written via SynTop into parameter "Fieldbus address" (Y-0-0125). It must be set in a range 2 to 63 for DeviceNet.

Note: After executing the "load base parameters" command, the address in parameter "Fieldbus address" (Y-0-0125) reads: 99! DNS03 does not accept invalid address and goes to the bus with DeviceNet default address 63.

Data formats The DeviceNet standard specifies the Intel format for data transmission. This can cause data format problems if the PLC is working with a Motorola processor and the master board has no knowledge of the current data structure.

For the DNS03 to be able to correct the data forms prior to relaying the data to the PPC, there are two possible data swapping settings:

- words in doubleword
- bytes in data exchange objects

SynTop can transmit both of these settings to bit 8 (words) and bit 9 (bytes) of parameter "Fieldbus - control bits" (Y-0-0129).

Note: After executing the command "load base parameters", both data format bits (swap bits) are deleted in parameter "Fieldbus - control bits" (Y-0-0129). This corresponds to the setting with a master board that conforms to standards.

Example for data format configuration

DeviceNet communication between SYNAX and an

- SLC 5/04 (Allen Bradley PLC) and a
- scanner 1747 SDN (master board)

requires both exchange settings (words and bytes) because the PLC works with a Motorola processor and the scanner works with an Intel word machine.

Note: The setting of both SWAP bits depends on the PLC and master board. It must be empirically determined when implementing the DeviceNet interface, if data format of the master board is not known.

4.4 Object structure

Class, instance and attribute

The DeviceNet specification denotes the data of a unit by class, instance and attribute. The data for the PLC are configured either as polled I/O or it accesses the fieldbus object or the SYNAX parameter with an Explicit Message using class/instance and attribute. The values for these are calculated as below:

S parameters (S-0-0001 - S-0-4095)

$$class = 101 + \left\lfloor \frac{S \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = S \text{ parameter} - ((class - 101) * 255)$$

$$attribute = \text{drive address}$$

Example for S parameters

SYNAX parameter	Class	Instance	Attribute
Axx: S-0-0001	101	1	xx
Ayy: S-0-0002	101	2	yy
...	...	...	...
Azz: S-0-0255	101	255	zz
Axx: S-0-0256	102	1	xx
...	...	...	...
Ayy: S-0-4094	117	14	yy
Azz: S-0-4095	117	15	zz

Fig. 4-1: Example for S parameters

P parameters (P-0-0001 - P-0-4095)

$$class = 118 + \left\lfloor \frac{P \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = P \text{ parameter} - ((class - 118) * 255)$$

$$attribute = \text{drive address}$$

Example for P parameters

SYNAX parameter	Class	Instance	Attribute
Axx: P-0-0001	118	1	xx
Ayy: P-0-0002	118	2	yy
...	...	...	...
Azz: P-0-0255	118	255	zz
Axx: P-0-0256	119	1	xx
...	...	...	...
Ayy: P-0-4094	134	14	yy
Azz: P-0-4095	134	15	zz

Fig. 4-2: Example for P parameters

A parameters (A-0-0001 - A-0-2047)

$$class = 135 + \left\lfloor \frac{A \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = A \text{ parameter} - ((class - 135) * 255)$$

$$attribute = drive \ address$$

Example for A parameters

SYNAX parameter	Class	Instance	Attribute
Axx: A-0-0001	135	1	xx
Ayy: A-0-0002	135	2	yy
...	...	...	...
Azz: A-0-0255	135	255	zz
Axx: A-0-0256	136	1	xx
...	...	...	...
Ayy: A-0-2046	143	6	yy
Azz: A-0-2047	143	7	zz

Fig. 4-3: Example for A parameters

C parameters (C-0-0001 - C-0-4095)

$$class = 144 + \left\lfloor \frac{C \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = C \text{ parameter} - ((class - 144) * 255)$$

$$attribute = master \ axis \ address$$

Example for C parameters

SYNAX parameter	Class	Instance	Attribute
Cxx: C-0-0001	144	1	xx
Cyy: C-0-0002	144	2	yy
...	...	...	...
Czz: C-0-0255	144	255	zz
Cxx: C-0-0256	145	1	xx
...	...	...	...
Cyy: C-0-4094	160	14	yy
Czz: C-0-4095	160	15	zz

Fig. 4-4: Example for C parameters

Y parameters (Y-0-0001 - Y-0-1647)

$$class = 144 + \left\lfloor \frac{Y \text{ parameter} + 4095}{255} \right\rfloor$$

$$instance = Y \text{ parameter} - ((class - 144) * 255) + 4096$$

$$attribute = 1$$

Example for Y parameters

SYNAX parameter	Class	Instance	Attribute
Y00: Y-0-0001	160	17	1
Y00: Y-0-0002	160	18	1
...	...	...	...
Y00: Y-0-0239	160	255	1
Y00: Y-0-0240	161	1	1
...	...	...	...
Y00: Y-0-1646	166	132	1
Y00: Y-0-1647	166	133	1

Fig. 4-5: Example for Y parameters

Data exchange objects (5E70 - 5FFE)

The following applies for the objects to the data exchange via SIS telegram, diagnosis and control of the multiplex channel:

Index	Subindex	Class	Instance	Attribute
5E70	0	100	1	100
5E71	0	100	2	100
5E72	0	100	3	100
5E73	0	100	4	100
5FF0	0	168	9	1
5FF2	0	168	11	1
5FF5	0	168	14	1
5FF6	0	168	15	1
5FFC	0	168	21	1
5FFD	0	168	22	1
5FFE	0	168	23	1

The DNS03 uses 8 bit values for class, instance and attribute.

4.5 Monitoring fieldbus transmission behavior with bus failure

As described in section 2.5 the DNS03 monitors bus behavior with a watchdog timer so that it can specifically respond to a bus failure.

Watchdog Function

The PLC indirectly sets watchdog time via Expected Packet Rate during boot up. The slave uses this to recognize a bus failure.

The time set in [ms] is computed based on the object

class 5, instance 1, attribute 9: expected packet rate

and multiplied by 4.

The watchdog is switched off with 0 value.

Error reaction with bus failures

The DNS03 freezes the most recently received data in the event of a bus failure and deletes bit 3 in object 'Status fieldbus' (5FF2). This bit is displayed in operating mode on the PPC at output "Fieldbus - real time channel active" (A:C01.08). The user can use this bit in the I/O logic (VKL) for a specific error reaction.

With bus return, output "Fieldbus - real time channel active" (A:C01.08) is set.

4.6 DNS03 board hardware

Front view of the DNS03

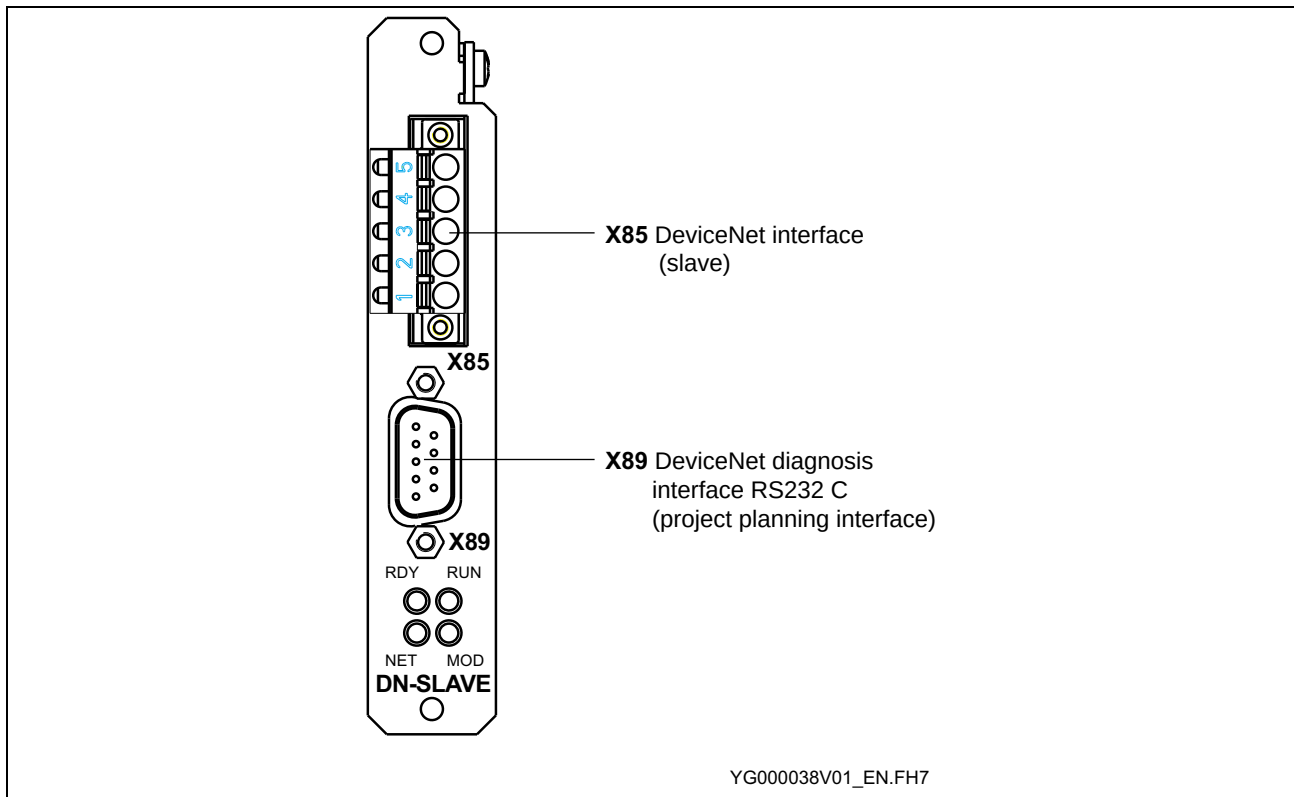


Fig. 4-6: Front view of the DNS03

DNS03 structure

The DNS03 board has been designed as a PC-104 board for direct mounting on the control board.

Interface to the PPC The PC-104-bus is the power supply for the DNS03 card and also the data exchange between DNS03 card and PPC card.

The DNS03 has the following interfaces:

DeviceNet interface The transceiver of the DNS03 module receive power in accordance with the ODVA specification via the bus cable. This does not, however, mean that the entire unit is supplied via the bus. The PPC control only works if it is supplied with voltage separately.

The DeviceNet interface is galvanically completely isolated in accordance with the ODVA specification via a Phoenix combicon 5 pin connector.

Signal configuration X85, DeviceNet

X85	Signal	Designation	Color
1	V-	supply voltage	black
2	CAN_L	CAN LOW signal	blue
3	Drain	reference potential	blank
4	CAN_H	CAN HIGH signal	white
5	V+	supply voltage +	red

Fig. 4-7: Signal configuration X85, DeviceNet

Signal configuration X89

Is not supported at the moment.

DNS03 diagnoses

LED diagnoses on the front panel

The DNS03 panel has a total of 4 diagnostic LED on the front. These enable the diagnosing of states between the bus communication and the communication between board DNS03 and PPC.

Arrangement of the diagnostic - LED

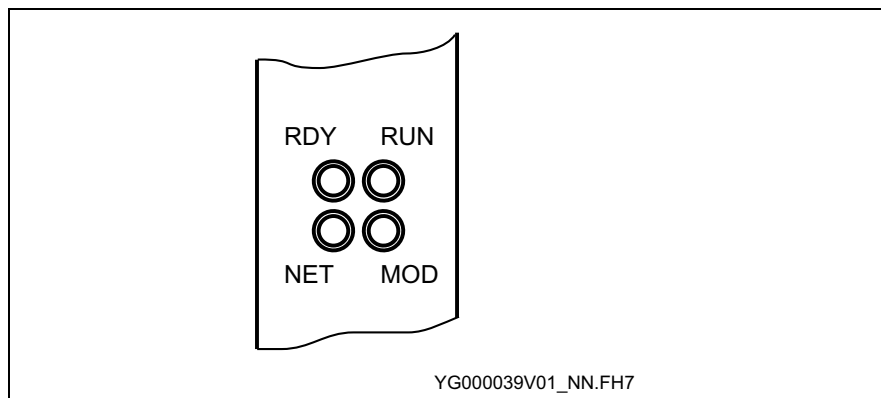


Fig. 4-8: Arrangement of the diagnostic - LED

LED diagnostic definitions

LED		Status	Definitions
RDY		on	DNS03 ready, hardware test and software test completed successfully
Ready		flashing cyclic	bootstraploader active, there is no firmware on the DNS03
Yellow		flashing acyclic	hardware or system error
		off	hardware defective
RUN		on	bus communication is active
Run		flashing cyclic	no bus communication
Green		flashing acyclic	missing bus configuration
		off	no communication
NET		on	bus: communication is set up
network status		flashing cyclic	bus: no communication
R-Red G-Green		on	
		flashing acyclic	
		off	no operating voltage, not at the bus
Mod		on	assembly works
Module Status		flashing cyclic	assembly at work
R-Red G-Green		on	error not correctable
		flashing cyclic	subordinated error
		off	no operating voltage

Example:

RDY			RUN	hardware error
NET			MOD	
RDY			RUN	bootstraploader active, there is no firmware on the DNS03
NET			MOD	
RDY			RUN	no configuration, the DNS03 was not yet configured via PPC
NET			MOD	
RDY			RUN	communication runs without an error
NET			MOD	

5 The fieldbus interface Ethernet/IP

5.1 Introduction

In section 2 the fundamentals have been described which are applicable to all supported fieldbus and Ethernet interfaces of SYNAX.

SYNAX mostly is connected to Ethernet via the assembly BGR RECO CONTROL T2 LK ETH01. This will be abbreviated as ETH01 below. At the PPC-R2x an Ethernet interface exists on-board too.

The ETH01 assembly can be operated in two modes as of firmware version 02VRS. These are the "Ethernet/IP mode" and the "Ethernet mode". At the on-board interface of the PPC-R2x the same holds too.

This chapter describes the operation of the Ethernet interface in function mode Ethernet/IP.

5.2 Functional features

The Ethernet slave connection is provided with the following functional features in Ethernet/IP mode:

- Autonegotiation (Ethernet user in accordance to IEEE802):
The auto recognition of the data transfer rate and the duplex mode is **possible at the on-board interface of the PPC-R2x only**.
- The manual setting of the data transfer rate to 100-Mbaud or 10-MBaud is **possible at the on-board interface of the PPC-R2x only**.
The ETH01 slave is a 10-Mbaud Ethernet user in accordance to IEEE802.3 in any case.
- Half and full duplex operation possible
- Ethernet interface with completely DC-decoupled interface; connection via RJ45 connector
- Diagnosis LEDs incorporated in the front panel
- Ethernet/IP level 2 server
- Cyclic data exchange via Ethernet/IP I/O Messaging
- Freely configurable cyclic process image with up to 64 words (16 bits) input and output data each
- Acyclic data exchange via Ethernet/IP Explicit Messaging (UCMM)
- Acyclic data exchange with up to 7 HMI clients at the same time, by means of Bosch Rexroth SIS protocol via TCP
- Acyclic access to all available SYNAX parameters
- Monitoring of the process data channel and the SIS channel
- Data exchange with the PPC control assembly via the dual-port memory
- Hardware and software synchronization with the PPC control assembly

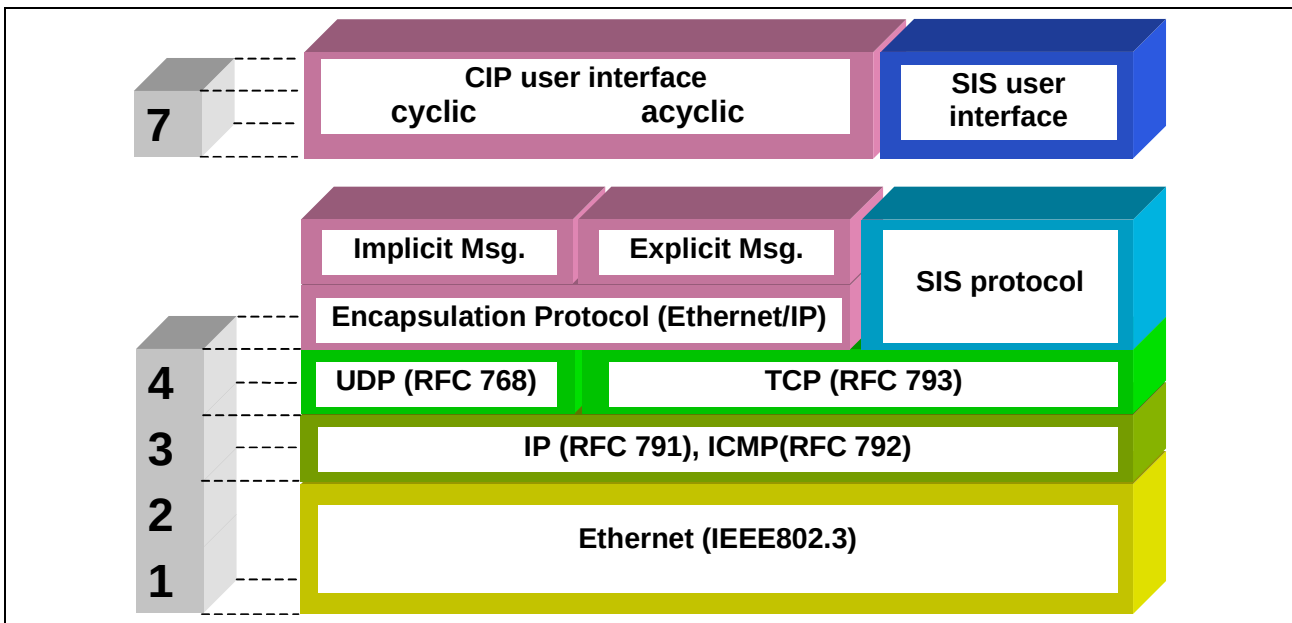


Fig. 5-1: Bosch Rexroth Ethernet/IP interfaces in the OSI layer model

A cyclic Ethernet/IP channel (Implicit Message) is supported at "Ethernet/IP mode". At the same an acyclic Ethernet/IP channel (Explicit Message) and up to 7 acyclic SIS channels can be operated.

5.3 Setting IP address and data format

IP address The originator (the master PLC) informs the target PPC-R at set-up of the cyclic connection, with which type of connection it should be communicated between the two Ethernet/IP users. This is often a point-to-point connection from the originator to the target and from the target to the originator of a multicast connection. The multicast address is allocated to the PPC with an algorithm that depends on the IP address of the PPC-R (see Ethernet/IP specification).

Data formats The Ethernet/IP specification determines the Intel format for the data transmission. This can cause data format problems, if the PLC is working with a Motorola processor and the master interface has no knowledge about the current data structure.

For the Ethernet/IP slave connection to be able to correct the data forms prior to relaying the data to the PPC, there are two possible data swapping setting:

- words in doubleword
- bytes in data exchange objects.

SynTop can transmit both of these settings to bit 8 (words) and bit 9 (bytes) of parameter "Fieldbus - control bits" (Y-0-0129).

Note: After executing the command "Load base parameters", both data format bits (swap bits) are deleted in parameter "Fieldbus - control bits" (Y-0-0129). This corresponds to the setting with a master interface (Intel format) that conforms to standard.

5.4 Ethernet/IP communication

Cyclic communication

The ETH01 and the on-board interface of the PPC-R2x support a cyclic Ethernet/IP channel. The cyclic communication with Ethernet/IP takes place in "cyclic" mode. Each user acts as a Producer and as a Consumer. It produces data cyclically and consumes data cyclically.

Note: At cyclic communication it is recommended from the Ethernet/IP specification to disconnect the Ethernet/IP network from other networks. A router can be used for this. Thus it should be avoided that the Ethernet/IP network is overcharged by Ethernet packets of other networks. This could stop the time monitored, cyclic communication. Switches are normally not suitable, because they don't filter broadcasts and multicasts. As it is worked with multicast telegrams at Ethernet/IP communication, also other networks can be overcharged.

Note: It is recommended to operate the users in full duplex mode to avoid collisions. If this is not done, it could cause failure in the cyclic communication. Because the ETH01 assembly does not support autonegotiation, it is necessary to configure the switchport, it is connected to, manual to full duplex operation. Otherwise it could cause communication failures.

Acyclic communication via direct object access

At acyclic communication of Ethernet/IP (Explicit Messaging) the services Set/GetAttributeSingle serve for writing or reading of object data. It can be accessed to SYNAX parameters at both slave connections (ETH01 and the on-board interface of the PPC-R2x) with them.

Class, instance and attribute

The Ethernet/IP specification denotes the data of a unit by class, instance and attribute. The data for the PLC are configured either in I/O channel (Implicit Message) or it accesses the fieldbus object or the SYNAX parameter with an Explicit Message using class/instance and attribute. The values for these are calculated as below:

S parameters (S-0-0001 - S-0-4095)

$$class = 101 + \left\lfloor \frac{S \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = S \text{ parameter} - ((class - 101) * 255)$$

$$attribute = drive \text{ address}$$

Example for S parameters

SYNAX parameter	Class	Instance	Attribute
Axx: S-0-0001	101	1	xx
Ayy: S-0-0002	101	2	yy
...	...	...	...
Azz: S-0-0255	101	255	zz
Axx: S-0-0256	102	1	xx
...	...	...	...
Ayy: S-0-4094	117	14	yy
Azz: S-0-4095	117	15	zz

Fig. 5-2: Example for S parameters

P parameters (P-0-0001 - P-0-4095)

$$class = 118 + \left\lfloor \frac{P \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = P \text{ parameter} - ((class - 118) * 255)$$

$$attribute = drive \text{ address}$$

Example for P parameters

SYNAX parameter	Class	Instance	Attribute
Axx: P-0-0001	118	1	xx
Ayy: P-0-0002	118	2	yy
...	...	...	...
Azz: P-0-0255	118	255	zz
Axx: P-0-0256	119	1	xx
...	...	...	...
Ayy: P-0-4094	134	14	yy
Azz: P-0-4095	134	15	zz

Fig. 5-3: Example for P parameters

A parameters (A-0-0001 - A-0-2047)

$$class = 135 + \left\lfloor \frac{A \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = A \text{ parameter} - ((class - 135) * 255)$$

$$attribute = \text{drive address}$$

Example for A parameters

SYNAX parameter	Class	Instance	Attribute
Axx: A-0-0001	135	1	xx
Ayy: A-0-0002	135	2	yy
...	...	...	...
Azz: A-0-0255	135	255	zz
Axx: A-0-0256	136	1	xx
...	...	...	...
Ayy: A-0-2046	143	6	yy
Azz: A-0-2047	143	7	zz

Fig. 5-4: Example for A parameters

C parameters (C-0-0001 - C-0-4095)

$$class = 144 + \left\lfloor \frac{C \text{ parameter} - 1}{255} \right\rfloor$$

$$instance = C \text{ parameter} - ((class - 144) * 255)$$

$$attribute = \text{master axis address}$$

Example for C parameters

SYNAX parameter	Class	Instance	Attribute
Cxx: C-0-0001	144	1	xx
Cyy: C-0-0002	144	2	yy
...	...	...	...
Czz: C-0-0255	144	255	zz
Cxx: C-0-0256	145	1	xx
...	...	...	...
Cyy: C-0-4094	160	14	yy
Czz: C-0-4095	160	15	zz

Fig. 5-5: Example for C parameters

Y parameters (Y-0-0001 - Y-0-1647)

$$class = 144 + \left\lfloor \frac{Y \text{ parameter} + 4095}{255} \right\rfloor$$

$$instance = Y \text{ parameter} - ((class - 144) * 255) + 4096$$

$$attribute = 1$$

Example for Y parameters

SYNAX parameter	Class	Instance	Attribute
Y00: Y-0-0001	160	17	1
Y00: Y-0-0002	160	18	1
...	...	...	...
Y00: Y-0-0239	160	255	1
Y00: Y-0-0240	161	1	1
...	...	...	...
Y00: Y-0-1646	166	132	1
Y00: Y-0-1647	166	133	1

Fig. 5-6: Example for Y parameters

The following applies for the objects to the data exchange via SIS telegram, diagnosis and control of the multiplex channel:

Index	Subindex	Class	Instance	Attribute
5E70	0	100	1	100
5E71	0	100	2	100
5E72	0	100	3	100
5E73	0	100	4	100
5FF0	0	168	9	1
5FF2	0	168	11	1
5FF5	0	168	14	1
5FF6	0	168	15	1
5FFC	0	168	21	1
5FFD	0	168	22	1
5FFE	0	168	23	1

The ETH01 and the on-board interface of the PPC-R2x use 8 bit values for class as well as for instance and attribute.

Acyclic data exchange via SIS-HMI channels

The acyclic SIS channel permits access to all SYNAX parameters. In case of the PPC, access to the SYNAX parameters is implemented via the parameter write and read services of the SIS protocol (see section 2.5). The SIS protocol is directly transmitted via the TCP data field. The maximum length of an SIS telegram can amount to 254 bytes (including header).

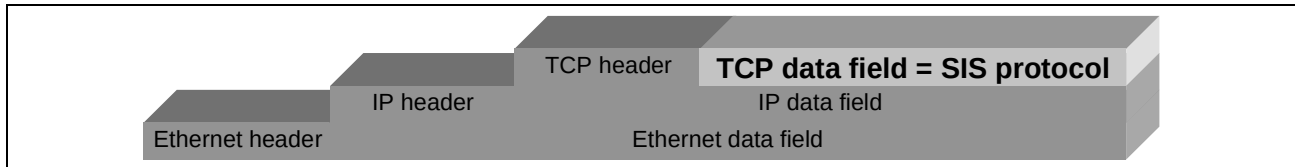


Fig. 5-7: Transmission of an SIS telegram

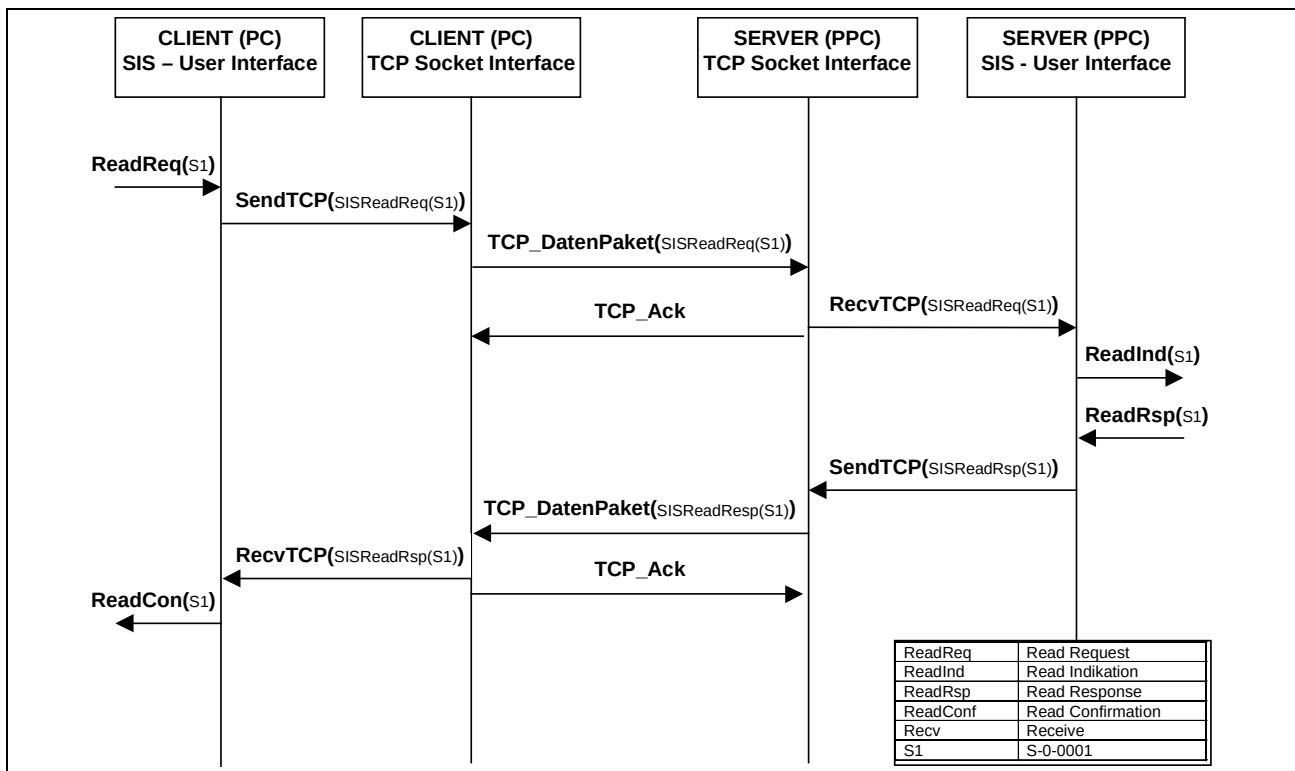


Fig. 5-8: SIS protocol – encapsulation in the TCP protocol (in the example, the drive parameter S-0-0001 is read)

7 HMI channels

The SIS server of the Ethernet/IP slave connection is able to process up to 7 SIS-HMI-TCP connections at the same time.

As compared with the HMI-SIS data packets, the Ethernet/IP data packets are treated with higher priority.

The HMI (human machine interface) port is provided as visualization, diagnosis and configuration interface which can, for instance, be used by SynTop to communicate with the PPC.

The HMI channels can be addressed via the port configured in the parameter "PPC - TCP port for HMI channels" (Y-0-0196, default: 5002).

Note: The TCP port for the HMI channels is automatically active as soon as the PPC, when powering up, detects an ETH01 assembly or an on-board assembly at the PPC-R2x. It does not have to be activated by parameterization.

5.5 Monitoring fieldbus transmission behavior with bus failure

As described in section 2.5 both Ethernet/IP slave connections monitor bus behavior with a watchdog timer so that it can specifically respond to a bus failure.

Watchdog Function

The PLC indirectly sets watchdog time via RPI and the Multiplier during boot up. The slave uses this to recognize a bus failure.

(see Ethernet/IP specification)

Error reaction with bus failures

The slave connection freezes the most recently received data in the event of a bus failure and deletes bit 3 in object 'Status fieldbus' (5FF2). This bit is displayed in operating mode on the PPC at output "Fieldbus - real time channel active" (_A:C01.08). The user can use this bit in the I/O logic (VKL) for a specific error reaction.

With bus return, output "Fieldbus - real time channel active" (_A:C01.08) is set.

5.6 Configuration, Ethernet/IP parameters

The network parameters are configured via the configuration tool SynTop. Refer to the SYNAX 200 parameter description (DOK-SYNAX*-SY*-12VRS**-PA01-EN-P) for a more detailed description of these parameters. The PPC configures the Ethernet assembly with the Ethernet parameters when powering up, before it reaches phase 0. Changed Ethernet parameters do not become active immediately. The Ethernet assembly is reconfigured with the new parameters only when the PPC switches over to the operating mode.

Diagnosis

Motion control parameter	Parameter name e	Example
Y-0-0296	Physical Ethernet address (MAC-ID)	00-02-A2-04-00-91
Y-0-0297	Fieldbus - diagnostic message	INIT: OK
Y-0-0298	Ethernet-HMI - diagnostic message	INIT: OK
Y-0-0299	Ethernet-HMI - firmware version	V02.001 25.06.02, V05T04__16102003

Fig. 5-9: Diagnosis parameters

Network

Motion control parameter	Parameter name	Example
Y-0-0191	PCP - IP address	192.168.10.152
Y-0-0192	PCP – subnet mask	255.255.255.000
Y-0-0193	PCP - standard gateway	192.168.10.1

Fig. 5-10: Network parameters

Cyclic channel via UDP

Motion control parameter	Parameter name	Example
Y-0-0127	Fieldbus/Ethernet 1 Configuration list of process input data	Y00:Y-0-0250, Y00:Y-0-0251, C01:C-0-0066, Y00:Y-0-0197
Y-0-0128	Fieldbus/Ethernet 1 Configuration list of process output data	Y00:Y-0-0200, Y00:Y-0-0201, Y00:Y-0-0202, Y00:Y-0-0199, Y00:Y-0-0197

Fig. 5-11: Configuration parameters for the cyclic channel (Cyclic-Modus)

Acyclic channel via TCP

Motion control parameter	Parameter name	Example
Y-0-0196	PPC - TCP port for HMI channels	5002

Fig. 5-12: Configuration parameters for the acyclic TCP channel

5.7 ETH01 board hardware

Front view of the ETH01

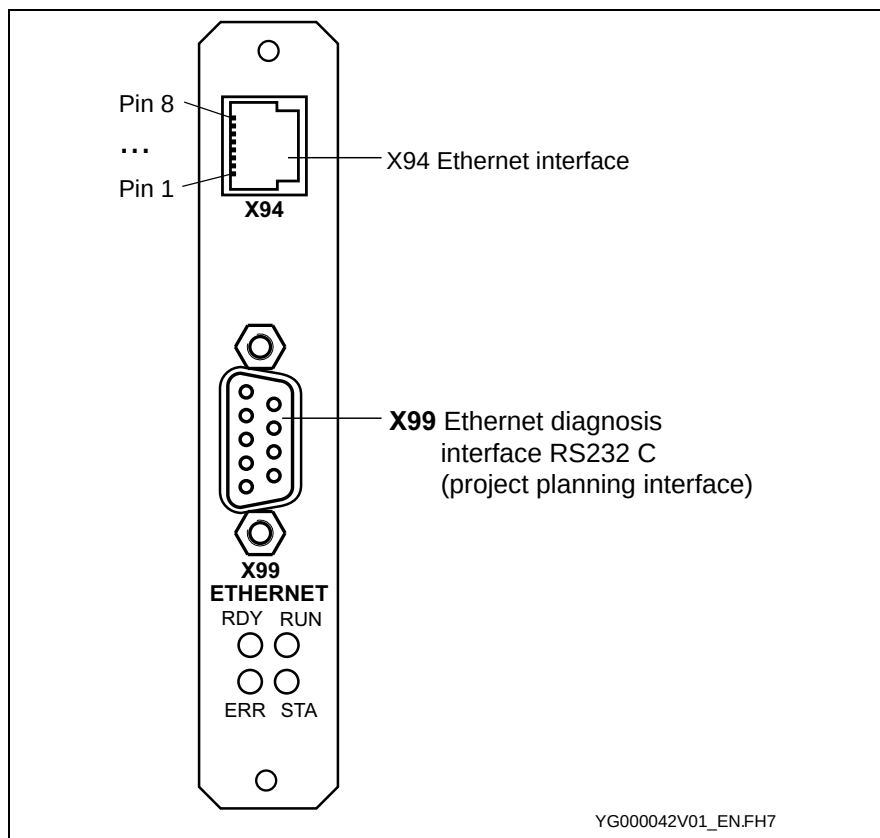


Fig. 5-13: Front view of the ETH01

ETH01 structure

The ETH01 board has been designed as a PC-104 board for direct mounting on the control board.

Interface to the PPC

The PC104-bus is the power supply for the ETH01 card and also the data exchange between ETH01 and the PPC card.

Ethernet interface RJ45

The signal circuits of the 8-pin Ethernet interface are completely DE-coupled.

The ETH01 assembly is provided with two interfaces to external equipment:

Signal configuration X94, Ethernet

PIN	Signal	Symbol
1	Transmitter data P	TX+
2	Transmitter data N	TX-
4	Receiver data P	RX+
6	Receiver data N	RX-

Fig. 5-14: Signal configuration X94, Ethernet

Interface X99

Diagnosis interface, not supported at the moment.

ETH01 diagnosis

LED diagnoses on the front panel

The ETH01 panel has a total of 4 diagnosis LED on the front. These enable the diagnosing of states between the Ethernet communication and the communication between the ETH01 assembly and the PPC.

Arrangement of the diagnoses - LED

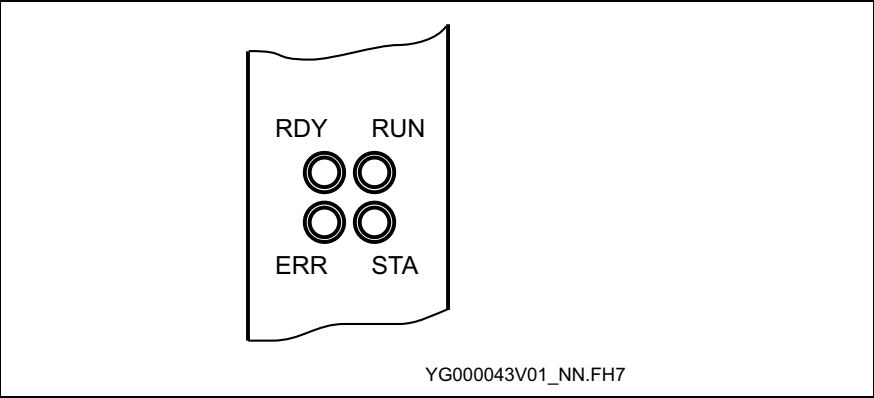


Fig. 5-15: Arrangement of the diagnoses - LED

LED diagnosis definitions

LED		Status	Definition
RDY Ready Yellow	●	on	ETH01 ready; hardware test and software test completed successfully.
	◐	flashing cyclic	Bootstraploader active, there is no firmware on the ETH01.
	◑	flashing acyclic	A hardware or system error has occurred.
	○	off	The hardware is defective.
RUN Green	●	on	Cyclic communication active
	◐	flashing cyclic	No cyclic communication active.
	◑	flashing acyclic	The Ethernet configuration is missing.
ERR Error Red	●	on	Hardware error
	○	off	
STA Status Yellow	●	on	Ethernet packets are received.
	○	off	Not Ethernet packets are received.

Fig. 5-16: Definition of the diagnoses LED

6 Ethernet interface

6.1 Introduction

In section 2 the fundamentals have been described which are applicable to all supported SYNAX fieldbusses and Ethernet interfaces.

SYNAX mostly is connected to Ethernet via the assembly BGR RECO CONTROL T2 LK ETH01. This will be abbreviated as ETH01 below. At the PPC-R2x an Ethernet interface exists on-board too.

The ETH01 assembly can be operated in two modes as of firmware version 02VRS. These are the "Ethernet/IP mode" and the "Ethernet mode". At the on-board interface of the PPC-R2x the same holds too.

This chapter describes the operation of the Ethernet interface in function mode Ethernet.

6.2 Functional features

The Ethernet slave connection is provided with the following functional features:

- Autonegotiation (Ethernet user in accordance to IEEE802):
The auto recognition of the data transfer rate and the duplex mode is **possible at the on-board interface of the PPC-R2x only**.
- The manual setting of the data transfer rate to 100-Mbaud or 10-MBaud is **possible at the on-board interface of the PPC-R2x only**.
The ETH01 slave is a 10-Mbaud Ethernet user in accordance with IEEE802.3 in any case.
- Ethernet interface with completely DC-decoupled interface; connection via RJ45 connector
- Diagnostic LEDs incorporated in the front panel
- TCP, UDP servers
- Cyclic data exchange with up to four clients via UDP
- Freely configurable cyclic process image with up to 256 words (16 bits) of input and output data each
- Acyclic data exchange with up to 11 clients (4 PLC clients + 7 HMI clients) at the same time, by means of Bosch Rexroth SIS protocol via TCP
- Acyclic access to all available SYNAX parameters
- Monitoring of the process data channel and the SIS channel
- Data exchange with the PPC control assembly via the dual-port memory
- Hardware and software synchronization with the PPC control assembly

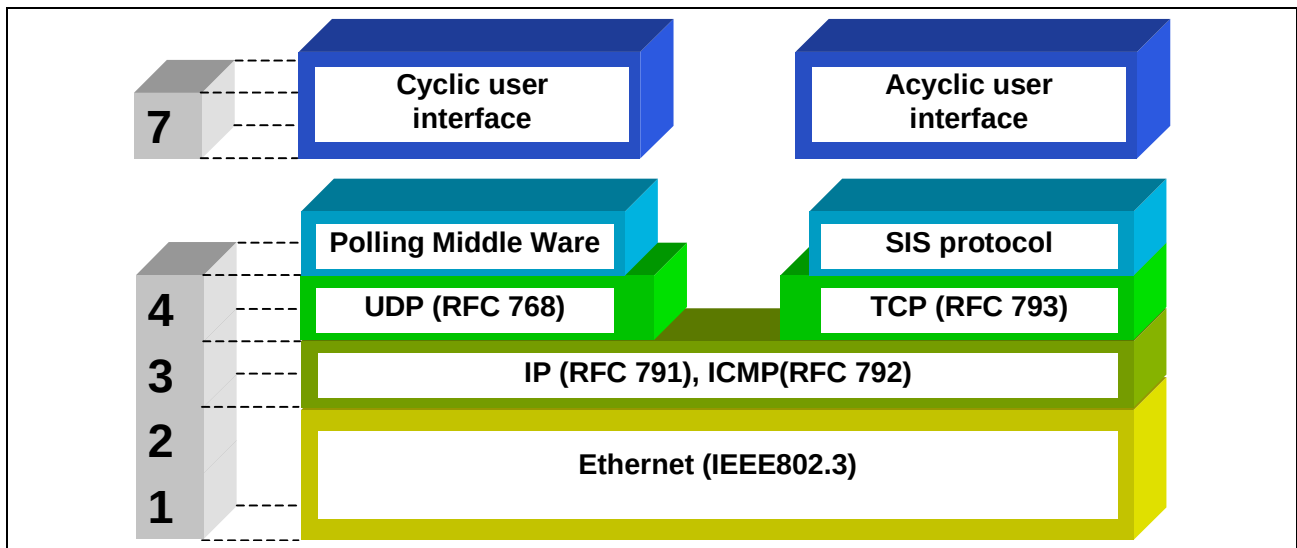


Fig. 6-1: Bosch Rexroth Ethernet interfaces in the OSI layer model

6.3 Cyclic data exchange, UDP

In case of the Ethernet slave connection, the cyclic data exchange is implemented via the UDP protocol. This type of data exchange is used for cyclic transmission of process data. The transmission of process data with the Ethernet slave connection can be compared with the master-slave polling communication if the Profibus-DP is used.

The master sends process output data to the slave and receives the process input data for acknowledge from the slave.

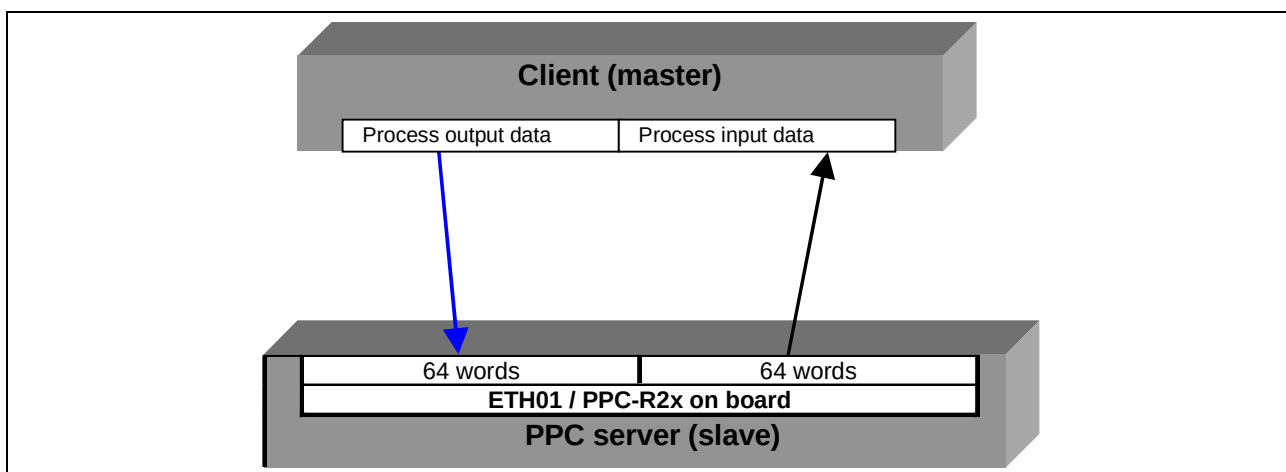


Fig. 6-2: UDP polling

The server of the slave connection is able to exchange process data with up to four clients (masters) at the same time. Up to 64 words of process input data and 64 words of output data can be exchanged with each client. That means, that the Ethernet slave connection is able to process a process image of $(4 \times 64) = 256$ words of process input data or $(4 \times 64) = 256$ words of output data. Cyclic data exchange can only be implemented in the local IP network.

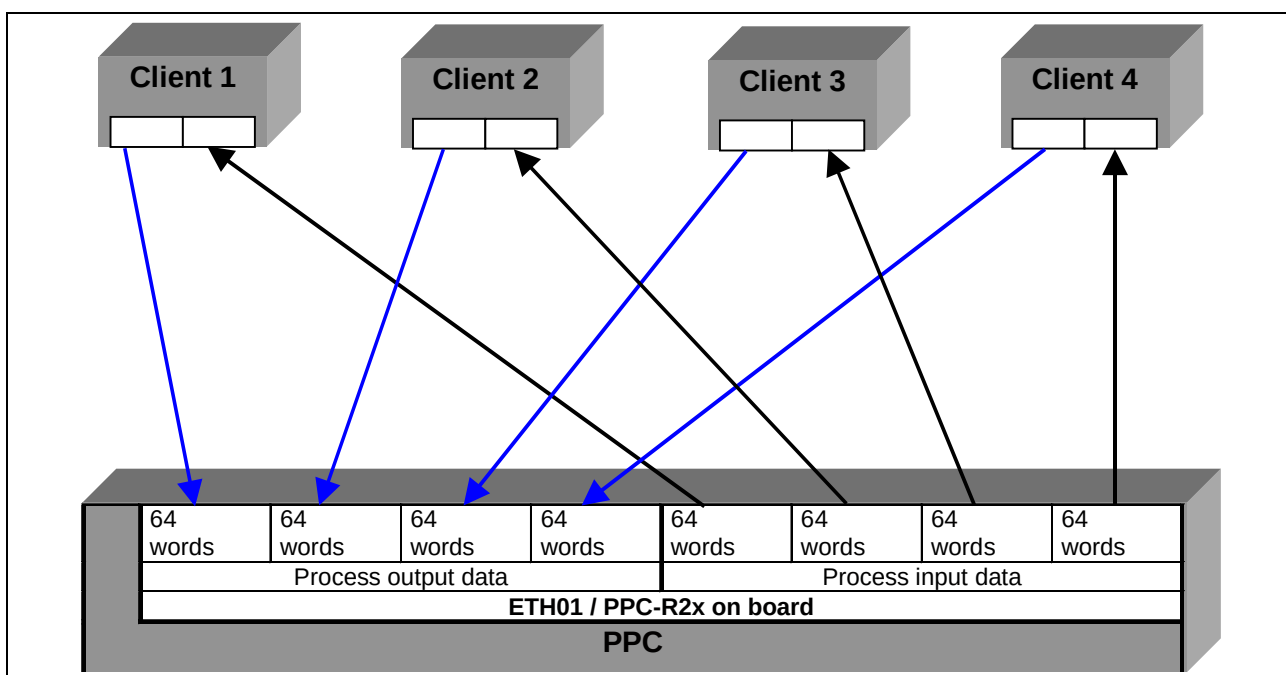


Fig. 6-3: UDP polling with 4 clients

With the Ethernet slave connection, the UDP polling communication with Client1 is activated analogously to the field busses in the "Host communication - control word" (Y-0-0033). The interfaces with the clients 2 to 4 are activated in the "Fieldbus - control bits" (Y-0-0129) (see Fig. 6-4).

Y-0-0129		Activated UDP polling client interfaces
Bit 20	Bit 21	
0	0	Client1
1	0	Client1 + Client2
0	1	Client1 + Client2 + Client3
1	1	Client1 + Client2 + Client3 + Client4

Fig. 6-4: Activation of the UDP polling client interfaces

Any client exchanges process data with the server PPC independently of the remaining clients. To simplify matters, the exchange of process data is described below with one client only.

Process data

The cyclic data exchange is used for the transmission of process data. The structure of the process data can be configured. The structural elements may be single SYNAX parameters.

Data type	Word (16 bit)	Double Word (32 bit)	Word (16 bit)
PPC, drive parameter	Y00:Y-0-0250	A01:S-0-0051	Y00:Y-0-0048

Fig. 6-5: Example of a process input data structure (8 bytes)

Data type	Word (16 bit)	Word (16 bit)
PPC, drive parameter	Y00:Y-0-0200	A01:A-0-0013

Fig. 6-6: Example of a process output data structure (4 bytes)

The process output data is transmitted from the client (PC, PLC) to the server (PPC); the process input data is transmitted from the server (PPC) to the client (PC, PLC). The parameter values of the structures are updated cyclically.

Data transmission

Transmission of the process data is achieved via the UDP protocol. The process data resides directly in the UDP data field. An additional header is not used in the UDP data field.

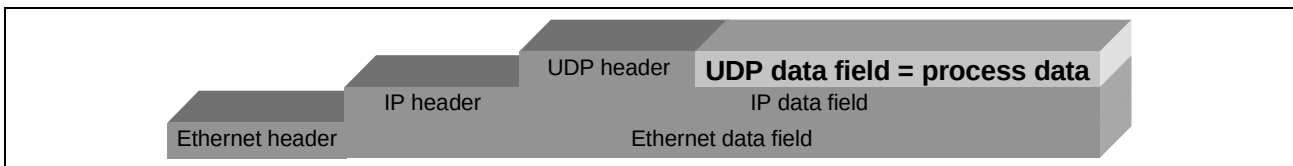


Fig. 6-7: Encapsulation of the process data in the UDP protocol

The cyclic data exchange is always activated by the client.

Data transmission cycle

A data transmission cycle comprises four steps:

1. The client (PC, PLC) transmits its process output data to the server (slave connection) in the UDP packet.
2. The server (slave connection) copies the data received into its input buffer.
3. The server reacts to the reception of the process output data with copying the process input data from the internal output buffer into the UDP transmission data field. This UDP packet is transmitted from the server to the client.
4. The client copies the UDP data received into its process input image.

Consistent data processing

Using the Dual-Port-RAM of the Ethernet assembly, the PPC copies the process input and output data acyclically to the Ethernet data transmission cycle. The process input data is consistently copied and processed across the entire data area both in the PPC and the Ethernet assembly.

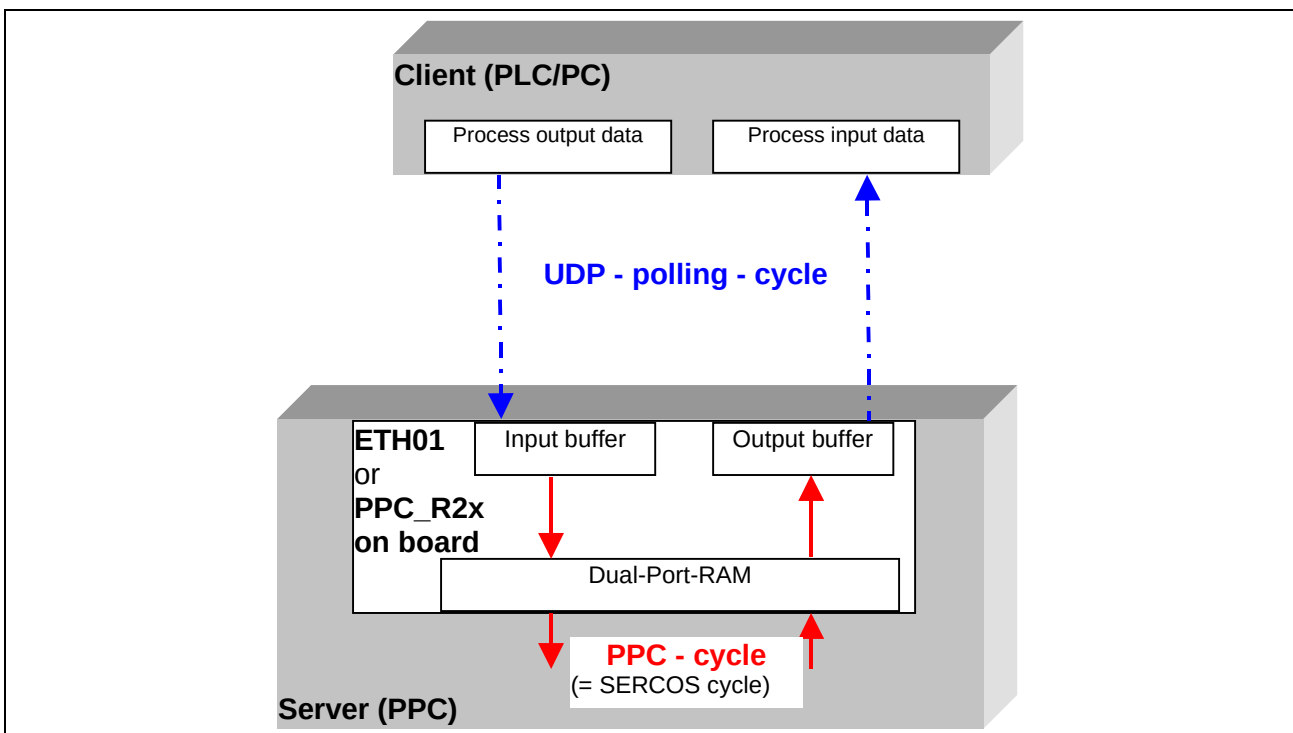


Fig. 6-8: PPC cycle and UDP polling cycle

Data format

Ethernet data are defined in the Big-Endian (Motorola) format. In the basic configuration of the PPC, the process data is transmitted in the Big-Endian format. Using parameter "Fieldbus - control bits" (Y-0-0129), the bytes can be exchanged in the word or the words in the doubleword.

Y-0-0129		Effect	Example
Bit 8	Bit 9		
0	0	Process data in the Big-Endian format	0x01020304, 0x0102 (default)
1	1	Process data in the Little-Endian (Intel) format	0x04030201, 0x0201

Fig. 6-9: Swapping between the Big-Endian and Little-Endian formats

State machine of the slave server

The **"non-existent"** state describes ETH01 in the de-energized state. After switch on, the Ethernet assembly enters the **"not configured"** state, i.e. the PPC has not yet assigned either the IP address or other communication parameters to the assembly. The PPC configures the Ethernet assembly using the Dual-Port-RAM. After completed configuration, the Ethernet interface enters the **"wait"** state. As a result, the assembly is ready for data exchange. If it receives an UDP datagram with the length and port number configured in the **"wait"** state, the assembly responds with an UDP data packet. Thereafter, the server (slave connection) enters the **"active"** state. In this state, the cyclic "inactive watchdog" is reset with each valid UDP data packet. After the "inactive watchdog" has elapsed, the assembly returns to the **"wait"** state. This is indicated at the ETH01 by the diagnostic LEDs (see "ETH01 diagnoses", page 6-12). The Ethernet assembly's response to a valid UDP data packet is an UDP data packet which contains the process input data.

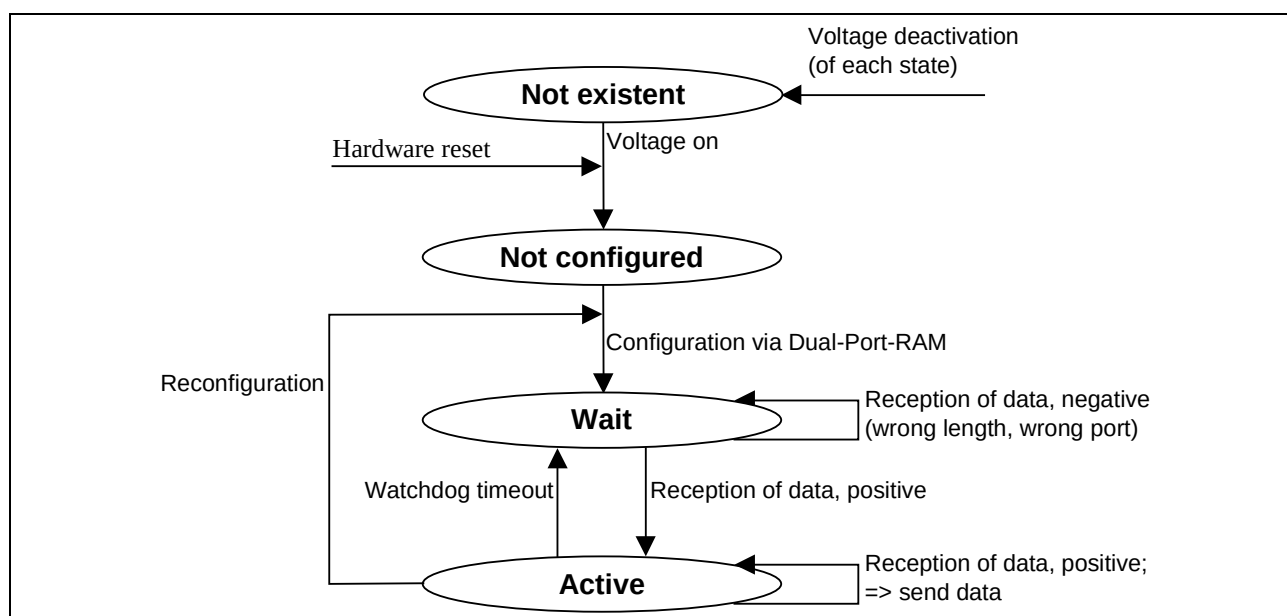


Fig. 6-10: State machine of the Ethernet assembly (UDP polling server)

Event	UDP polling server state			
	Non-existent	Not configured	Wait	Active
Voltage deactivation	-	Move to non-existent state	Move to non-existent state	Move to non-existent state
Hardware reset	Move to not configured state	-	Move to not configured state	Move to not configured state
Voltage on	Move to not configured state	-	-	-
Configuration	-	Move to wait state	-	-
Reconfiguration	-	-	-	Move to wait state
"ping with PC" (example: "ping 192.168.10.152")	No response	No response	Response (if no response: wrong IP address, subnet mask, gateway address)	Response (if no response: wrong IP address, subnet mask, gateway address)
Reception of data with incorrect data length or wrong port number	No UDP response datagram	No UDP response datagram	No UDP response data packet	No UDP response data packet
Reception of data with configured data length and port number	No UDP response data packet	No UDP response data packet	UDP polling response data packet; move to active state	UDP polling response data packet
"Inactive watchdog" responds	-	-	-	Move to wait state (see LED description)

Fig. 6-11: State event matrix of the slave assembly (UDP polling server)

UDP polling middleware

An UDP polling middleware should be implemented on the client side. The task of this software is to check the length of the received UDP data. If this length is not equal to the length configured, an error message should be emitted to the user interface. The configurable timer of the middleware should monitor the time the UDP response telegram takes to respond to an UDP request telegram. If this time exceeds a configurable value, the application should be informed accordingly.

Multiplexing

In case of the Ethernet interface, the multiplexing process described in section 2 is executed separately for each UDP channel. Up to 16 layers can be processed per UDP channel. The multiplexing process described in section 2 can be completely applied to the Ethernet assembly.

Monitoring of the Ethernet transmission behavior

Watchdog function As described in the section "Cyclic data exchange, UDP", the UDP channel is monitored by means of a watchdog, so as to permit a selective reaction to a failure in communication.

Error reaction in the event of an UDP channel failure

In the event of an UDP failure, the Ethernet slave connection freezes the previously received data and deletes the bit 3 in the object "field bus state" (0x5FF2). In the operating mode, this bit is mapped by the PPC to the "Fieldbus - real time channel active" (_A:C01.08). The user can use this bit in the linkage logic (VKL) for a selective error reaction.

When the bus traffic is restored, the output "Fieldbus - real time channel active" (_A:C01.08) is set.

6.4 Acyclic Data Exchange, TCP

The acyclic channel permits access to all SYNAX parameters. In case of the PPC, access to the SYNAX parameters is implemented via the parameter write and read services of the SIS protocol (see section 2.10). The SIS protocol is directly transmitted via the TCP data field. The maximum length of an SIS telegram can amount to 254 bytes (including header).

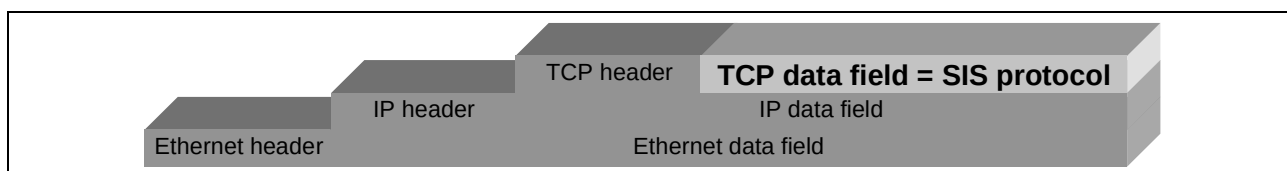


Fig. 6-12: Transmission of an SIS telegram

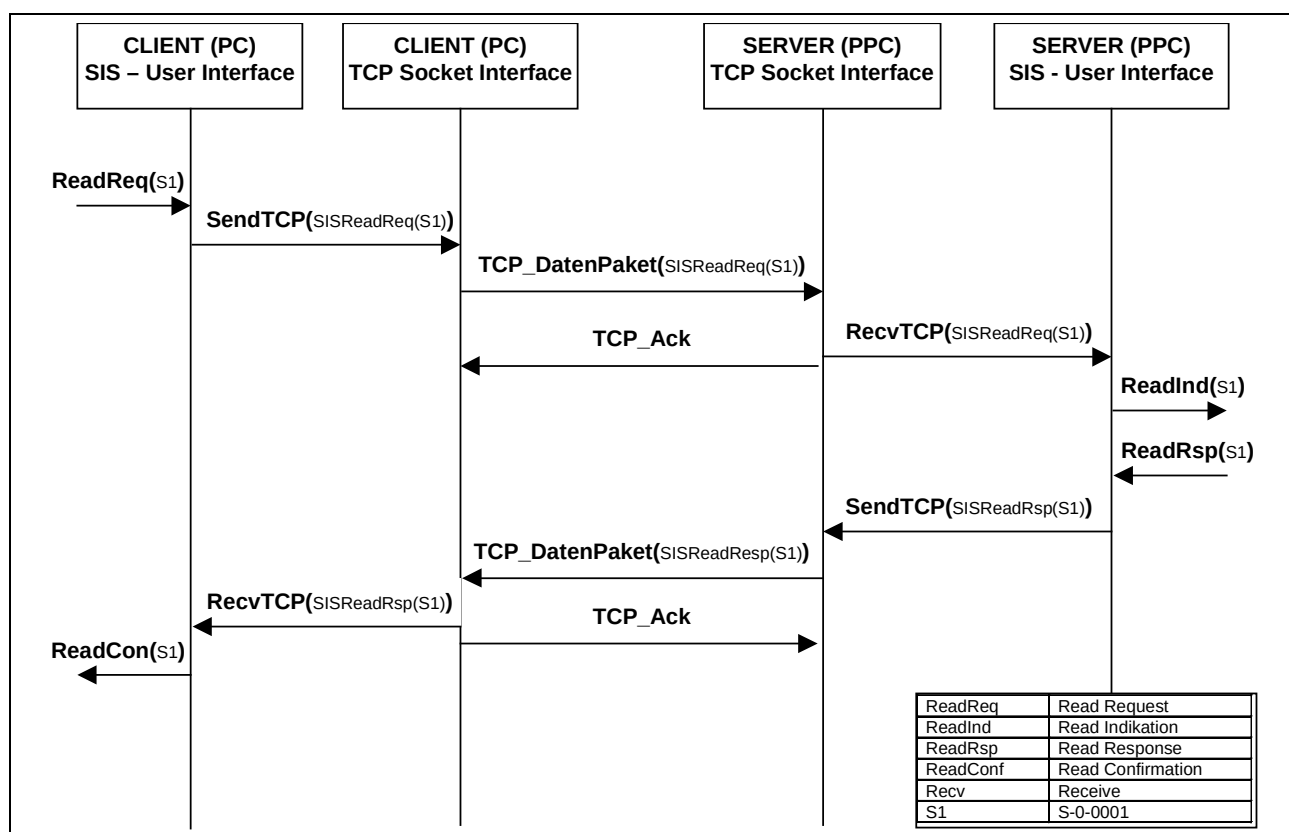


Fig. 6-13: SIS protocol – encapsulation in the TCP protocol (in the example, the drive parameter S-0-0001 is read)

Altogether, the SIS server of the slave connection is able to process up to 11 TCP connections at the same time. These 11 connections are subdivided into

- 4 PLC channels and
- 7 HMI- (human machine interface) channels.

These two channel types are differentiated via their port number. There is no difference in the transmission of data. As compared with the HMI data packets, the PLC data packets are treated with a higher priority.

4 PLC channels If the PLC channels are used, up to four "Ethernet masters" can communicate acyclically with the PPC, each via an UDP channel and via a TCP connection. For all TCP connections, use is made of the port number configured in the parameter "PPC - TCP port for acyclic PLC channels" (Y-0-0195, default: 5001).

Note: The 4 TCP PLC channels are active only if Ethernet is configured as interface in the "Host communication - control word" (Y-0-0033).

7 HMI channels The HMI (human machine interface) port is provided as visualization, diagnosis and configuration interface which can, for instance, be used by SynTop to communicate with the PPC.

The HMI channels can be addressed via the port configured in the parameter "PPC - TCP port for HMI channels" (Y-0-0196, default: 5002).

Note: The TCP port for the HMI channels is automatically active as soon as the PPC, when powering up, detects an ETH01 assembly or an on-board assembly at the PPC-R2x. It does not have to be activated by parameterization.

6.5 Configuration, Ethernet parameters

The network parameters are configured via the configuration tool SynTop. Refer to the SYNAX 200 parameter description (DOK-SYNAX*-SY*-12VRS**-PA01-EN-P) for a more detailed description of these parameters. The PPC configures the Ethernet assembly with the Ethernet parameters when powering up, before it reaches phase 0. Changed Ethernet parameters do not become active immediately. The Ethernet assembly is reconfigured with the new parameters only when the PPC switches over to the operating mode.

Network

Motion control parameter	Parameter name	Example
Y-0-0191	PCP - IP address	192.168.10.152
Y-0-0192	PCP - Subnet mask	255.255.255.000
Y-0-0193	PCP - Standard gateway	192.168.10.1

Fig. 6-14: Network parameters

Diagnosis

Motion control parameter	Parameter name	Example
Y-0-0296	Physical Ethernet address (MAC-ID)	00-02-A2-04-00-91
Y-0-0297	Fieldbus – diagnostic message	INIT: OK
Y-0-0298	Ethernet-HMI - diagnostic message	INIT: OK
Y-0-0299	Ethernet-HMI - firmware version	V02.001 25.06.02, V05T04__16102003

Fig. 6-15: Diagnosis parameters

Cyclic channel via UDP

Motion control parameter	Parameter name	Example
Y-0-0194	PPC - UDP ports	7001, 7002, 7003, 7004
Y-0-0127	Fieldbus/Ethernet 1 configuration list of process input data	Y00:Y-0-0250, Y00:Y-0-0251, C01:C-0-0066, Y00:Y-0-0197
Y-0-0188	Ethernet 2 - configuration list of process input data	"
Y-0-0189	Ethernet 3 - configuration list of process input data	"
Y-0-0190	Ethernet 4 - configuration list of process input data	"
Y-0-0128	Fieldbus/Ethernet 1 configuration list of process output data	Y00:Y-0-0200, Y00:Y-0-0201, Y00:Y-0-0202, Y00:Y-0-0199, Y00:Y-0-0197
Y-0-0185	Ethernet 2 - configuration list of process output data	"
Y-0-0186	Ethernet 3 - configuration list of process output data	"
Y-0-0187	Ethernet 4 - configuration list of process output data	"

Fig. 6-16: Configuration parameters for the cyclic UDP channel

Acyclic channel via TCP

Motion control parameter	Parameter name	Example
Y-0-0196	PPC - TCP port for HMI channels	5002
Y-0-0195	PPC - TCP port for acyclic PLC channels	5001

Fig. 6-17: Configuration parameters for the acyclic TCP channel

6.6 ETH01 board hardware

Front view of the ETH01

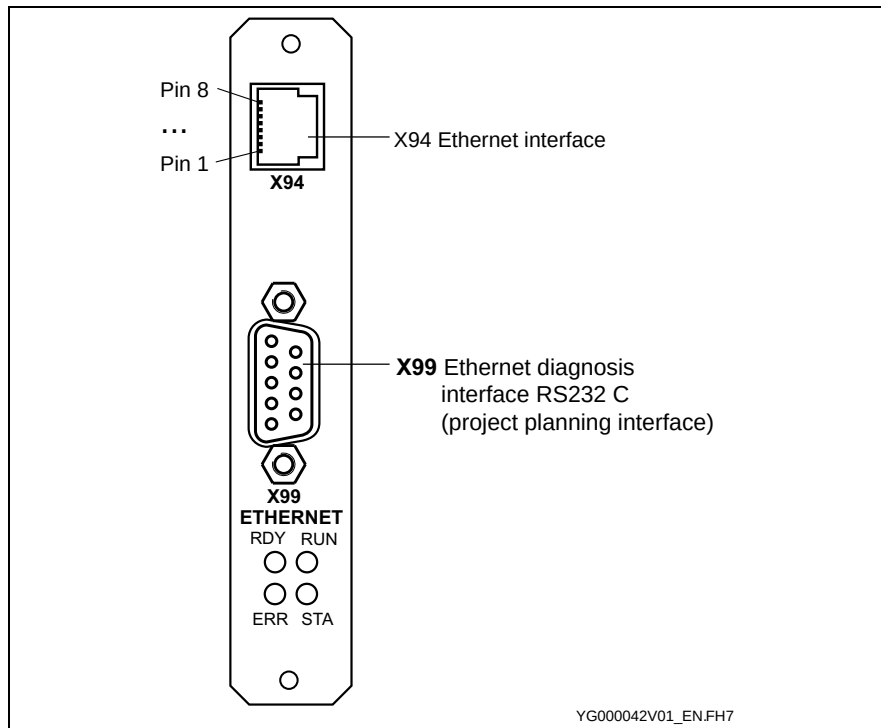


Fig. 6-18: Front view of the ETH01

ETH01 structure

The ETH01 board has been designed as a PC-104 board for direct mounting on the control board.

Interface to the PPC

The PC-104-bus is the power supply for the ETH01 card and also the data exchange between ETH01 card and PPC card.

The ETH01 assembly is provided with two interfaces to external equipment:

Ethernet interface RJ45

The Ethernet interface with 8-pin RJ45 connector socket is completely DC-decoupled.

PIN	Signal	Symbol
1	Transmitter data P	TX+
2	Transmitter data N	TX-
4	Receiver data P	RX+
6	Receiver data N	RX-

Fig. 6-19: Signal assignment X94, Ethernet

Interface X99

Diagnostic interface, not supported at the moment.

ETH01 diagnoses

The ETH01 assembly is provided with 4 diagnostic LEDs on the front panel. These LEDs permit diagnosis of the state of the Ethernet communication as well as the communication between the ETH01 and PPC assemblies.

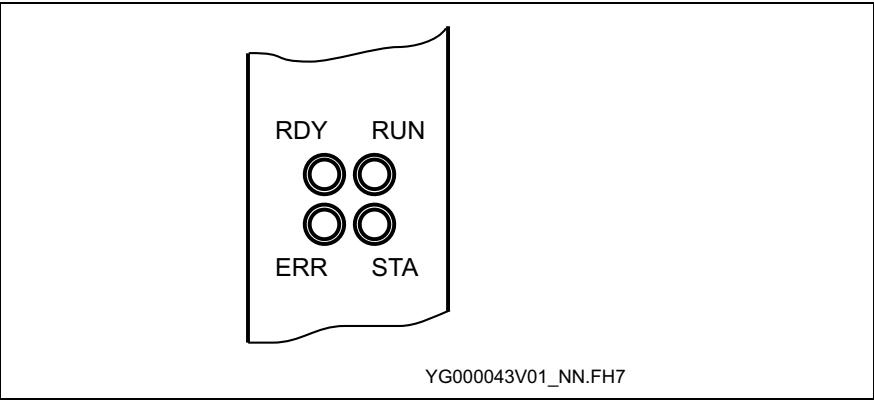


Fig. 6-20: Arrangement of the diagnostic - LED

LED diagnostic definitions

LED		Status	Definition
RDY Ready Yellow	●	On	ETH01 ready; hardware test and software test completed successfully.
	◐	Flashing cyclic	Bootstraploader active, there is no firmware on the ETH01.
	◑	Flashing acyclic	A hardware or system error has occurred.
	○	Off	The hardware is defective.
RUN Green	●	On	At least one UDP communication is running.
	◐	Flashing cyclic	None of the 4 UDP channels is still communicating (all 4 watchdogs have elapsed).
	◑	Flashing acyclic	The Ethernet configuration is missing.
ERR Error Red	●	On	At least one of the 4 UDP channels is not communicating any longer (watchdog elapsed).
	○	Off	No watchdog has elapsed.
STA Status Yellow	◑	On	Ethernet packets are received.
	○	Off	No Ethernet packets are received.

Fig. 6-21: Definition of the diagnostic LEDs

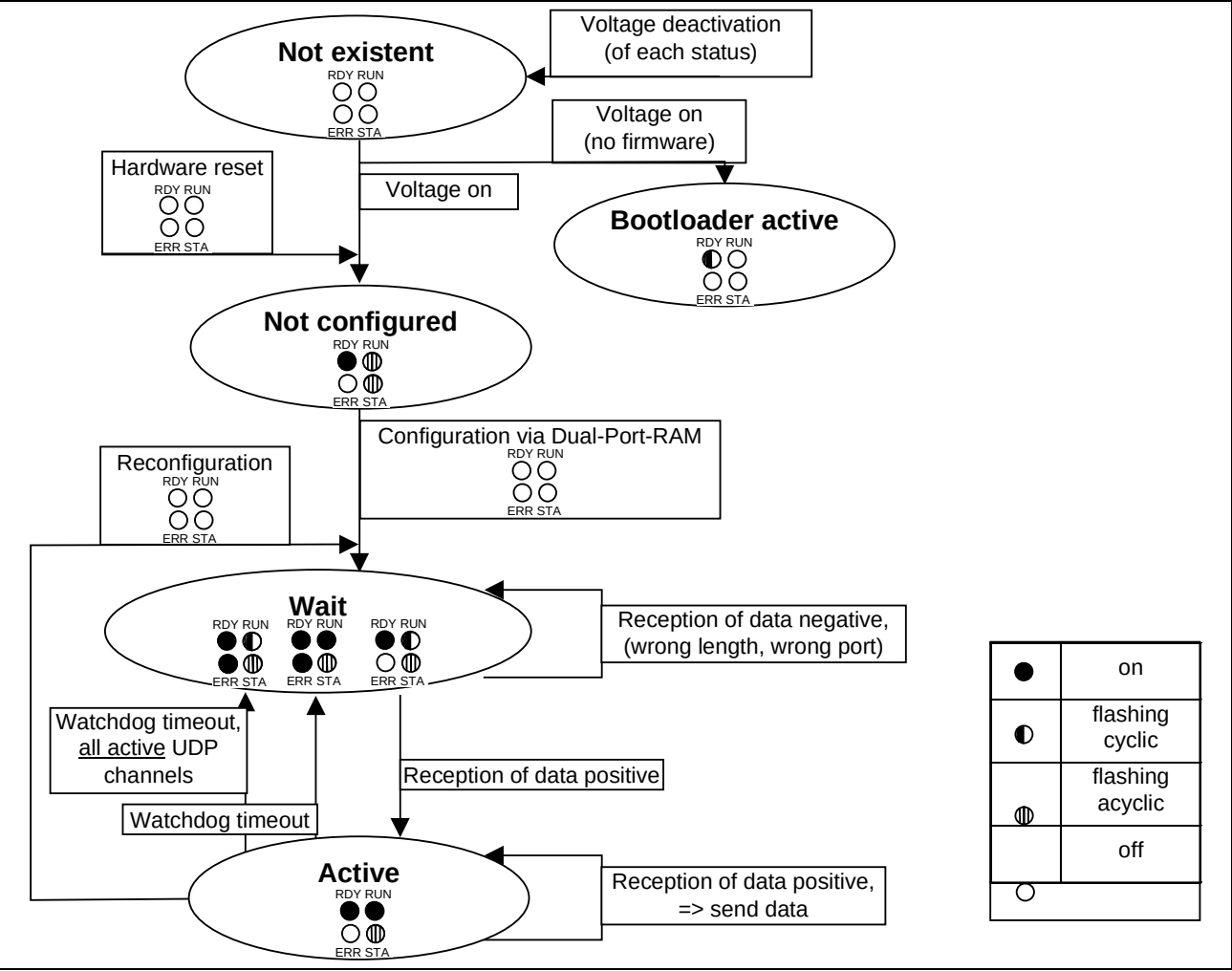


Fig. 6-22: State machine of the diagnostic LEDs

Event	UDP polling server status, LED status				
	Not existent All LEDs off	Not configured RDY on, RUN is flashing acyclic	Wait RDY on, RUN is flashing cyclic	Active RDY on, RUN on	Bootloader active RDY is flashing cyclic
Voltage deactivation	Move to status Not existent => ¹ All LEDs off	Move to status Not existent =>All LEDs off	Move to status Not existent =>All LEDs off	Move to status Not existent =>All LEDs off	Move to status Not existent =>All LEDs off
Hardware reset	Move to status Not configured => RDY on, RUN is flashing acyclic	Move to status Not configured => RDY on, RUN is flashing acyclic	Move to status Not configured => RDY on, RUN is flashing acyclic	Move to status Not configured => RDY on, RUN is flashing acyclic	Move to status Bootloader active =>All LEDs off =>RDY is flashing cyclic
Voltage on	Move to status Not configured => RDY on, RUN is flashing acyclic	-	-	-	-
Configuration	-	Move to status Wait =>All LEDs off =>RDY on, RUN is flashing cyclic	-	-	
Reconfiguration	-	-	Move to status Wait =>All LEDs off =>RDY on, RUN is flashing cyclic	Move to status Wait =>All LEDs off =>RDY on, RUN is flashing cyclic	
Reception of data with configured data length and port number	No UDP response datagram	No UDP response datagram	UDP polling response datagram. Move to status "active" RDY on RUN on ERR off	UDP polling response datagram	
"Inactive watchdog" responds	-	-	-	Move to status Wait RDY on, RUN on, RUN is flashing cyclic (if the watchdogs for all UDP channels have elapsed) ERR on	

Fig. 6-23: LED state matrix

¹ =>... means: LEDs move to status...;
=>... =>... means: =>LEDs move to status..., => then to status...

7 ARCNET interface

7.1 Introduction

This section describes the serial user interface ARCNET of the Bosch Rexroth PPC-R control.

With this interface

- parameter transmissions
- switching between operating and parameter mode
- and digital input/output transmissions are possible.

The serial interface can be used as a point-to-point connection via option card DAQ, as a bus.

The serial user interface supports the user data transmitted in a telegram as well as the Motorola and Intel formats. These data formats differentiate in terms of the serial sequence of the individual bytes of 2 byte and 4 byte data.

Example: The logical order of the data:

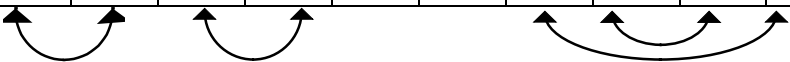
00	08	00	1F	00	02	1D	00	01	64
word		word		byte	byte	long (double word)			

The order of the bytes sent in **Motorola format**

00	08	00	1F	00	02	1D	00	01	64
----	----	----	----	----	----	----	----	----	----

The order of the bytes sent in **Intel format**

08	00	1F	00	00	02	64	01	00	1D
----	----	----	----	----	----	----	----	----	----



Note: Parameter "Host communication - control word" (Y-0-0033) is available to configure the serial interface.

7.2 ARCNET coupling with data exchange protocol

The following significant points are outlined in the general ARCNET specifications:

- ARCNET is a token-passing oriented network based on a coaxial cable with a gross data transmission rate of 2.5 Mbits/second.
- An ARCNET network can be made up of up to 255 interface connections with equality of access whereby each interface connection can communicate with all others.
- ARCNET interface connections are realized with ARCNET controllers which process the elementary ARCNET protocol without the need for a host computer.
- ARCNET controllers independently check whether packages sent were actually transmitted without any errors. This means that it is not necessary to run an additional check at a higher protocol level.

- Two different package types can be transmitted in ARCNET:
 - the length of short packages ranges between 1 and 253 bytes,
 - long packages range between 25 and 508 bytes.
 Packages with a length between 254 and 256 bytes must be sent as long packages with the corresponding number of fill bytes.
- Packages can be sent to a specific or all ARCNET switchons (broadcast news). This broadcast message is not, however, acknowledged and is therefore not secured.
- Each ARCNET controller, and thus every physical ARCNET interface connection has - comparable with an Ethernet address - a definite ARCNET address.

SYNAX realizes an ARCNET coupling with the settings:

Data rate	Data rate is set at 2.5 Mbits/second.
Address setting	The address is set with parameter "ARCNET participant number" (Y-0-0180).
Extended timeout	Extended timeout is set at the default value. Reconfiguration time thus equals 840 ms.
Package length	Short and long packages are allowed. The PPC only sends short packages.
Messages	Broadcast messages are not supported.
Number of NACKs	128 NACKs must be set.

Above and beyond the basic network functions which ARCNET controllers themselves perform, additional demands must be met. These are of described in the following sections.

The ARCNET interface is realized as a master/slave system on the PPC. The ARCNET partner is always the master. It sends command telegrams. This means that the PPC is always the slave. It sends response telegrams.

There are two forms of command telegrams:

- TO telegrams - data is sent
- FETCH telegrams, data is prompted

The PPC then responds:

- with a response telegram with data after a FETCH telegram
- or no response telegram after a TO telegram or a FETCH telegram with a logical error.

Every logical error occurring during the processing of a telegram is stored in parameter "Serial interface error number" (Y-0-0057). The current status must always be read. Successful telegram transmissions clear this parameter!

Note: The ARCNET coupling with data exchange protocol is activated in parameter "Host communication - control word" (Y-0-0033).

Transmission order

Data transmission with TO telegrams looks like this:

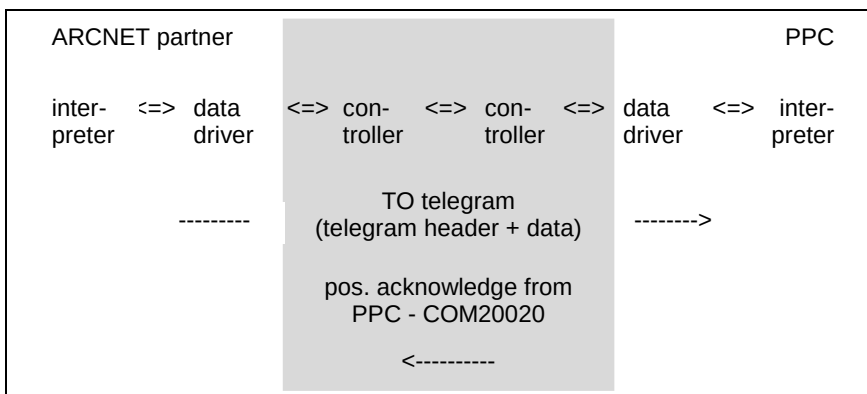


Fig. 7-1: Data transmission with TO telegrams

The PPC does not send response telegrams. The success of a TO telegram can be checked with a following FETCH telegram.

The transmitter of a telegram must be able to recognize the following:

- If the ARCNET controller signals successful transmission, then it is certain that the receiver has actually received the telegram (see the sequence described above).
- If the controller signals that the transmission contained an error because the receiver did either not respond or an error was detected in the checksum of the package (TMA¹), then the transmission is started once more. If another TMA error occurs, then it is useless to try again since either the receiver apparently does not exist or does not function, and an error thus ensues.
- If the controller signals that the transmission was completed with an error because there was no available space in the receiver buffer of the receiver (EXCNAK²), then the transmission must be repeatedly restarted over a specific period of time. This is necessary because the EXCNAK error is issued directly by the token rotation time and therefore is dependent upon the number of connected ARCNET controllers as well as network capacity. PPC internal processes with priority can also lock the receiver buffer of the controller. A constant send timeout is what is wanted, however.

The receiver cannot detect transmission errors because ARCNET controller sends no message with an error.

¹ **TMA** is a flag in an internal register of the ARCNET controller. This flag indicates that the ARCNET address which was entered as receiver does not exist.

² **EXCNAK** is a flag in an internal register of the ARCNET controller. This flag indicates that the ARCNET controller has tried 128 times to send the message to the ARCNET address which was entered as receiver. The receiver never had a free buffer, however.

Data transmission with FETCH telegrams looks like this:

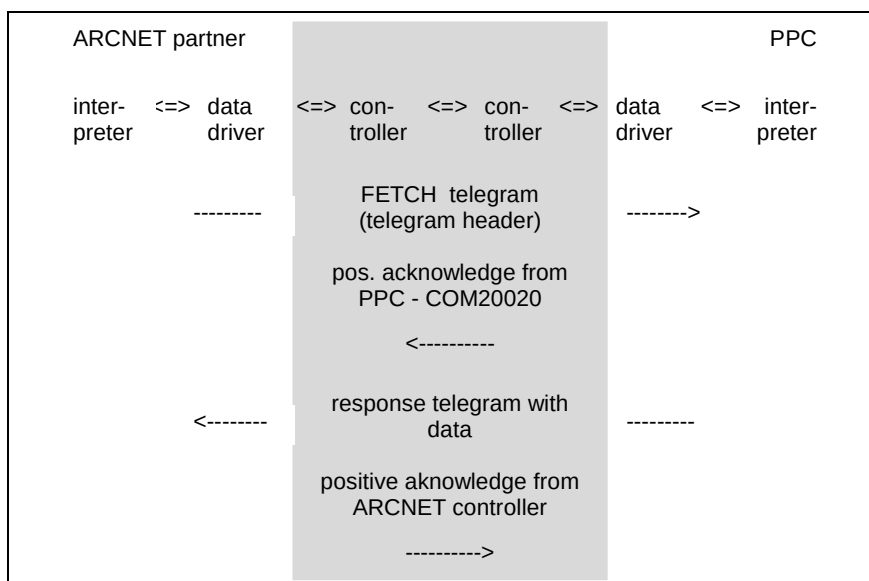


Fig. 7-2: Data transmission with FETCH telegrams

The PPC only sends a response telegram if the interpreter has not detected a logical error in the FETCH telegram.

In the event of a logical error in the command telegram, the PPC discards this telegram and makes a note of the error number in parameter Y-0-0057. These errors can be caused by the following:

Error no.	Cause
4	The following is valid for the prompt specified in the user data header: the prompt is unknown, or it is not yet supported.
5	The prompt cannot presently be completed as the required data queue is full.
6	The PPC has signaled an error for the Y/C/A parameters.
7	The drive has signaled an error for the S/P parameters.
8	Switch mode failed.
9	Data block specified not available.
10	The length of the data block does not agree with the configuration.
14	Error when writing I/O data (too many inputs).
21	The telegram header does not agree with the specification.
22	A following telegram was received but not expected, or a following telegram was expected but a normal telegram received. (Note: presently no support of following telegrams.)
23	The number of the user data does not agree with anticipated number.
24	The ARCNET partner is sending excessive NAKs.
25	The partner selected is not an ARCNET participant.

Fig. 7-3: Possible causes of errors

Note: Even error free telegrams are noted in Y-0-0057 with error number 0. This means that the status of the last telegram can always be read in Y-0-0057. For evaluating a logical error, a FETCH access to Y-0-0057 is always thus needed after every TO command or unanswered FETCH telegram.

Telegram structure and content of the data exchange protocol (7F protocol)

The data exchange protocol specified for the ARCNET interface integrates the Indramat protocol expansion (see "Bosch Rexroth protocol expansion", section 8.4). It is laid out in such a way that a message oriented data exchange can take place with user data in different data types. It has its own protocol ID to permit the other data exchange protocols via the same physical bus.

Command telegram A TO telegram is made up of a telegram header and data, a FETCH telegram only has a telegram header.

The telegram header is made up of 13 bytes and has two sections:

- The first seven bytes form the actual telegram header. This is comparable to the telegram header for the S5 interface.
- The final six bytes are the integral Indramat protocol expansion (see "Bosch Rexroth protocol expansion", section 8.4). They also support the transmission of data blocks. It is placed ahead of the user data of each telegram that is not the following telegram of a data transmission.

This part of the telegram header is dropped with following telegrams (see "Definition of the data blocks", page 7-14).

1	2	3	4	5	6	7
PK	logical NN LB HB		KB	DT	length LB HB	

8	9	10	11	12	13	... n
info access LB HB		drive-ad- dress	parameter ID Byte LB HB			data only with TO-Tel.

1	protocol ID	(always 0x7f)
2	log. message no. low byte	(0x20 -0xFF in following telegrams 0x10 otherwise)
3	log. message no. high byte	(0x1C in following telegrams 0x80 otherwise)
4	command byte	(0 = TO telegram) (1 = FETCH telegram)
5	data type byte	(always 0x05 = Word-Array)
6	length in [Word] low byte	(telegram dependent)
7	length in [Word] high byte	(telegram dependent)
8	info access low byte	(see section 8.4)
9	info access high byte	(see section 8.4)
10	drive address	(with Y parameter:0) (with C parameter ≤2) (with A, S and P parameter: ≤40) (for data blocks > 40)
11	parameter ID (Byte)	(see section 8.4)
12	parameter ID (LB)	(see section 8.4)
13	parameter ID (HB)	(see section 8.4)
14 ff	data only with TO telegrams	

Note: In conjunction with ARCNET, bit 1 of info access (in progress transmission - see section 8.4) plays an important part in Indramat protocol expansion as the data length of the telegram only relates to this telegram, not to the total data transmission.

If following telegrams still have to be sent in the case of long data blocks, then the first telegram of the data transmission must be designated a "transmission in progress". The transmission of the following telegram is thus controlled via the logical message number.

The telegram data are always interpreted as words per the protocol ID PK which is always located on an uneven address in the ARCNET RAM. The data length of the pure user data is limited to 123 words, following telegrams are not specified.

Reaction telegram

The prompted data are only sent back in response telegrams with FETCH telegrams. It is also made up of 13 bytes and contains the same information as does a command telegram.

1	2	3	4	5	6	7
PK	logical NN		KB	DT	length	
	LB	HB			LB	HB

8	9	10	11	12	13	... n
info access		drive- addre ss	parameter ID			data
LB	HB		Byte	LB	HB	

1	protocol ID	(always 0x7f)
2	log. message no. low byte	(0x20 .. 0xFF in following telegrams 0x10 otherwise)
3	log. message no. high byte	(0x1C in following telegrams 0x80 otherwise)
4	command byte	(always 0 = TO telegram)
5	data type byte	(always 0x05 = Word-Array)
6	length in [Word] low byte	(telegram dependent)
7	length in [Word] high byte	(telegram dependent)
8	info access low byte	(as in FETCH telegram)
9	info access high byte	(as in FETCH telegram)
10	drive address	(as in FETCH telegram)
11	parameter ID (Byte)	(as in FETCH telegram)
12	parameter ID (LB)	(as in FETCH telegram)
13	parameter ID (HB)	(as in FETCH telegram)
14 ff	data	

The telegram data are always interpreted as words per the protocol ID PK which is always located on an uneven address in the ARCNET RAM. The data length of the pure user data is limited to 123 words, following telegrams are not specified.

Data structure as a short package

Every telegram is sent from one ARCNET controller to the other in a short package in the ARCNET configuration supported by SYNAX. A short package always has 256 bytes and contains the required ARCNET information in addition to the telegram data.

- SID** (Source ID) ARCNET address of the participant sending
- DID** (Destination ID) ARCNET address of the participant receiving
- Count** Data offset in short package as of where telegram data start.

An ARCNET package is always organized in such a way that the ARCNET specific section is located at the beginning of the package, the telegram data end at precisely the 256th byte of the package. The intermediate range is invalid. The data offset "count" references that byte of a short package where the data once again are valid. The data structure of a telegram with length n looks like this:

D0	SID (ARCNET address of the sender)
D1	DID (ARCNET address of the receiver)
D2	count = 256-n (short package) = data offset in ARCNET-RAM
D3	
D4	
...	
D Count	protocol ID: always 7Fh (always uneven address)
D Count+1	logical message number LB: 10h
D Count+2	logical message number HB: 80h
D Count+3	command byte
D Count+4	data type byte: always 5h (Word-Array)
D Count+5	data length LB
D Count+6	data length HB: always 0h
D Count+7	info access LB
D Count+8	info access HB: always 0h
D Count+9	address byte (drives, data blocks)
D Count+10	master ID byte
D Count+11	parameter ID LB
D Count+12	parameter ID HB
D Count+13	1. data word LB
D Count+14	1. data word HB
...	
D 254	m th data word LB
D 255	m th data word HB (end short package)

Fig. 7-4: ARCNET table 1

The fixed values of the data exchange protocol are highlighted. The following applies to the values which can be changed:

- logical NN LB: (0x20-0xFF in following telegrams, 0x10 otherwise)
- logical NN HB: (0x1C in following telegrams, 0x80 otherwise)
- command byte: 0 => TO telegram, 1 => FETCH telegram
- data length LB: length in [word] as of date "info access LB"
- info access LB: 1Fh => data write
1Eh => data read
other possible values see section 8.4
- address byte 0 => with Y parameter,
> 40 => with data blocks (as DB number),
> 2 => drive address with A, S or P parameters
> 0 => master axis or drive address with C, A, S or P parameters
- master ID: 4 => Y parameters
2 => C parameters
1 => A parameters
0 => others
- parameter ID: Identity number of the parameters; with P parameters, add value 8000 h
- data words:

Example 1: The ARCNET partner wants to change the adjustable parameter (parameter A-0-0013) of drive 2.

D0	0A	(station number of the section computer)
D1	0B	(station number of the PPC)
D2	F1	(data offset: 241)
D3		
D4		
...		
D241	7F	(protocol ID)
D242	10	(logical message number LB)
D243	80	(logical message number HB)
D244	00	(command:"TO")
D245	05	(data type: word-array)
D246	04	(data length in word: LB)
D247	00	(data length in word HB)
D248	1F	(info access LB: operating data write)
D249	00	(info access HB: not used)
D250	02	(address byte: drive 2)
D251	01	(master ID byte: A parameters on the PPC)
D252	0D	(parameter ID LB: parameter number 13)
D253	00	(parameter ID HB: parameter number 13)
D254	01	(1 = speed offset A-0-0031)
D255	00	(1 = speed offset A-0-0031)

Fig. 7-5: ARCNET table 2

The data D0 to D3 are ARCNET specific. The telegram header of the data exchange protocol starts with D241. From D254, the operating data follows parameter A-0-0013 of drive 2.

The variable data bytes are highlighted. Up to drive address D250, this protocol data block remains the same for all write access to the adjustable parameter of the drives.

The PPC sends no response to "TO" commands. The success of the transmission can be checked by the ARCNET partner by reading back the parameters.

Example 2: The ARCNET partner wants to read the data status (see example "Reading the data status", section 8.4).

D0	0A	(station number of the section computer)
D1	0B	(station number of the PPC)
D2	F1	(data offset: 241)
D3		
D4		
...		
D241	7F	(protocol ID)
D242	10	(logical message number LB)
D243	80	(logical message number HB)
D244	00	(command:"TO")
D245	05	(data type: word-array)
D246	04	(data length in word: LB)
D247	00	(data length in word HB)
D248	07	(info access LB: identity number write)
D249	00	(info access HB: not used)
D250	01	(address byte: drive 1)
D251	00	(master ID byte: drive parameter)
D252	0C	(parameter ID LB: parameter number 12)
D253	80	(parameter ID HB: P parameter)
D254	0C	(identity number LB: 12)
D255	80	(identity number HB: P parameter)

Fig. 7-6: ARCNET table 3

The data D0 to D3 are ARCNET specific. The telegram header of the data exchange protocol starts with D241. From D254 follows the identity number of drive 1.

The variable data bytes are highlighted. Up to drive address D250, this protocol data block remains the same for all write access to the identity number of the drive parameters.

Note: Contrary to all other "TO" commands the PPC sends a reaction telegram in case of write access to the identity number (with the identity number). This telegram contains the data status.

Response: The PPC sends the requested data status to the ARCNET partner.

D0	0A	(station number of the section computer)
D1	0B	(station number of the PPC)
D2	F1	(data offset: 241)
D3		
D4		
...		
D241	7F	(protocol ID)
D242	10	(logical message number LB)
D243	80	(logical message number HB)
D244	00	(command:"TO")
D245	05	(data type: word-array)
D246	04	(data length in word: LB)
D247	00	(data length in word HB)
D248	07	(info access LB: identity number write)
D249	00	(info access HB: not used)
D250	01	(address byte: drive 1)
D251	00	(master ID byte: drive parameter)
D252	0C	(parameter ID LB: parameter number 12)
D253	80	(parameter ID HB: P parameter)
D254	03	(data status LB: 3)
D255	00	(data status HB: 0)

Fig. 7-7: ARCNET table 4

The data D0 to D3 are ARCNET specific. The telegram header of the data exchange protocol starts with D241. From D254 follows the data status.

For the evaluation of the data status see example "Reading the data status", section 8.4.

Example 3: The ARCNET partner wants to know the general diagnostic flag (parameter Y-0-0046) which is the only parameter configured in data block 101.

D0	0A	(station number of the section computer)
D1	0B	(station number of the PPC)
D2	F3	(data offset: 243)
D3		
D4		
...		
D243	7F	(protocol ID)
D244	10	(logical message number LB)
D245	80	(logical message number HB)
D246	01	(command:"FETCH")
D247	05	(data type: Word-Array)
D248	03	(data length in Word: LB)
D249	00	(data length in Word HB)
D250	1E	(info access LB: read operating data)
D251	00	(info access HB: not used)

D252	65	(address byte: data block 101)
D253	00	(master ID byte: not used)
D254	00	(parameter ID LB: not used)
D255	00	(parameter ID HB: not used)

Fig. 7-8: ARCNET table 5

The data D0 to D3 are ARCNET specific. The telegram header of the data exchange protocol starts at D239. It contains no user data. The contents of the data block 101 are prompted in the header.

Response: The PPC sends the requested diagnosis to the ARCNET partner.

D0	0B	(station number of the PPC)
D1	0A	(station number of the section computer)
D2	EF	(data offset: 239)
D3		
D4		
...		
D239	7F	(protocol ID)
D240	10	(logical message number LB)
D241	80	(logical message number HB)
D242	00	(command:"TO")
D243	05	(data type: Word-Array)
D244	05	(data length in Word: LB)
D245	00	(data length in Word HB)
D246	1E	(info access LB: as specified in prompt)
D247	00	(info access HB: not used)
D248	65	(address byte: as specified in prompt)
D249	00	(master ID byte: as specified in prompt)
D250	00	(parameter ID LB: as specified in prompt)
D251	00	(parameter ID HB: as specified in prompt)
D252	00	(operating data LW LB: 65536,)
D253	00	(operating data LW HB: means)
D254	01	(operating data HW LB: error on)
D255	00	(operating data HW HB: of the PPC)

Fig. 7-9: ARCNET table 6

Data D0 to D3 are ARCNET specific. The telegram header of the data exchange protocol starts at D239. The user data are made up of operating data. It is in the ARCNET RAM starting at D2523. Only the value of the operating data (**highlighted**) can change. Everything else remains the same.

How the SYNAX internal command processing affects ARCNET

The internal data driver of the PPC polls the interfaces and conducts the entered prompts to the PPC parameter handler or the SERCOS required data handler or the corresponding entities. These data handler complete their data management in a purely sequential fashion. This means that the data prompts are processed in order on the FIFO buffer. In this FIFO buffer, the data requests which arrive via the serial interface as well as the PPC internal prompts, queue.

The data handlers have a low-task priority, lower than the tasks of the control unit of the machine axis or the run up of the PPC. This means that the processing of data prompts by SYNAX is a non-deterministic process.

The following numbers should help clarify the processing times:

Transaction	Maximum processing time in [ms]
read I/O data	30
read drive parameters (max. 4 bytes)	30 + 10*SERCOS cycle time
read a S/P list of 246 bytes	30 + 132*SERCOS cycle time
read current phase	30
write I/O data	20
write drive parameters (max. 4 bytes)	30
write a S/P list of 246 bytes	30
change into parameter mode	(2000)
change to parameter mode	(500*(#drives))

Fig. 7-10: Clarification of processing times

Switching into operating mode requires the most time since the number of drives is directly involved in the time required. Because of the high priority of the run up task, it cannot be expected that a different task is performed during run up, e.g., checking current phase. Only with the I/O data can a deterministic behavior respective time be expected.

The way which telegrams are handled internally does not correlate with bus communications. The goal of communications via a bus, such as is the case with ARCNET, is to exchange data as quickly as possible with all participants (equal access).

Two aspects of this matter are relevant. On the one hand, the telegram should be processed as quickly as possible by the receiver. The limits are set by the above table in this case.

On the other hand, the telegram should be immediately accepted, the receiver buffer of the ARCNET controller must be released immediately.

In order to meet both requirements, an internal FIFO buffer is provided for up to 32 telegrams. Controlled by interrupts, incoming telegrams are copied into the buffer and the receiving buffer in the COM20020 is released again. The telegrams collected in the buffer are then processed sequentially.

Behavior in case of a telegram buffer overflow

An overflow of the ARCNET telegram buffer is displayed like before with the operating system error 5414. This has the disadvantage that the output "PPC ready" is cleared and the error cannot be reset by the user.

SYNAX 200 now optional supports up from version 11VRS the output of the diagnostic message "ARCNET telegram buffer: Overflow error" (F 146) that can be reset by the input "PPC error – clear external communication error" (_E:C01.03).

The mode of the diagnosis in case of a buffer overflow is selectable in the "Host communication – control word" (Y-0-0033) using bit 14.

Definition of the data blocks

The following conditions apply to the data blocks:

- It is only possible to access data blocks in their entirety. It is not possible to access individual elements of the data blocks.
- The length of the data blocks is limited to the following bytes:

	short package	long package
data type 04/05 (byte/word array)	246	500
data type 06 (long array)	244	498

Predefined data blocks There are predefined data blocks on the PPC destined for specific tasks:

DB no.98: Terminate transmission with following telegrams

This data block is used to inform the PPC that a data transmission begun by the ARCNET partners with following telegrams is being terminated. The PPC then finishes off internal "clean-up work" and discards the incomplete data block and is then ready to start a new data transmission.

The data block can only be used in a SEND-telegram and contains no data.

ARCNET partner

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7F	0x10	0x80	0x00	0x05	0x00	0x00	0x00	0x00	0x62	0x00	0x00	0x00
telegram header												

It makes sense to use this if the data line has physically broken down during a data transmission with following telegrams, if transmission is not yet running with stability or if the ARCNET unit has crashed.

DB no. 99 I/O data

Using this data block it is possible to set inputs via the serial interfaces and read outputs. The user data have a width of two bytes each. They are thus similar to the I/O data of a DEA. A maximum of 32 input or output words can be accessed.

Note: The use of data block 99 only makes sense when in operating mode (phase 4).

The mapping of the I/O data to the data block is explained in section "I/Os in serial protocols" (section 7 "Control signals (binary inputs/outputs)", SYNAX 200 - Functional Description, DOK-SYNAX*-SY*-12V*1/2-FK01-EN-P).

The following special features apply when writing inputs:

- If not all words are accessed, user data only must be transmitted to the word changed as the highest input word (see following example). The telegram length in the telegram header must be adjusted.
- If the data record is made up of more input words as specified in the mapping of the I/O data to the telegram, then the superfluous data is ignored.
- Word access is used for I/O handling via the serial interface. If only one bit is changed, then 15 others must be transmitted with their old values.
- The PPC always positively acknowledges a write access in parameter mode. Data is ignored, however.
- It is only possible to write to inputs, not to read them.

1. Example: write the PPC inputs

ARCNET partner:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x01	0x05	0x03	0x00	0x43	0x00	0x63	0x00	0x00	0x00
telegram header												

14	15	16	17
0xa5	0x5a	0x12	0x34
data			

In this example, we see two input words for I/O handling via serial interface. Both input values can be set as described above.

The following applies to writing inputs:

- If only the first word is accessed, then only one input value (bytes 11 and 12) need be accessed.
- If the data record is made up of more than two input values, then the superfluous data are ignored.
- I/O handling via serial/parallel interfaces uses word access. If only one bit is to be changed, then the other 15 must be transmitted with their old values.
- A write access in parameter mode is always positively acknowledged by the PPC. The data is ignored, however.
- It is only possible to write to inputs, not to read them.

Second example: Read PPC outputs

The PPC outputs look like this:

ARCNET partner:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x01	0x05	0x03	0x00	0x42	0x00	0x63	0x00	0x00	0x00
telegram header												

PPC:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x00	0x05	0x05	0x00	0x42	0x00	0x63	0x00	0x00	0x00
telegram header												

14	15	16	17
0xed	0xcb	0x5a	0xa5
data			

Two output words are configured in this example, via the serial interface, for I/O handling.

The following applies when reading outputs:

- When reading PPC outputs, all configured outputs are sent as data records. This means it is not possible to read an individual PPC output or a single output word.
- A read access when in parameter mode always results in an empty data record.
- Outputs can only be read. It is not possible to write to them.

Note: With the support of the fieldbus I/O parameters Y-0-0200..Y-0-0231 and Y-0-0250..Y-0-0281 by the 3964R interface now the user has the possibility to configure the transmission of binary input/output data via data blocks 101..116 in his application (look at point "Configurable data blocks"). The advantages for the user can be:

- I/O data are scalable and freely configurable (in the data blocks 101...116)
- the configuration of the I/O data can be changed dynamically
- the binary input data can be read back

This mode of the binary data transmission is selected in the parameter "Host communication – control word" (Y-0-0033) if bit 13 is set.

DB no. 100 switching modes

This data block supports the switching between operating, parameter and initialization modes. It has a length of one word. The parameter can be read and written to. The data contents specify the current mode on the PPC, or it sets the mode to be switched into. The following data contents are possible:.

0: initialization mode

2 : parameter mode

4 : operating mode

Note: The PPC will only permits a mode switch when in operating mode if the master axis is standing and no drive enable signal is being applied at any drive.
A mode switch is not executed if the phase specified is not permissible or a different switch progression is still active.

Example:

Switching to operating mode:

ARCNET partner:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x00	0x05	0x04	0x00	0x83	0x00	0x64	0x00	0x00	0x00
telegram header												

14	15
0x04	0x00
data	

Reading current operating mode:

ARCNET partner:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x01	0x05	0x03	0x00	0x82	0x00	0x64	0x00	0x00	0x00
telegram header												

PPC:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x00	0x05	0x04	0x00	0x82	0x00	0x64	0x00	0x00	0x00
telegram header												

14	15
0x04	0x00
data	

Configurable data blocks

On the PPC 16 data blocks can be freely configured for the ARCNET interface.

The configuration is effected via the 8 parameters "Configuration list data block 101" (Y-0-0058) to "Configuration list data block 108" (Y-0-0065). This corresponds to data block addresses 101 to 108. The 8 parameters "Configuration list data block 109" (Y-0-0078) to "Configuration list data block 116" (Y-0-0085) configure the data blocks with the addresses 109 to 116.

Y-0-0058 "Configuration list data block 101"	————>	data block no. 101
		...
Y-0-0065 "Configuration list data block 108"	————>	data block no. 108
Y-0-0078 "Configuration list data block 109"	————>	data block no. 109
		...
Y-0-0085 "Configuration list data block 116"	————>	data block no. 116

These parameters are lists of variable length. Data width equals four bytes. Every element in the list designates a parameter in the associated data block and has the following structure (SYNTAX format).

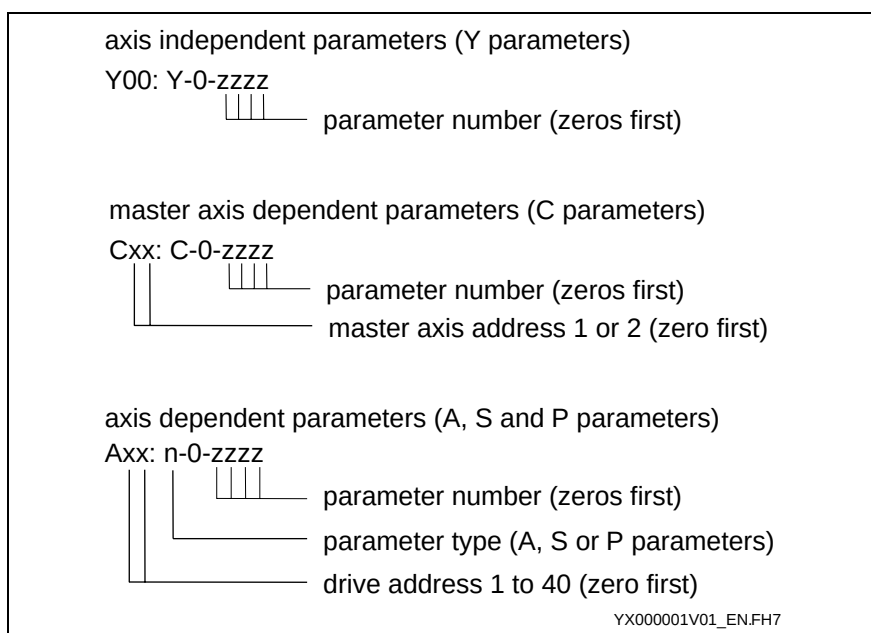


Fig. 7-11: Parameter structure in SYNAX identity number format

Note: Data blocks must be transmitted in one ARCNET telegram.
The maximum user data length of all parameters is 240 bytes.

Example:

Configuration list data block 1	Axis no.	Parameter-ID	data length	Parameter name	Weighting	Example of data contents
A01:A-0-0004	1	A-0-0004	4 bytes	Angle offset target value	0.0001°	10°
A02:A-0-0004	2	A-0-0004	4 bytes	Angle offset target value	0.0001°	0.1°
A03:A-0-0013	3	A-0-0013	2 bytes	Adjustable parameter selection	--	0 (fine adjustment, additive)
C01:C-0-0006	1	C-0-0006	4 bytes	VM - speed command 1	0.0001rpm	300 rpm

Fig. 7-12: Example of configurable data blocks

The resulting contents of data block 101:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
A-0-0004 axis 1				A-0-0004 axis 2				A-0-0013 axis 3		C-0-0006 master axis 1				

Example of a transmission: a) write data: (See example above, data block 101 = 0x65)

Master writes:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x00	0x05	0x0a	0x00	0x1f	0x00	0x65	0x00	0x00	0x00
telegram header												

14	15	16	17	18	19	20	21	22	23	24	25	26	27	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
user data														

b) read data: (See example above)

Master prompts data:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x01	0x05	0x03	0x00	0x1e	0x00	0x65	0x00	0x00	0x00
telegram header												

Slave sends data:

1	2	3	4	5	6	7	8	9	10	11	12	13
0x7f	0x10	0x80	0x00	0x05	0x0a	0x00	0x1e	0x00	0x65	0x00	0x00	0x00
telegram header												

14	15	16	17	18	19	20	21	22	23	24	25	26	27	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
user data														

Notes on compiling configuration lists

The following must be noted when compiling a configuration list for transmitting data both efficiently and without error.

- The parameters are divided into five groups, viz., the Y, C, A, S and P parameters. The Y, A and C parameters are on the PPC, while the S and P parameters are on the drives. This means that accessing the S and P parameters will take longer as the PPC must first access these parameters via the SERCOS required data channel.
- Parameter groups can be mixed in any way desired within one data block. The number of S and P parameters should, with an eye towards the response time, be limited to what is necessary. The S and P parameters of an axes should also, considering response time, not be arranged in sequence.
- There are three classes of parameters within each group:
 - Parameters that can only be read, actual values, for example (**S**: write protected)
 - Parameters that can always be read but only in parameter mode can these be written into, e.g., synchronization mode (**B**: write protected in operating mode)
 - Parameters that can always be read and written into, e.g., command values (**K**: parameter not write protected)

The data blocks must be configured in such a way that no communications error can occur.

Data block access mode	Legal parameter classes within a data block
only in read mode	S, B, K see above
read in all modes and write only in parameter mode	B, K see above
read and write in all modes	K see above

Fig. 7-13: Configuration of the data blocks

Handling S/P parameters when deactivating drives

In data blocks it is possible to configure Y, A and C parameters of the PPC control as well as S and P parameters of the drive in any combination. To administer the data contents of a data block the data length of all configured parameters must be known PPC internally. Data lengths of S and P parameters are stored in the drives. If a drive is deactivated then it does not participate in communications via the SERCOS interface. The PPC cannot read out the length information.

Data blocks that access parameters on a deactivated drive must be transmittable without having to change the configuration. To support this, a list of "preferred parameters" is on the PPC. The list contains preselected S and P parameters and their lengths. The administration of a data block is thus basically also possible if not all addressed drives are active or present.

Note: Given write accessing of parameters of a deactivated drive, the relevant data of the transmitted data block are discarded. Given read accessing, the PPC sends the value 0 for these parameters in the reaction telegram.

If a drive parameter is to be transmitted that is not in this preferred list, then it is possible to manually expand the list. Using parameters "Data blocks - configurable S and P parameters, ID number" (Y-0-0157) and "Data blocks - configurable S and P parameters, data length" (Y-0-0158) it is possible to expand up to 7 drive parameters.

The commissioning program SynTop offers a dialog for the settings with "PPC host communications". Use it to select additional S and P parameters.

List of the deposited PPC internal parameters see "Data blocks - configurable S and P parameters, ID number" (Y-0-0157) in SYNAX 200 Parameter Description, DOK-SYNAX*-SY*-12VRS**-PA01-EN-P.

Data transmission with following telegrams (lists of variable length)

As data transmission with data blocks are not supported by following telegrams and parameters with fixed lengths are always transmitted in a telegram, the transmission of following telegrams is restricted to lists, i.e., parameters with variable lengths.

All lists mutually share the trait that the actual length (= programmed length, one word) and the maximum length, one word, are placed ahead of the actual list elements.

The length information supports the administration of the lists and offers both the PPC and the ARCNET partner the capability of allocating a sufficiently sized storage for the collection of all list data.

So as not to have to differentiate between short and long lists with respect to following telegrams, only the two long words are transmitted in the first telegram. As such it should be clear that each list transmission, even the short ones, necessitates at least one following telegram.

Note: The ARCNET partner can select the data length of the following telegram, depending only on the application. In other words, the maximum number of user data must not be exhausted in the data package. The PPC, however, in its telegram always transmits the maximum number of user data.

Announcing following telegrams

The PPC interprets the parameter number received on an ever higher level and thus cannot know at a telegram level and without additional information whether it can expect write access with following telegrams.

Since the data length in the telegram header only refers to a single telegram and not the entire length, and the value 0x8010 has been agreed to for a logical message number in the first telegram of a data transmission, it is necessary to announce following telegrams in the Indramat protocol expansion.

An ARCNET partner must know with each list access at an interpreter level that it is dealing with a list so as to be able to handle the long words in the first data telegram. It is also necessary with list transmissions in info access, of the Indramat protocol expansion (see "Bosch Rexroth protocol expansion", section 8.4), to treat the bit for "transmission in progress/final transmission" in the following manner:

- write a list ==> info access = 0x001D, 'transmission in progress', as further data telegrams are still to be sent.
- read a list ==> info access = 0x001E, 'final transmission', as only data are requested.

Logical message number of following telegrams

The Indramat protocol expansion is dropped with following telegrams. Only user data are transmitted. As a result, the following telegrams are designed via the logical message number. The final following telegram always contains the value 0x1CFF, the first has the value 0x1C20, the others result from the incrementation of the previous number:

- logical message number = 0x1C20 ==> 1st following telegram
- logical message number = 0x1C21 ==> 2nd following telegram
- logical message number = 0x1C22 ==> 3rd following telegram
- ...
- ...
- logical message number = 0x1C28 ==> 9. following telegrams (penultimate telegram)
- logical message number = 0x1CFF ==> final following telegram

A jump in the value of the logical message number can therefore only occur between the penultimate and the ultimate .

An incorrect sequence of the following telegram or an unacceptable logical message number value are recognized by the PPC as an error with write access.

Examples of a list transmission

The following two examples depict the sequence of a list transmission by illustrating the individual ARCNET packages that result for the respective data transmission for the PPC and the ARCNET partner. The data are prepared in Intel format (see "Introduction", page 7-1).

Example 1: The ARCNET partner wants to load parameter Y-0-0013 (a type list of 128 elements) into the PPC.

First, it must be marked that further telegrams are to follow. Then the actual and maximum lengths of the list are entered as user data in the first telegram. This results in five data words as permanent data length. The **bold values** are the important data for the first telegram for writing a list.

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	EF	(data offset: 239)
D 3		
D 4		
...		
...		
D 239	7F	(protocol ID)
D 240	10	(logical message number LB)
D 241	80	(logical message number HB)
D 242	00	(command: 'TO')
D 243	05	(data type: Word-Array)
D 244	05	(data length in Word: LB)
D 245	00	(data length in Word: HB)
D 246	1D	(info access LB: write access with following telegrams)
D 247	00	(info access HB: not used)
D 248	00	(address byte: no axis)
D 249	04	(master ID byte: Y parameter on the PPC)
D 250	0D	(parameter ID LB: Y-0-0013)
D 251	00	(parameter ID HB: Y-0-0013)
D 252	00	(actual length in bytes LB: 256 bytes = 128 Word —>)
D 253	01	(actual length in bytes HB: as agreed in list)
D 254	10	(maximum length in bytes LB: 10000 Word ->)
D 255	27	(maximum length in bytes HB: see SYNAX)

Fig. 7-14: ARCNET table 7

The first following telegram with 123 data words can now be sent. It has logical message number 0x1C20. The **bold data** indicates the relevant data for the following telegrams.

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	03	(data offset: 3)
D 3	7F	(protocol ID)
D 4	20	(logical message number: LB)
D 5	1C	(logical message number: HB)
D 6	00	(command: 'TO')
D 7	05	(data type: Word-Array)
D 8	7B	(data length in Word: LB)
D 9	00	(data length in Word: HB)
D 10	00	(1st list element LB)
D 11	00	(1st list element HB)
D 12	00	(2nd list element LB)
D 13	02	(2nd list element HB)

...		
...		
D 254	0D	(123 rd list element LB)
D 255	00	(123 rd list element HB)

Fig. 7-15: ARCNET table 8

The final following telegram with the last five data words looks like what is shown below (the **bold data** indicates the relevant data for the final following telegram):

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	EF	(data offset: 239)
...		
...		
D 239	7F	(protocol ID)
D 240	FF	(logical message number: LB)
D 241	1C	(logical message number: HB)
D 242	00	(command: 'TO')
D 243	05	(data type: Word-Array)
D 244	05	(data length in Word: LB)
D 245	00	(data length in Word: HB)
D 246	00	(124 th list element LB)
D 247	02	(124 th list element HB)
...		
...		
D 254	0D	(128 th list element LB)
D 255	00	(128 th list element HB)

Fig. 7-16: ARCNET table 9

By means of the logical message number 0x1CFF, the PPC recognizes that it is the final data telegram and writes parameter Y-0-0013 after accepting the data.

Example 2: The ARCNET partner wants to read parameter Y-0-0013 (a type word list with 128 elements, for example).

The command telegram is not different from a read access without following telegram, with which response telegrams must be requested until the final following telegram has been entered.

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	F3	(data offset: 243)
D 3		
D 4		
...		
...		

D 243	7F	(protocol ID)
D 244	10	(logical message number: LB)
D 245	80	(logical message number: HB)
D 246	01	(command: 'FETCH')
D 247	05	(data type: Word-Array)
D 248	03	(data length in Word: LB)
D 249	00	(data length in Word: HB)
D 250	1E	(info access LB: read parameters)
D 251	00	(info access HB: not used)
D 252	00	(address byte: no axis)
D 253	04	(master ID byte: Y parameter on the PPC)
D 254	0D	(parameter number LB: Y-0-0013)
D 255	00	(parameter number HB: Y-0-0013)

Fig. 7-17: ARCNET table 10

The PPC first sends the long words of the request list. The **bold values** mark the important data for the first response telegram when reading a list.

D 0	0B	(station number of the PPC)
D 1	0A	(station number of the receiver)
D 2	EF	(data offset: 239)
D 3		
D 4		
...		
...		
D 239	7F	(protocol ID)
D 240	10	(logical message number: LB)
D 241	80	(logical message number: HB)
D 242	00	(command: 'TO')
D 243	05	(data type: Word-Array)
D 244	05	(data length in Word: LB)
D 245	00	(data length in Word: HB)
D 246	1E	(info access LB: read parameters as requested)
D 247	00	(info access HB: not used)
D 248	00	(address byte: no axis)
D 249	04	(master ID byte: Y parameter on the PPC)
D 250	0D	(parameter number LB: Y-0-0013)
D 251	00	(parameter number HB: Y-0-0013)
D 252	00	(actual length in bytes LB: 256 bytes = 128 Word —>)
D 253	01	(actual length in bytes HB: as agreed to in the list)
D 254	10	(maximum length in bytes LB: 10000 Word ->)
D 255	27	(maximum length in bytes HB: see SYNAX)

Fig. 7-18: ARCNET table 11

The ARCNET partner can now request the first following telegram with 123 data.

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	F3	(data offset: 243)
D 3		
D 4		
...		
...		
D 243	7F	(protocol ID)
D 244	10	(logical message number: LB)
D 245	80	(logical message number: HB)
D 246	01	(command: 'FETCH')
D 247	05	(data type: Word-Array)
D 248	03	(data length in Word: LB)
D 249	00	(data length in Word: HB)
D 250	1E	(info access LB: read parameters)
D 251	00	(info access HB: not used)
D 252	00	(address byte: no axis)
D 253	04	(master ID byte: Y parameters on the PPC)
D 254	0D	(parameter number LB: Y-0-0013)
D 255	00	(parameter number HB: Y-0-0013)

Fig. 7-19: ARCNET table 12

The PPC sends the first following telegram with 123 data words with the logical message number 0x1C20. The **bold values** indicate relevant data for the following telegrams.

D 0	0B	(station number of the PPC)
D 1	0A	(station number of the receiver)
D 2	03	(data offset: 3)
D 3	7F	(protocol ID)
D 4	20	(logical message number: LB)
D 5	1C	(logical message number: HB)
D 6	00	(command: 'TO')
D 7	05	(data type: Word-Array)
D 8	7B	(data length in Word: LB)
D 9	00	(data length in Word: HB)
D 10	00	(1st list element LB)
D 11	00	(1st list element HB)
D 12	00	(2nd list element LB)
D 13	02	(2nd list element HB)
...		
...		
D 254	0D	(123rd list element LB)
D 255	00	(123rd list element HB)

Fig. 7-20: ARCNET table 13

Finally, the ARCNET partner can request the final following telegram with the last 5 data words.

D 0	0A	(station number of the sender)
D 1	0B	(station number of the PPC)
D 2	F3	(data offset: 243)
D 3		
D 4		
...		
...		
D 243	7F	(protocol ID)
D 244	10	(logical message number: LB)
D 245	80	(logical message number: HB)
D 246	01	(command: 'FETCH')
D 247	05	(data type: Word-Array)
D 248	03	(data length in Word: LB)
D 249	00	(data length in Word: HB)
D 250	1E	(info access LB: read parameters)
D 251	00	(info access HB: not used)
D 252	00	(address byte: no axis)
D 253	04	(master ID byte: Y parameters on the PPC)
D 254	0D	(parameter number LB: Y-0-0013)
D 255	00	(parameter number HB: Y-0-0013)

Fig. 7-21: ARCNET table 14

The PPC sends the following telegram with the last 5 data words (the **bold values** indicate relevant data for the final following telegram:

D 0	0B	(station number of the PPC)
D 1	0A	(station number of the receiver)
D 2	EF	(data offset: 239)
...		
...		
D 239	7F	(protocol ID)
D 240	FF	(logical message number: LB)
D 241	1C	(logical message number: HB)
D 242	00	(command: 'TO')
D 243	05	(data type: Word-Array)
D 244	05	(data length in Word: LB)
D 245	00	(data length in Word: HB)
D 246	00	(124 th element LB)
D 247	02	(124 th list element HB)
...		
...		
D 254	0D	(128 th list element LB)
D 255	00	(128 th list element HB)

Fig. 7-22: ARCNET table 15

By means of the logical message number 0x1CFF the ARCNET partner recognizes that it has received the entire list with this telegram.

Data structure as a long package

A long telegram is sent from one ARCNET controller to the other in a long package in the ARCNET configuration supported by SYNAX. A long package always has 512 bytes and contains the required ARCNET information in addition to the telegram data.

- SID** (Source ID) ARCNET address of the participant sending
DID (Destination ID) ARCNET address of the participant receiving
Count Data offset in long package as of where telegram data start.

An ARCNET package is always organized in such a way that the ARCNET specific section is located at the beginning of the package, the telegram data end at precisely the 512th byte of the package. The intermediate range is invalid. The data offset "count" references that byte of a long package where the data once again are valid. The data structure of a telegram with length n looks like this:

D0	SID (ARCNET address of the sender)
D1	DID (ARCNET address of the receiver)
D2	0 (ID for long packages)
D3	Count = 512-n (long package) = data offset in ARCNET-RAM, n>256.
D4	
...	
D Count	protocol ID: always 7Fh (always uneven address)
D Count+1	logical message number LB: 10h
D Count+2	logical message number HB: 80h
D Count+3	command byte
D Count+4	data type byte: always 5h (Word-Array)
D Count+5	data length LB
D Count+6	data length HB: always 0h
D Count+7	info access LB
D Count+8	info access HB: always 0h
D Count+9	address byte (drives, data blocks)
D Count+10	master ID byte
D Count+11	parameter ID LB
D Count+12	parameter ID HB
D Count+13	1. data word LB
D Count+14	1. data word HB
...	
D 510	m th data word LB
D 511	m th data word HB (end short package)

Fig. 7-23: ARCNET table 16

The fixed values of the data exchange protocol are highlighted. The following applies to the values which can be changed:

- logical NN LB: (0x20 .. 0xFF in following telegrams, 0x10 otherwise)
- logical NN HB: (0x1C in following telegrams, 0x80 otherwise)
- command byte: 0 => TO telegram, 1 => FETCH telegram
- data length LB: length in [word] as of address byte
- info access LB: 1Fh => data write,
1Eh => data read;
other possible values see section 8.4
- address byte
 - 0 => with Y parameter,
 - > 40 => with data blocks (as DB number),
 - > 2 => drive address with A, S or P parameters
 - > 0 => master axis or drive address with C, A, S or P parameters
- master ID:
 - 4 => Y parameters
 - 2 => C parameters
 - 1 => A parameters
 - 0 => others
- parameter ID: Identity number of the parameters, with P parameters, add value 8000h
- data words:
 - D 509 is fill byte at telegram length n=254
 - D 510 is fill byte at telegram length n=254, 255
 - D511 is fill byte at telegram length n=254, 255, 256

Example: The transmission of Y-0-0013 (see example 1 for the transmission of a list transmission with short data packages) can be shortened to two telegrams.

The ARCNET partner wants to load parameter Y-0-0013 (a list of the type 'Word' with, e.g., 128 elements) into the PPC.

It is necessary to first mark that other telegrams are following. Then, in the first telegram, only the actual lengths and the maximum lengths of the list of user data are entered. The results are five data words as a fixed data length. The **bold values** are the important data for the first telegram when writing a list.

This telegram must always be sent in a short data package.

D0	0A	(station number of the section computer)
D1	0B	(station number of the PPC)
D2	EF	(data offset: 241)
D3		
D4		
...		
D239	7F	(protocol ID)
D240	10	(logical message number LB)
D241	80	(logical message number HB)
D242	00	(command: "TO")
D243	05	(data type: word-array)
D244	05	(data length in word: LB)

D245	00	(data length in word HB)
D246	1D	(info access LB: write access with following telegrams)
D247	00	(info access HB: not used)
D248	00	(address byte: no axis)
D249	04	(master ID byte: Y parameter on the PPC)
D250	0D	(parameter ID LB: Y-0-0013)
D251	00	(parameter ID HB: Y-0-0013)
D252	00	(actual length in byte LB: 256 byte = 128 word —>)
D253	01	(actual length in byte HB: as agreed in the list)
D254	10	(maximum length in byte LB: 10000 word ->)
D255	27	(maximum length in byte HB: see SYNAX)

Fig. 7-24: ARCNET table 17

The following telegram with the 128 data words can now be sent in a long data package. It has the logical message number 0x1CFF. The **bold data** represent the relevant data for the following telegrams.

D 0	0A	(station number of the receiver)
D 1	0B	(station number of the PPC)
D 2	EF	(data offset: 249)
...		
...		
D 249	7F	(protocol ID)
D 250	FF	(logical message number: LB)
D 251	1C	(logical message number: HB)
D 252	00	(command: 'TO')
D 253	05	(data type: word-Array)
D 254	05	(data length in word: LB)
D 255	00	(data length in word: HB)
D 256	00	(1. list element LB)
D 257	02	(1. list element HB)
...		
...		
D 510	0D	(128. list element LB)
D 511	00	(128. list element HB)

Fig. 7-25: ARCNET table 18

Note: Telegrams with a length of less than 253 bytes must always be sent in a short data package. Telegrams with a length of 254-256 bytes can be sent in a long data package with 3-1 filler bytes at the end of the telegram.

Hardware

The ARCNET coupling with the PPC uses the DAQ option card. It has the following important components.

COM 20020

is the ARCNET controller which processes the basic ARCNET protocol (ANSI 878.1).

HYC 9088/R

is the HIT module (High Impedance Transceiver). It realizes the electrical bus connection. The DAQ can thus be placed anywhere in the ARCNET bus.

BNC coaxial bushing

This establishes the connection for the transmission medium. The connection to the bus neighbors is established via the T-BNC adapter by means of the coaxial RG-62.

A final resistance (93 Ω) must be mounted in the ARCNET bus or in the case of a wye-system topology.

Topology	Bus
transmission medium	coaxial cable RG-62(/U) 93 Ω
bus connection	BNC (room for T-adapter)
bus interface	HIT (High Impedance Transceiver)
participant number	1 ... 255 configurable

Fig. 7-26: Prerequisites for SYNAX hardware

8 SYNAX with 3964R Interface Protocol

8.1 Introduction

This section describes the Siemens 3964R interface (RS232 and RS422) of the Bosch Rexroth PPC-R motion control (connection of the interface to the motion control see section 8.6, "Hardware").

Interface This interface enables:

- the transmission of parameters,
- switching between operating and parameterization mode
- and the transmission of digital inputs and outputs.

The serial interface is operated as a point-to-point connection between the serial ports of the PPC and a PLC.

Formats The serial user interface supports the Motorola or Intel formats for the user data transmitted in a telegram. These data formats are differentiated in terms of the serial order of the individual bytes of 2 and 4 byte data.

User data sequence Example:

The logical order of the data:

00	08	00	1F	00	02	1D	00	01	64
word		word		byte	byte	long (double word)			

The order of bytes sent in **Motorola format**

00	08	00	1F	00	02	1D	00	01	64
----	----	----	----	----	----	----	----	----	----

The order of bytes sent in **Intel format**

08	00	1F	00	00	02	64	01	00	1D
----	----	----	----	----	----	----	----	----	----

Note: To configure the serial interface, parameter "Host communication - control word" (Y-0-0033) is available.

8.2 General transmission structure

In general, the following transmissions are possible:

- transmission via data block (DB),
- transmission via Bosch Rexroth protocol expansion.

Transmission between transmitter (master) and receiver (slave) is structured as follows:

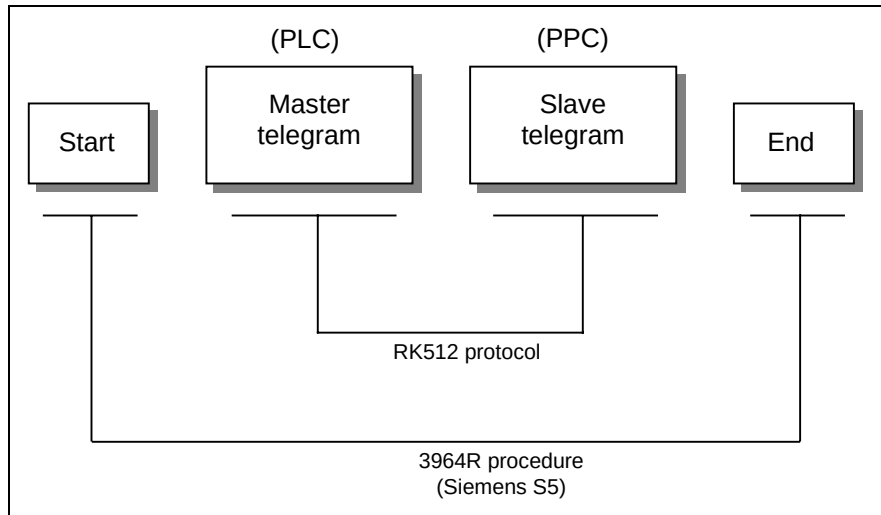


Fig. 8-1: Transmission between transmitter and receiver

Establishing and clearing connections of the 3964R procedure form the structure of the communication. The data are transmitted via RK512 protocol as telegrams. The communication from master to slave is operated according the following 3964R specification.

8.3 Siemens S5 coupling 3964R

General information The S5 interface is realized on the PPC in the form of a master/slave system. The S5 partner is always the master and sends command telegrams. The PPC is always the slave and only sends response telegrams.

Telegram types There are two forms of command telegrams:

- **SEND telegrams** - data is sent
- **FETCH telegrams** - data is prompted.

Then the PPC sends

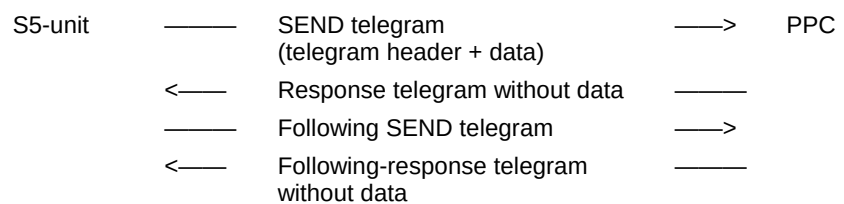
- **response telegrams without data** - after a SEND telegram
- **response telegrams with data** - after a FETCH telegram.

Note: If the number of user data exceeds 128 bytes, then following telegrams are sent with all type of telegrams.

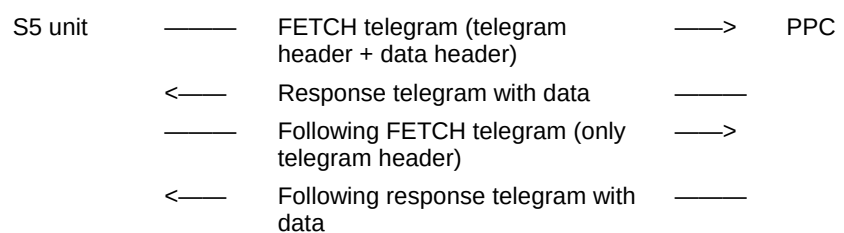
The Siemens S5 3964R coupling is activated in parameter "Host communication - control word" (Y-0-0033).

Order of transmission

Send telegram The order of data transmission with SEND telegrams is as follows:



Fetch telegram Data transmission with FETCH telegrams is as follows:



Note: It is necessary to wait for a response telegram after a command telegram. Only then a new telegram can be executed.

Telegram structure and content

A SEND telegram is made up of a telegram header and data. A FETCH telegram is only made up of a telegram and data header¹.

The telegram header is made up of ten bytes and, with a SEND telegram, contains information about the target of data. There is information about the source of the data with FETCH telegrams.

Master

1	2	3	4	5	6	7	8	9	10	11	
0x00	0x00	command		DB no. DW-no. target/source		high low number		CPU no.KM		data header ¹	
telegram header										with SEND only	

The telegram header in following telegrams is reduced to the first four bytes:

1	2	3	4	5	
0xff	0x00	command		user data	
telegram header				with SEND only	

- Byte definition**
- 1 Telegram ID (0x00, or 0xff with following telegrams),
 - 2 Telegram ID (always 0x00),
 - 3 Command ==> 'A' = SEND, 'E' = FETCH,
 - 4 Command type, i.e., type of data to be transmitted ==> 'D' = data blocks, all other types are not used
 - 5,6 target address with SEND telegrams
source address with FETCH telegrams
byte 5 = data block number (DB no.)
byte 6 = data word number (DW no.) **is not evaluated or interpreted as 0 by default**
 - 7,8 number of user data to be sent in the unit [word], **is not evaluated with read access by default**
 - 9 byte number of the coordination flag (with S5 only),
 - 10 byte number of the coordination flag and CPU no. (only with S5).

Bytes 9 and 10 of the telegram header are not evaluated in the PPC.

Slave The header of the response telegram is made up of four bytes and contains information about the course of the assignment.

1	2	3	4	5	
0x00 (0xff)	0x00	0x00	F no.	user data	
telegram header				only with FETCH	

¹ The data header is only relevant to Bosch Rexroth protocol expansion. It does not apply to data blocks.

- Definition of the bytes:**
- 1 Telegram ID (0x00, or 0xff with following telegrams),
 - 2 Telegram ID (always 0x00),
 - 3 command (always 0x00),
 - 4 command number => 0 = no error, otherwise error no. (See table)

Example: Data block no. 101 sent with a length of 128 bytes (0x40 words)

S5 unit:

1	2	3	4	5	6	7	8	9	10	11	...	138
0x00	0x00	'A'	'D'	0x65	0x00	0x00	0x40	0x00	0x00	...		
telegram header										data		

PPC:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

Error recognition

Byte 4 of the response telegram of the PPC supplies data for error recognition at serial transmission.

Note: The following error numbers are displayed in parameter "Serial interface error number" (Y-0-0057).

Error number table:

Error no.	Cause
1	The telegram header does not agree with the specification.
2	A following telegram was received but not expected or a following telegram was expected but a normal telegram received.
3	The number of user data does not agree: the expected number (with following telegrams) or the signaled block length (with normal telegrams).
4	Prompts specified for the user data header or prompt unknown or not yet supported.
5	Prompt cannot presently be executed as the required data queue is full.
6	The PPC has signaled an error for the Y/C/A parameters.
7	The drive has signaled an error for the S/P parameters.
9	The data block specified is not available.
10	The length of the data block does not agree with the configuration.
14	Error during writing I/O data (too many inputs).

Fig. 8-2: Error number table

Data block definitions

The following conditions apply to the data blocks (deviates from specification "computer coupling with RK 512"):

- After switching on the control the individual data blocks can only be accessed in their entirety. It is not possible to access individual parameters or parts of the data blocks.
- The support of data block segments must be controlled with DB no. 97.
- The length of the data blocks is limited to 128 bytes.

Pre-defined data blocks There are pre-defined data blocks on the PPC intended for specific tasks:

DB no. 97 controlling the support of data block segments

With this data block the support of the data block segments can be activated or deactivated.

If the support of the data block segments is activated, parts of the data blocks can be accessed.

In this mode byte 6 (DW no. = data word number) as well as 7 and 8 (data length) in the telegram header of the RK512 protocol must also be written by the user with permissible values (according to the data structure); also with access on the complete data block.

Note: The data block can only be used in a SEND telegram and contains exactly one data word.

S5 unit:

1	2	3	4	5	6	7	8	9	10	11	12
0x00	0x00	'A'	'D'	0x61	0x00	0x00	0x00	0x00	0x00	0x00	0xnn
telegram header										user data	

PPC:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

nn = 00: no support of data block segments (deactivation, default)

nn > 0: support of data block segments (activation)

DB no 98 following telegram transmission broken off

Purpose This data block is used to tell the PPC that a data transmission, started with following telegrams, has been broken off by the S5 partner. It can be sensibly implemented if the data line physically breaks down during data transmission with following telegram, if the transmission is not yet stable during the test phase, or if the S5 unit is in an error condition.

Function The PPC then still completes an internal "clean up operation", does not pass the garbled data record on, and then sends a positive response telegram. The PPC is now ready to start a new transmission.

Note: The data block can only be used in a SEND telegram and contains no data.

S5 unit:

1	2	3	4	5	6	7	7	9	10
0x00	0x00	'A'	'D'	0x62	0x00	0x00	0x00	0x00	0x00
telegram header									

PPC:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

DB no. 99 I/O data

Purpose With this data block inputs can be set via interface and outputs can be read. User data has a data width of two bytes each and thus corresponds to the I/O data of a DEA. A maximum of 32 input or output words can be accessed.

Note: The use of data block 99 only makes sense in operating mode (phase 4).

Rules The following rules apply when writing inputs:

- If not all words are accessed, user data only must be transmitted to the word changed as the highest input word (see following example). The telegram length in the telegram header must be adjusted.
- If the data record is made up of more input words as specified in the mapping of the I/O data to the telegram, then the superfluous data is ignored.
- Word access is used for I/O handling via the serial interface. If only one bit is changed, then 15 others must be transmitted with their old values.
- The PPC always positively acknowledges a write access in parameterization mode. Data is ignored, however.
- It is only possible to write inputs, not to read them.

First example: writing the PPC inputs

This example uses two input words via the serial interface for I/O handling. The two input values can be set as described above.

The following applies when writing inputs:

- If only the first word is accessed, then only one input value must be set (bytes 11 and 12).
- If the data record is made up of more than two input values, then the superfluous data is ignored.
- Word access is used for I/O handling via the serial interface. If only one bit is changed, then the 15 others must be transmitted with their old values.
- The PPC always positively acknowledges a write access in parameterization mode. Data is ignored, however.
- It is only possible to write inputs, not to read them.

S5 unit:

1	2	3	4	5	6	7	8	9	10	11	12	13	14
0x00	0x00	'A'	'D'	0x63	0x00	0x00	0x02	0x00	0x00	0xA5	0x5A	0x12	0x34
telegram header										data			

PPC:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

Second example: reading PPC outputs

In this example two output words are configured for I/O handling via the serial interface.

The following applies when reading outputs:

- When reading PPC outputs, all configured outputs of the data record are sent. This means it is not possible to read only one single PPC output or one single output word.
- Read access during parameterization mode always results in an empty data record.
- It is only possible to read outputs, but not to write them.

The outputs of the PPC look like this:

S5 unit:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	'E'	'D'	0x63	0x00	0x00	0x00	0x00	0x00
telegram header									

PPC:

1	2	3	4	1	2	3	4
0x00	0x00	0x00	0x00	0xED	0xCB	0x5A	0xA5
telegram header				data			

Note: With the support of the fieldbus I/O parameters Y-0-0200..Y-0-0231 and Y-0-0250..Y-0-0281 by the 3964R interface now the user has the possibility to configure the transmission of binary input/output data via data blocks 101..116 in his application (look at point "Configurable data blocks"). The advantages for the user can be:

- I/O data are scalable and freely configurable (in the data blocks 101...116)
- the configuration of the I/O data can be changed dynamically
- the binary input data can be read back

This mode of the binary data transmission is selected in the parameter "Host communication – control word" (Y-0-0033) if bit 13 is set.

DB no. 100 switching modes

Purpose This data block supports mode switches between operating, parameterization and initialization mode. It is one word long. It is possible to read and write into this parameter. Data content indicates the current PPC mode, or specifies the mode to be switched into. The following data contents is possible:

0: initialization mode

2: parameterization mode

4: operating mode

Function The PPC will only permit a mode change in operation mode if the master axis is standing and no drive enable signal is being applied to the drive. A mode switch will not be conducted if the phase specified is not permissible or a different change procedure is still active.

Note: SEND telegrams are acknowledged with a positive response telegram before mode changeover is activated because the change between the different modes can take many seconds depending on the number of axes (e.g., 30 seconds). The success of a phase switching must, therefore, be checked by polling the current phase, e.g., read access with DB100.

Example: Switching to operating mode:

S5 unit:

1	2	3	4	5	6	7	8	9	10	11	12
0x00	0x00	'A'	'D'	0x64	0x00	0x00	0x01	0x00	0x00	0x00	0x04
telegram header										user data	

PPC:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

Read current operating mode:

S5 unit:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	'E'	'D'	0x64	0x00	0x00	0x00	0x00	0x00
telegram header									

PPC:

1	2	3	4	5	6
0x00	0x00	0x00	0x00	0x00	0x04
telegram header				data	

DB101-116 configurable data blocks

Purpose On the PPC 16 data blocks can be freely configured for the 3964R interface.

Function The configuration is effected via the 8 parameters "Configuration list data block 101" (Y-0-0058) to "Configuration list data block 108" (Y-0-0065). This corresponds to data block addresses 101 to 108. The 8 parameters "Configuration list data block 109" (Y-0-0078) to "Configuration list data block 116" (Y-0-0085) configure the data blocks with the addresses 109 to 116.

Y-0-0058 "Configuration list data block 101"	————>	data block no. 101
		...
Y-0-0065 "Configuration list data block 108"	————>	data block no. 108
Y-0-0078 "Configuration list data block 109"	————>	data block no. 109
		...
Y-0-0085 "Configuration list data block 116"	————>	data block no. 116

These parameters are lists of variable length. Data width equals four bytes. Every element in the list designates a parameter in the associated data block and has the following structure (SYNTAX format).

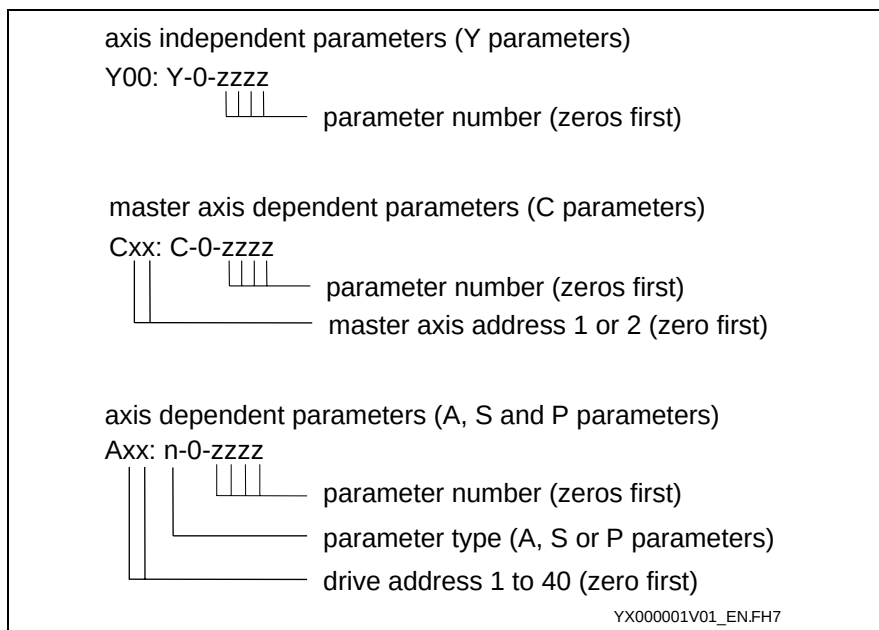


Fig. 8-3: Parameter structure in SYNAX identity number format

Note: Data blocks must be transmitted in one telegram. The maximum user data length of all parameters is 128 bytes.

Notes on compiling a configuration list

The following must be noted when compiling a configuration list for transmitting data both efficiently and without error.

- The parameters are divided into five groups: the Y, C, A, S and P parameters. The Y, A and C parameters are on the PPC while the S and P parameters are on the drives. This means that accessing the S and P parameters will take longer because the PPC must first access these parameters via the SERCOS required data channel.
- Parameter groups can be mixed in any way desired within one data block. The number of S and P parameters should, with an eye towards the response time, be limited to what is necessary. The S and P parameters of an axes should also, with an eye towards the response time, not be arranged in sequence.

Example:

Disadvantageous		Advantageous	
S-0-0091	axis 1	S-0-0091	axis 1
S-0-0092	axis 1	S-0-0091	axis 2
S-0-0091	axis 2	S-0-0091	axis 3
S-0-0092	axis 2	S-0-0092	axis 1
S-0-0091	axis 3	S-0-0092	axis 2
S-0-0092	axis 3	S-0-0092	axis 3

Fig. 8-4: Example: Allocating parameter groups

- There are three classes of parameters within each group:
 - Parameters that can only be read, actual values, for example (**S**: write protected)
 - Parameters that can always be read but only be written in parameterization mode, e.g., synchronization mode. (**B**: write protected in operating mode)
 - Parameters that can always be read and written, e.g., command values (**K**: Parameter not write protected)

The data blocks must be configured in such a way that no communications error can occur.

Data block access mode	Legal parameter classes within a data block	
Only read in all modes	S, B, K	see above
Read in all modes, write only in parameterization mode	B, K	see above
Read and write in all modes	K	see above

Fig. 8-5: Data block configurations

Example:

Configuration list data block 1	Axis no.	Parameter ID	Data length	Parameter name	Weighting	Example of data contents
A01:A-0-0004	1	A-0-0004	4 bytes	Position command offset	0.0001°	10°
A02:A-0-0004	2	A-0-0004	4 bytes	Position command offset	0.0001°	0.1°
A03:A-0-0013	3	A-0-0013	2 bytes	Adjustable parameter selection	--	0 (fine adjustment, additive)
C01:C-0-0006	1	C-0-0006	4 bytes	VM - speed command 1	0.0001 rpm	300 rpm

Fig. 8-6: Example of configurable data blocks

The resulting contents of data block 101:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
A-0-0004 axis 1				A-0-0004 axis 2				A-0-0013 axis 3		C-0-0006 master axis 1				

Example of transmissions a) Write data: (see example: data block 101 = 0x65)

Master writes:

1	2	3	4	5	6	7	8	9	10
00	00	'A'	'D'	65	00	00	07	00	00
telegram header									

11	12	13	14	15	16	17	18	19	20	21	22	23	24	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
user data														

Slave acknowledges:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

b) Read data: (see example above)

Master prompts data:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	'E'	'D'	0x65	0x00	0x00	0x00	0x00	0x00
telegram header									

Slave sends data:

1	2	3	4
00	00	00	00
telegram header			

5	6	7	8	9	10	11	12	13	14	15	16	17	18	
00	01	86	a0	00	00	03	e8	00	00	00	2d	c6	c0	hex
user data														

Handling S/P parameters when deactivating drives

In data blocks it is possible to configure Y, A and C parameters of the PPC control as well as S and P parameters of the drive in any combination. To administer the data contents of a data block the data length of all configured parameters must be known PPC internally. Data lengths of S and P parameters are stored in the drives. If a drive is deactivated then it does not participate in communications via the SERCOS interface. The PPC cannot read out the length information.

Data blocks that access parameters on a deactivated drive must be transmittable without having to change the configuration. To support this, a list of "preferred parameters" is on the PPC. The list contains preselected S and P parameters and their lengths. The administration of a data block is thus basically also possible if not all addressed drives are active or present.

Note: Given write accessing of parameters of a deactivated drive, the relevant data of the transmitted data block are discarded. Given read access, the PPC sends the value 0 for these parameters in the reaction telegram.

If a drive parameter is to be transmitted that is not in this preferred list, then it is possible to manually expand the list. Using parameters "Data blocks - configurable S and P parameters, ID number" (Y-0-0157) and "Data blocks - configurable S and P parameters, data length" (Y-0-0158) it is possible to expand up to 7 drive parameters.

The commissioning program SynTop offers a dialog for the settings with "PPC host communications". Use it to select additional S and P parameters.

List of the deposited PPC internal parameters see "Data blocks - configurable S and P parameters, ID number" (Y-0-0157) in SYNAX Parameter Description, DOK-SYNAX*-SY*-12VRS**-PA01-EN-P.

8.4 Bosch Rexroth protocol expansion

Purpose Additional to the parameter transmission via data block (DB) the following possibilities are given:

- transmission of parameter value and attribute,
- transmission of long lists > 128 bytes, e.g., cam, I/O logic.

With the Siemens S5 interface and its inherent principle of the data block, it is only possible to transmit the operating data of parameters. Accessing other elements of a parameter is only possible with an additional data access specification.

This specification supplies the Bosch Rexroth protocol expansion. It is put in front of the user data as a user data header in the first (with short transmissions, the only) telegram of a data transmission and is subject of the data format specified for user data.

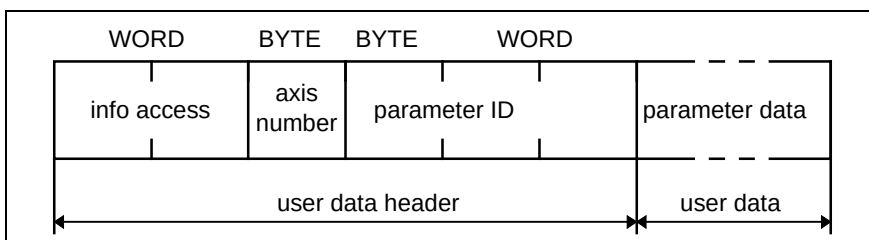
Function Without delimiting such standardized data blocks as DB99 or DB100, this protocol expansion can be used to select such functions as axis selection, I/O transmissions or mode changeovers.

The protocol expansion is made up of a data block with a six byte length and will be described in detail below for the support of

- parameter transmissions
- mode changeovers
- I/O transmissions
- axis selection for multiple transmissions.

The parameter transmission

The following user data header is used in this case



The user data header and the user data are transmitted in the data field of the communication telegram.

Access info (parameter transmission)

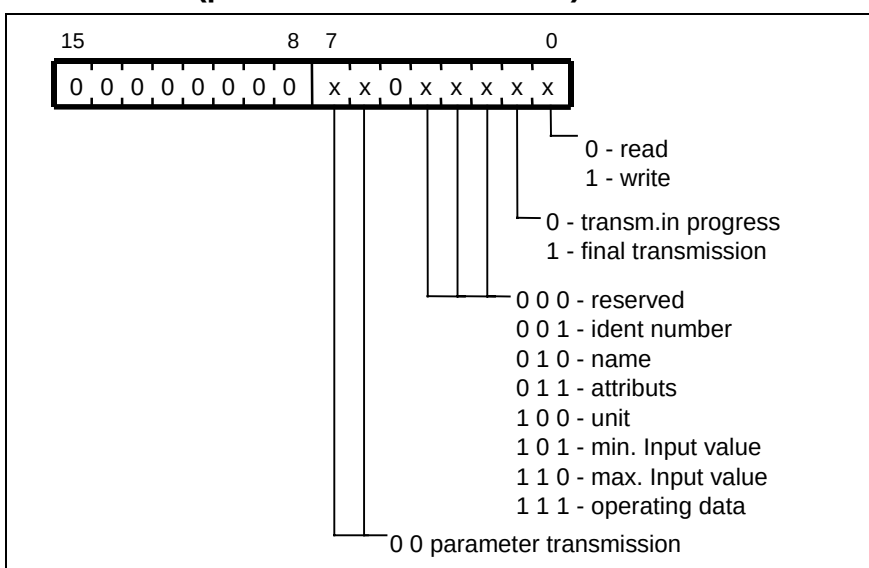


Fig. 8-7: Access info (parameter transmission)

Axis number

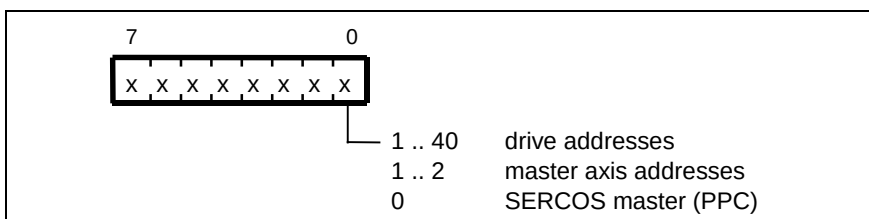


Fig. 8-8: Axis number

Parameter ID

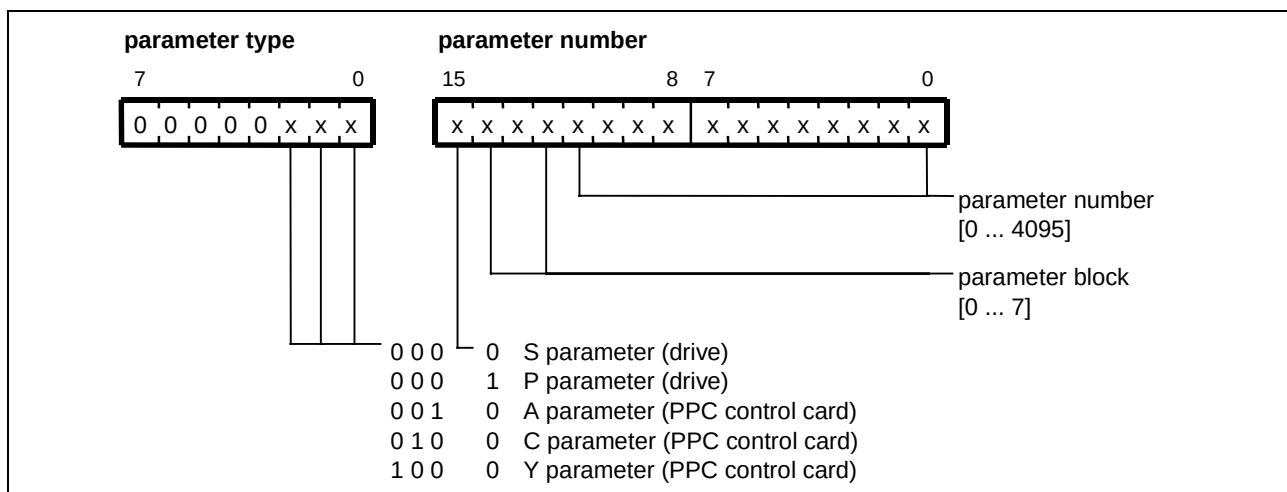


Fig. 8-9: Parameter ID (parameter transmission)

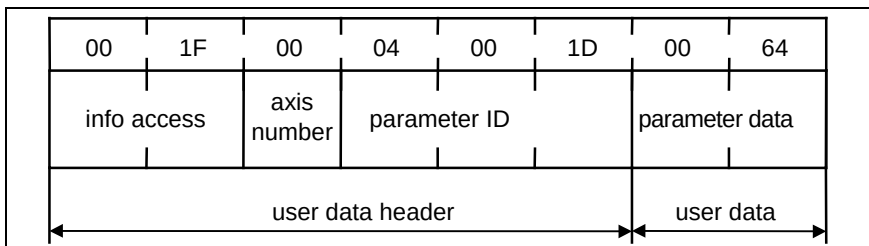
Parameter data

The length of the parameter data is outlined in the "SYNAX 200 Parameter Description", DOK-SYNAX*-SY*-12VRS**-PA01-EN-P.

Writing a parameter

When writing, the superior system transmits parameter data to the PPC.

Example: Transmit parameter "Jogging time constant" (Y-0-0029) = 100, parameter length: two bytes.



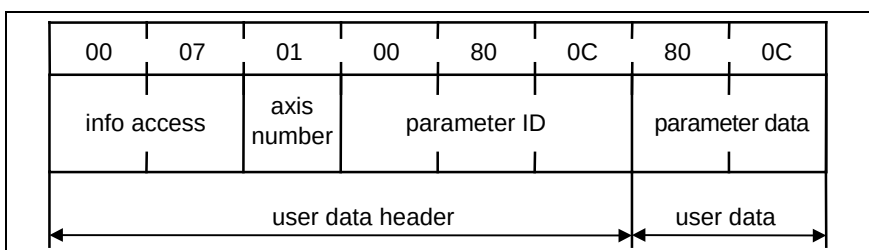
Reading the data status

To check a procedure command execution, it is necessary to read the data status. This is supplied as per SERCOS interface when write requesting the ID number of a parameter in the form of user data.

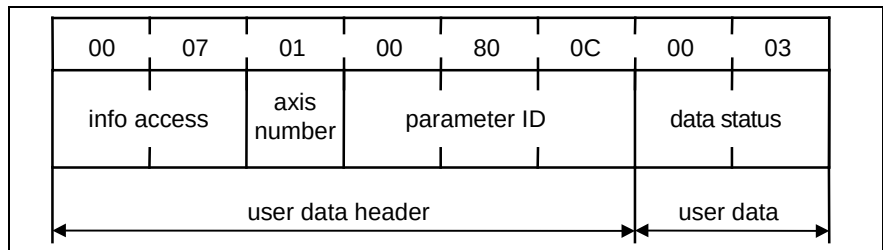
The write request is performed with the identity number (2-byte data) of the procedure command.

Example: The following is an example of a check of the data status of the procedure command "C300 command set absolute measurement" (P-0-0012, drive 1).

Command telegram:



The PPC now sends the data status:



Data status 0x0003 shows that the procedure command is set, enabled and executed successfully.

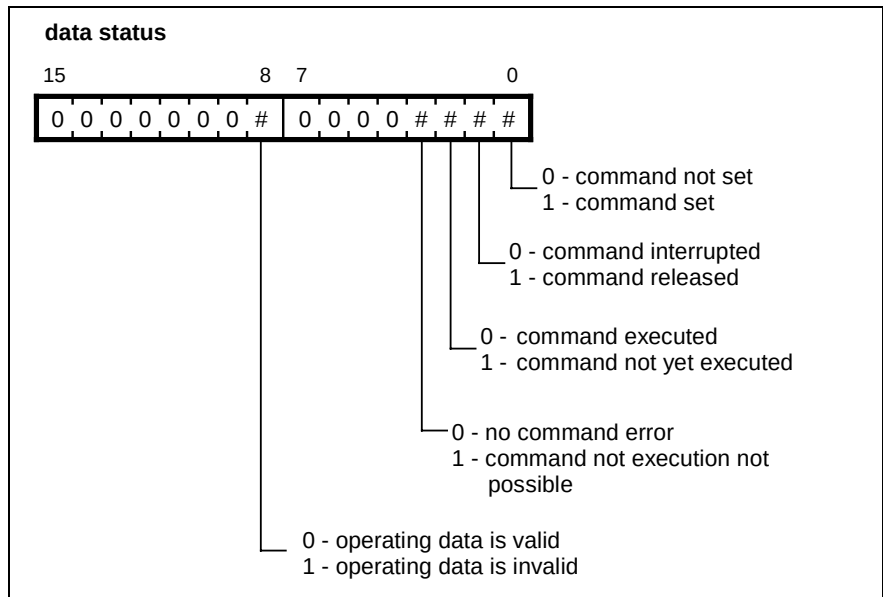
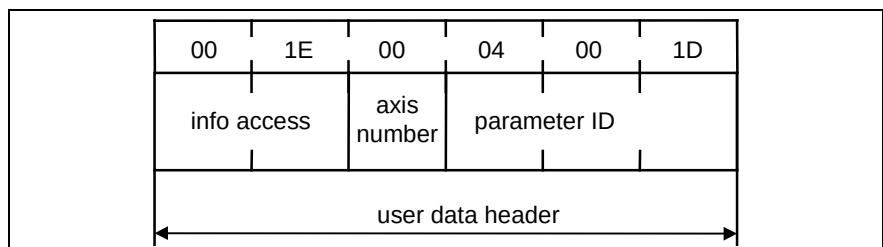


Fig. 8-10: Data status

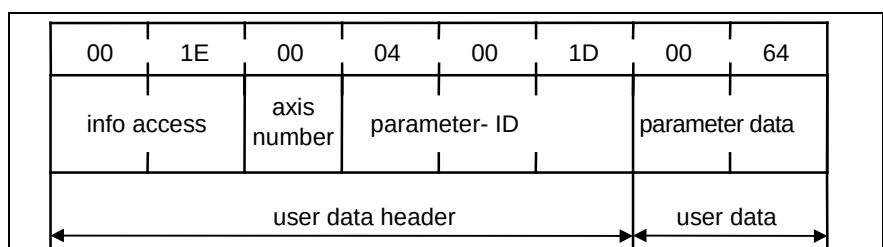
Reading a parameter

When reading, the superior system sends a transmission prompt to the PPC.

Example: Read parameter "Jogging time constant" (Y-0-0029)



The PPC now sends the parameter contents to the superior system.



If the telegram is longer than 1024 bytes, then the byte info access must contain the message "transmission in progress" (bit 1 = 0).

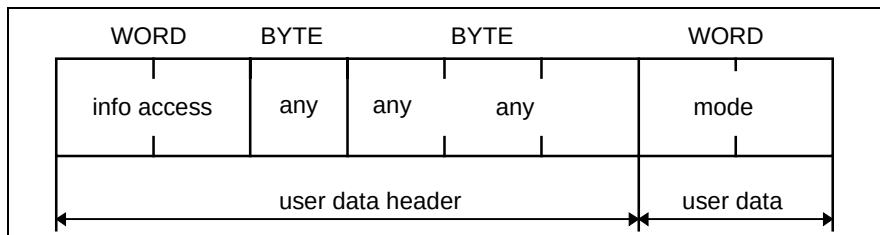
To send the remaining segments of the data it is necessary to repeat all processes until transmission has been completed. The completion is then signaled by the PPC with "final transmission" (bit 1 = 1).

Switch mode

With the help of switch mode, it is possible to switch the PPC between operating and parameterization mode. Otherwise, the current mode can be queried.

The switch mode protocol is largely identical to parameter transmission.

Switch mode protocol (user data)



Info access (switch mode)

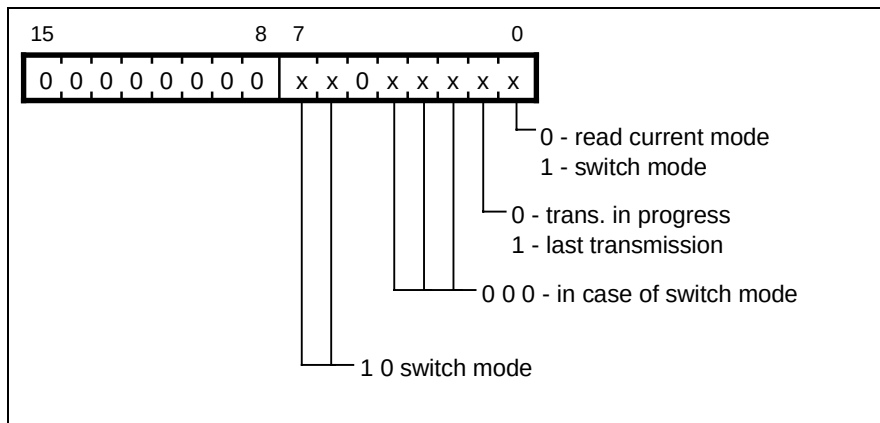


Fig. 8-11: Info access (switch mode)

Mode

= 2: parameterization mode

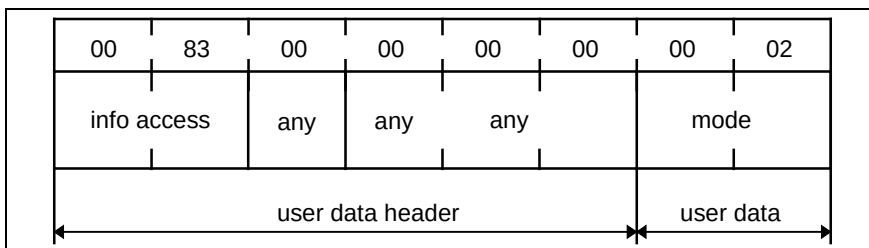
= 4: operating mode

Switch mode order

The superior system sends a switch sequence to the PPC.

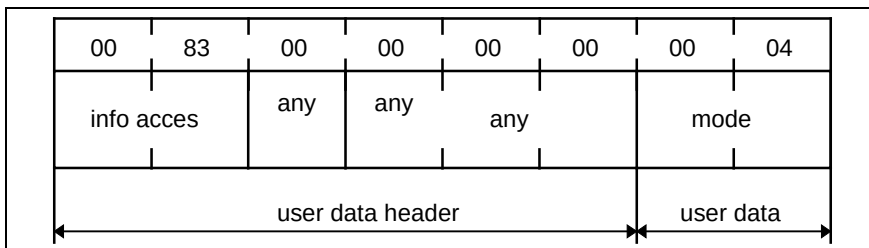
Switching into parameterization mode

To switch into parameterization mode, the superior system sends the following sequence to the PPC:



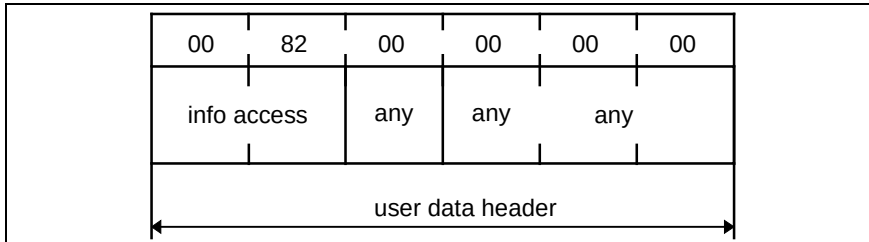
Switching to operating mode

To switch into operating mode, the superior system sends the following sequence to the PPC:

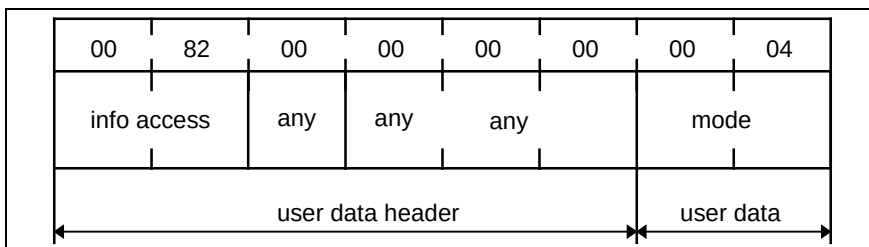


Reading current operating mode

To read the current operating mode, the superior system sends the following sequence to the PPC:



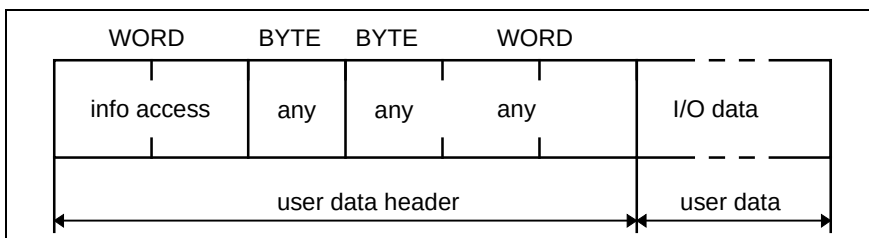
If the PPC is in operating mode, then the response is as follows:



I/O transmission

The assignment possibilities of the digital I/Os of the PPC are described in the SYNAX 200 functional description, DOK-SYNAX*-SY*-12V*1/2-FK01-EN-P, section 7, "Control signals (binary inputs/outputs)" in detail.

The following user data header is used for the I/O transmissions:



Telegram length is specified in bytes.

Info access (I/O transmission)

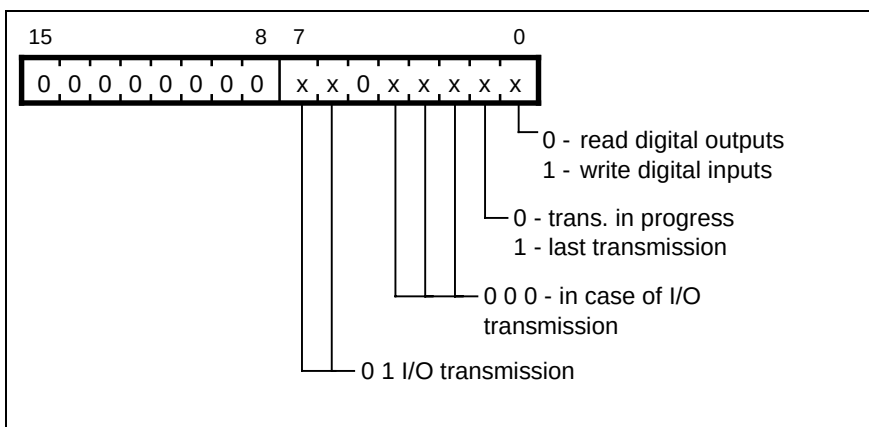


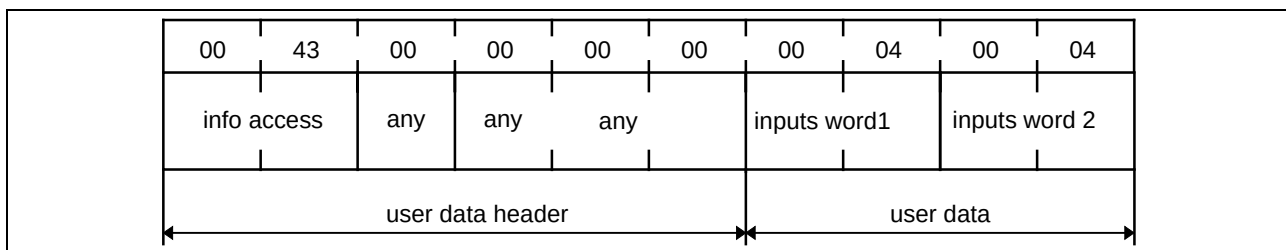
Fig. 8-12: Info access (I/O transmission)

The I/Os are organized in words, i.e., 2, 4 and 6 bytes are always transmitted.

Writing the digital inputs of the PPC

When writing, the superior system sends the digital inputs to the inputs of the PPC.

Example: Transmission of two word inputs 0A0F and B0E0

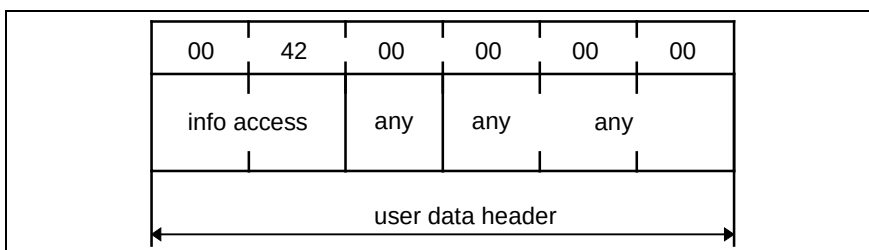


Reading the digital outputs of the PPC

When reading, the superior system sends a transmission request to the PPC.

Example: Reading the digital outputs.

Note: This sequence is always the same.

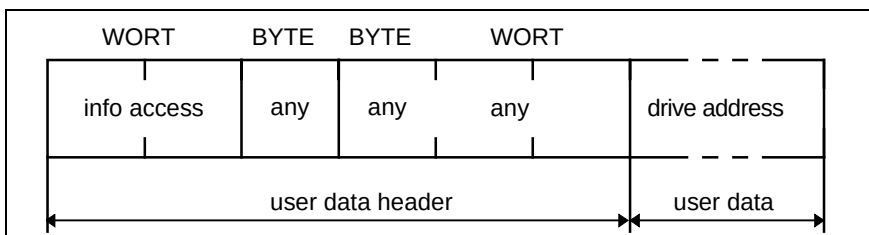


The PPC now sends its digital outputs to the superior system.

Select axis for multiple transmission

Considering the relatively slow transmission speeds of serial interfaces, it is advantageous to transmit a data record to several drives, only once to the PPC. Once the PPC has the information to which drive the data must be transmitted to, then a multiple transmission can be internally run.

The following is the user data header for axis selection:



Info access (select axis)

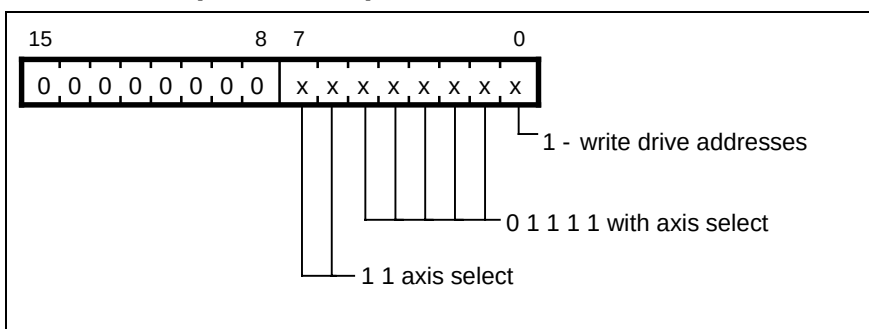


Fig. 8-13: Info access (axis select)

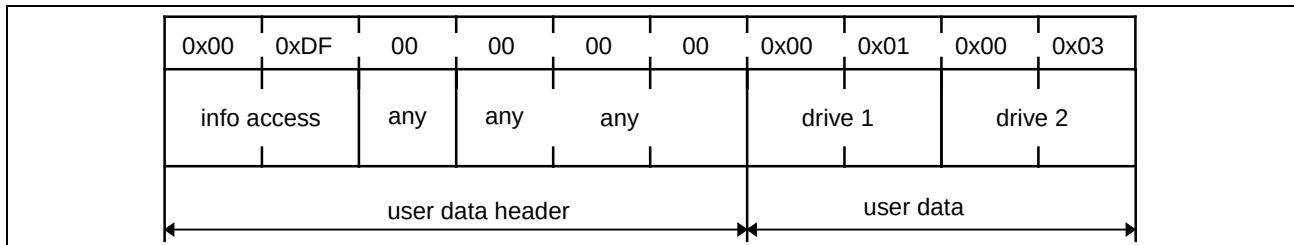
Axis select order

The drive addresses are organized in words. This means that 2, 4 and 6 bytes are always transmitted.

Writing the drive addresses

When writing, the superior system sends drive addresses to the PPC.

Example: Transmit drive addresses 1 and 3

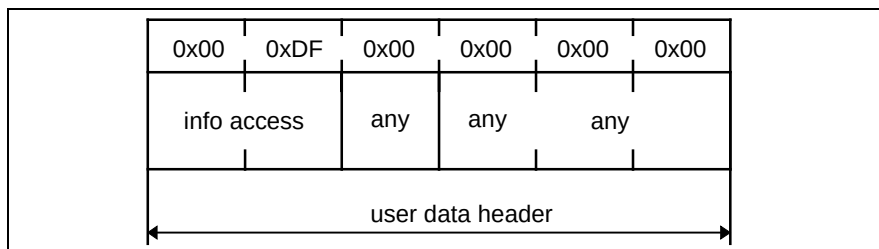


Once an axis has been selected, it is relevant for all write accesses of A, S or P parameters. This applies until the select list is changed or cleared.

The axis selection is cleared by means of any empty drive address list (no user data).

Example: Clearing the axis selection list.

Note: This sequence is always the same.



Note: A reading of axis selection is not intended. A read access will clear the list.

8.5 3964R with Bosch Rexroth protocol expansion

The protocol expansion resulting from the 3964R makes it possible to access every C, A, S or P parameter without preliminary definition of data blocks. In addition to the operating data, it is also possible to simultaneously read their unit, name, attribute, minimum and maximum values.

Note: 3964R with protocol expansion is supported by standard protocol drivers for write access only (SEND telegrams). A protocol driver for read access (FETCH telegram) is needed that can also issue the protocol expansion over and above the telegram header with FETCH telegrams. In conjunction with the 3964R, bit 1 of the information access of the protocol expansion (in progress/last transmission - see "Bosch Rexroth protocol expansion", [page 8-15](#)) is of no significance. It is automatically handled by the PPC via the telegram length.

Standard transmission with data block D0

There is no restriction as to the length of parameter tables.

The data field is divided into user data header and user data with protocol expansion. Both the user data header and the user data correspond to the specification outlined in section "Order of transmission", [page 8-3](#). The user data are only in SEND telegrams.

1	2	3	4	5	6	7	8	9	10
0x00 (0xff)	0x00	command		high.	low	high	low	CPU-no./KM	
				target/source		number		0	0
telegram header									

11	12	13	14	15	16	17	...
info access		axis no.	parameter ID			parameter data	
user data header						user data	

- Byte definition:**
- 1 telegram ID (0x00, or 0xff with following telegrams),
 - 2 telegram ID (always 0x00),
 - 3 command ==> 'A' = SEND, 'E' = FETCH,
 - 4 command type, i.e., type of data to be transmitted ==> 'D' = data block, all other types not used,
 - 5,6 target address with SEND telegrams (0,0),
source address with FETCH telegrams (0, 01)
 - 7,8 number of user data to be transmitted including user data header in the unit [word].
 - 9 byte number of the coordination flag (only in S5),
 - 10 bit number of the coordination flag and CPU no. (Only in S5).
 - 11...16 see section "Bosch Rexroth protocol expansion", [page 8-15](#)

Bytes 6, 9 and 10 of the telegram header have no meaning in the PPC.

The response telegram is structured as follows. The response telegram is made up of only one telegram header in SEND telegrams. In FETCH telegrams, only the user data follow. The user data header is not repeated.

Slave

1	2	3	4	5	...
0x00 (0xff)	0x00	comm and	error no.	parameter data	
telegram header				user data	

- Byte definition:**
- 1 telegram ID (0x00, or 0xff with following telegrams),
 - 2 telegram ID (always 0x00),
 - 3 command (always 0x00),
 - 4 error number ==> 0 = no error, otherwise error.

Examples for a transmission a) Writing data: (axis 3, parameter A-0-0170 = 13)

Master writes:

1	2	3	4	5	6	7	8	9	10
		command		target/source		number		CPU-no./KM	
0x00	0x00	'A'	'D'	0x00	0x00	0x0005		0x00	0x00
telegram header									
11	12	13	14	15	16	17	18	19	20
Info access		axis- no. 0x03	parameter ID 0x0100AA			parameter data 0x0000000D			
0x001F									
user data header						user data			

Slave acknowledges:

1	2	3	4
0x00	0x00	0x00	error no. 0x00
telegram header			

b) Read data: (axis 2, parameter A-0-0126= 7)

Master prompts data:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	command		target/source		number		CPU-no./KM	
		'E'	'D'	0x00	0x00	0x00	0x00	0x00	0x00
telegram header									

11	12	13	14	15	16
info access		axis no. 0x02	parameter ID		
0x00	0x00		0x01007E		
user data header					

Slave transmits data:

1	2	3	4	5	6	7	8
0x00	0x00	0x00	error no. 0x00	parameter data 0x00000007			
telegram header				user data			

c) Axis select for multiple transmission: (drive addresses 1 and 4)

Master writes:

1	2	3	4	5	6	7	8	9	10
		command		target/source		number		CPU no./KM	
0x00	0x00	'A'	'D'	0x00	0x00	0x0005		0x00	0x00
telegram header									

11	12	13	14	15	16	17	18	19	20
info access		axis-no.	parameter ID			drive address		drive address	
0x00DF		0x00	0x000000			0x0001		0x0004	
user data header									

Slave acknowledges:

1	2	3	4
0x00	0x00	0x00	error no. 0x00
telegram header			

The following applies when selecting an axis for multiple transmission:

- The list with the selected drives (drive list) is empty after the PPC is run up. Data is transmitted, as usual, in individual transmissions.
- A prepared drive list remains relevant for write access with the Bosch Rexroth user level until it is changed or cleared.
- The drive list is cleared by a select command without drive address.
- The choice of axis has no effect on data transmission with data blocks, on Y and C parameters or any type of read access.
- Transmission to the drive is done in the sequence set by the selection list.
- Multiple transmission of S and P parameters in parameterization mode requires much more time than in operating mode. This also applies to A parameters when in operating mode. The transmission of S and P parameters in this case runs parallel.
- A sequential data transmission is interrupted in the event of an error. This is not true in case of parallel transmission because the prompt is first sent to all drives and then the replies are evaluated.

Note:

When loading a long list (more than 1984 data words), the PPC sends, prior to the internal SERCOS transmission, a positive reaction telegram on the final partial telegram of the list. The results of the SERCOS transmission can be checked by reading parameter "Serial interface error number" (Y-0-0057).

Up to the end of the SERCOS transmission or the making available of the results on Y-0-0057, all command telegrams from the PPC will be responded with a BUSY signal. In other words, Y-0-0057 is polled until the PPC sends a positive reaction telegram again.

Using the Siemens S 5 to send long lists

Lists with a maximum of 2043 data words can be sent with the standard S5 control. This restriction does not apply to other master units.

Data blocks DB1 -DB5 are available to support the transmission of long parameter lists, even with a standard Siemens S5 control, e.g., cam tables P-0-0072. Like DB0, these also cannot be configured and only support Bosch Rexroth protocol expansion.

In contrast to DB0, a normal telegram is expected after 30 following telegrams. Several aspects of this normal telegram must correlate with the first normal telegram. The following applies to DB1 to DB5.

- They cannot be configured.
- They only support the Bosch Rexroth user level (at long lists).
- They are compatible with the Siemens S5.
- They are limited. This means that after a sequence of one normal telegram and 30 following telegrams, precisely one normal telegram, and maximum of 30 following telegrams, is now expected until all the data is transmitted.
- In the following normal telegrams, the DBs are counted down to DB1.
- The data length of the remaining data record must be contained in the telegram header of the last normal telegram (only values $\leq$ block length are allowed). The maximum block length of 1984 data words is always contained in the previous telegrams.
- Command (0x41) and command type (0x44) must always agree in the following normal telegrams and the DB number must be decremented. This is monitored.
- The following normal telegrams are treated in the PPC as a following telegram. This is why the data length in the following normal telegrams is not monitored.

The length of the data record determines which data block starts data transmission. The number nn of the data block results from the equation

$$nn = \frac{\text{number of data words} + 1983}{1984}$$

Fig. 8-14: Number of the data block

Example: With 2053 data words (cam),

$$nn = \frac{2053 + 1983}{1984} = \frac{4036}{1984} = 2$$

Fig. 8-15: Example: Number of the data block

as the relevant number of the data block. Loading a cam (P-0-0072) out of the Siemens S5 into PPC thus looks like this:

- First, 1984 data words are sent to data block DB2 of the PPC from the expanded DB of the S5,
- now the remaining 69 data words follow to the DB1 of the PPC from any DB of the S5.

With data blocks DB1 - DB5, a maximum of $(1984 \cdot 5 =) 9920$ data words (equals 19.375 Kbytes data) can be transmitted to the PPC.

Example for writing to a cam (P-0-0072), drive 1) with Siemens S5

The Siemens-S5 sends the SEND telegram for the DB2:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	0x41	0x44	0x02	0x00	0x07	0xc0	0x00	0x00
telegram header									

11	12	13	14	15	16	17	18	19	20
0x00	0x1f	0x01	0x00	0x80	0x48	0x10	0x00	0x10	0x00
user data header						list length (actual, maximum)			

21	22	23	24	25	26	27	28	29	30
0x00	0x00	0x00	0x01	0x00	0x00	0x00	0x02	0x00	0x00
user data (always 4 bytes per support point of the cam)									

31	32	33	34	..	..	135	136	137	138
0x00	0x03	0x00	0x00	..	..	0x00	0x1d	0x00	0x00
user data (a total of the first 118 bytes of the cam)									

The PPC sends response telegram:

1	2	3	4
0x00	0x00	0x00	0x00
telegram header			

The Siemens S5 sends the first following SEND telegram:

1	2	3	4
0xff	0x00	0x41	0x44
telegram header			

5	6	7	8	9	10	11	12	13	14
0x00	0x1e	0x00	0x00	0x00	0x1f	0x00	0x00	0x00	0x20
user data (always 4 bytes per support point of the cam)									

15	16	17	18	..	..	129	130	131	132
0x00	0x00	0x00	0x21	..	..	0x00	0x3d	0x00	0x00
user data (a total of the first 128 bytes of the cam)									

The PPC sends the first following response telegram:

1	2	3	4
0xff	0x00	0x00	0x00
telegram header			

The Siemens S5 sends the second following SEND telegram:

1	2	3	4
0xff	0x00	0x41	0x44
telegram header			

5	6	7	8	9	10	11	12	13	14
0x00	0x3e	0x00	0x00	0x00	0x3f	0x00	0x00	0x00	0x40
user data (always 4 bytes per support point of the cam)									

15	16	17	18	..	..	129	130	131	132
0x00	0x00	0x00	0x41	..	..	0x00	0x5d	0x00	0x00
user data (a total of the first 128 bytes of the cam)									

The PPC sends the second following response telegram:

1	2	3	4
0xff	0x00	0x00	0x00
telegram header			

The Siemens S5 ends the 30th following SEND telegram:

1	2	3	4
0xff	0x00	0x41	0x44
telegram header			

5	6	7	8	9	10	11	12	13	14
0x03	0xbe	0x00	0x00	0x03	0xbf	0x00	0x00	0x03	0xc0
user data (always 4 bytes per support point of the cam)									

15	16	17	18	..	..	129	130	131	132
0x00	0x00	0x03	0xc1	..	..	0x03	0xdd	0x00	0x00
user data (a total of the first 128 bytes of the cam)									

The PPC sends the 30th following response telegram:

1	2	3	4
0xff	0x00	0x00	0x00
telegram header			

The Siemens S5 sends the SEND telegram for DB1:

1	2	3	4	5	6	7	8	9	10
0x00	0x00	0x41	0x44	0x01	0x00	0x00	0x45	0x00	0x00
telegram header									
11	12	13	14	15	16	17	18	19	20
0x03	0xde	0x00	0x00	0x03	0xdf	0x00	0x00	0x03	0xe0
user data (always 4 bytes per support point of the cam)									
21	22	23	24	..	..	135	136	137	138
0x00	0x00	0x03	0xe1	..	..	0x03	0xfd	0x00	0x00
user data (a total of the first 128 bytes of the cam)									

The PPC sends the reaction telegram:

1	2	3	4
0xff	0x00	0x00	0x00
telegram header			

The Siemens S5 sends the first and simultaneously the last following SEND telegram:

1	2	3	4
0xff	0x00	0x41	0x44
telegram header			

The PPC sends the first and simultaneously the last following response telegram after transmission to the drive(s).

5	6	7	8	9	10	11	12	13	14
0x03	0xfe	0x00	0x00	0x03	0xff	0x00	0x00	0x04	0x00
user data (the final ten bytes of the cam)									

1	2	3	4
0xff	0x00	0x00	0x00
telegram header			

Note: When loading a long list (more than 1984 data words), the PPC sends, prior to the internal SERCOS transmission, a positive reaction telegram on the final partial telegram of the list. The results of the SERCOS transmission can be checked by reading parameter "Serial interface error number " (Y-0-0057).
Up to the end of the SERCOS transmission or the making available of the results on Y-0-0057, all command telegrams from the PPC will be responded to with a BUSY signal. In other words, Y-0-0057 is polled until the PPC sends a positive reaction telegram again.

If some of DB1 or DB5 are already assigned, then the previously described transmission of a cam can also be executed with the additional data blocks DB6 through DB10. The block numbers correlate as follows:

DB5	<————>	DB10
DB4	<————>	DB9
DB3	<————>	DB8
DB2	<————>	DB7
DB1	<————>	DB6

DB2 must be released to DB7 and DB7 and DB1 by DB6 in the telegrams in order to transmit a cam.

8.6 Hardware

Terminals X10 and X16 can be used for communications. The connector must be set in parameter "Host communication - control word" (Y-0-0033).

Note: The PROG interface (X10) is isolated and should preferably be used for permanent interfaces (e.g., 3964R, SynTop via RS485) or long cables (e.g., SynTop with long serial cable).

RS232

Connector assignment RS232

Ready-made cable IKB0005 is available for connecting those participants which have nine pin, D subminiature connectors.

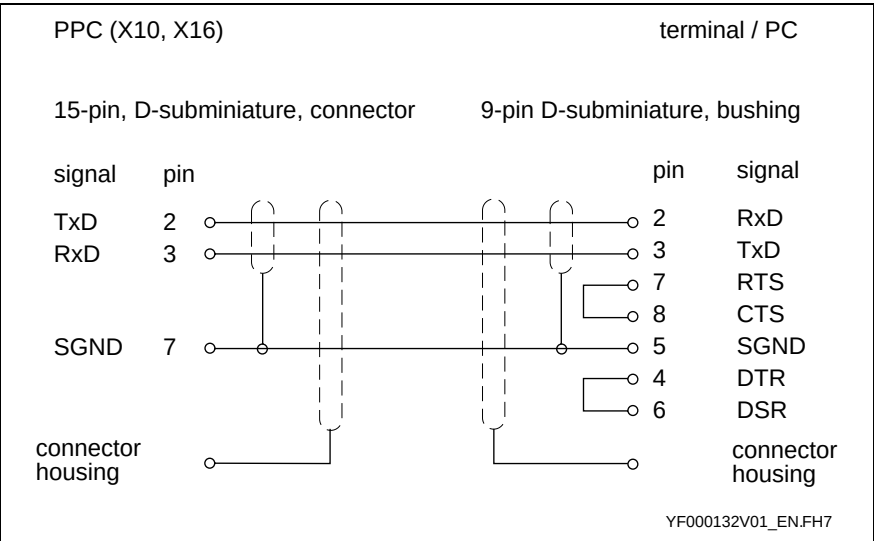


Fig. 8-16: Serial cable IKB0005

RS422

Cable map for RS422

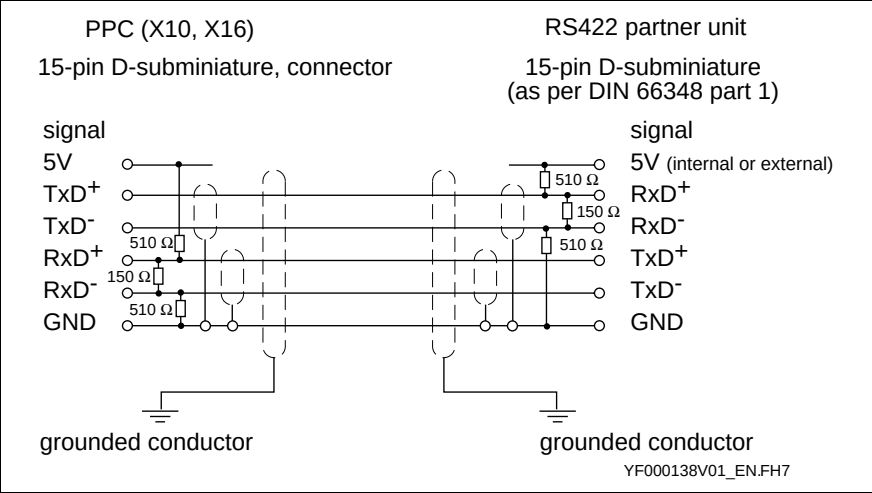


Fig. 8-17: Serial cable for RS422 from the PPC to the RS422 partner unit

Note: The baud rate of the 3964R is always 9600 baud (8 data bits, 1 stop bit and even parity).

Pin assignment on the PPC at RS422

Signal	Pin
5V	12
TxD ⁺	9
TxD ⁻	11
RxD ⁺	4
RxD ⁻	5
GND	7

Fig. 8-18: Pin assignment on the PPC at RS422

For the connection to participants with 15-pin D-subminiature connector (bushing) is the multiple shielded cable IKN0234 available. This cable must be ready-made with terminators at the receiver and balancing resistors as well as connectors (bushings) as described in the cable map for RS422 (Fig. 8-18).

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10 Service & Support

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